

Data visualization guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 1.0 (review draft) October 2026 All example data is synthetic

OVERVIEW

How to use this guide

One set of rules, three ways in. The guide teaches the thinking; the starter kit makes the right answer the easy one in every tool we use.

Every chart we make should do one job: answer a business question so clearly that the reader knows what to do next. The rules here are written to be checked, by a reviewer or by an agent, before a chart leaves the team.

Inspired by the structure of a leading consulting firm's public data-visualization guide, rebuilt for McKesson's brand and for the work of Workforce Innovation.

CONSULTANTS AND ANALYSTS

Decks, reports, workbooks

Read Fundamentals once, then use the chart pages as you build. Keep the cheat sheet next to you.

- PowerPoint template and starter deck
- Excel chart workbook with the palette
- One-page cheat sheet and chart chooser
- Power BI report theme

DEVELOPERS AND DATA SCIENTISTS

Python, HTML, R, Power BI

Each chart page links to tested code. Themes are generated from one token file, so every tool matches.

- `mckviz` for matplotlib and Plotly
- CSS variables, Plotly and Vega-Lite themes
- `theme_mck()` for ggplot2
- `mv.lint(fig)` to check a chart

CODING AGENTS

Claude, Copilot and others

Agents read `AGENTS.md` automatically in this repo. For other repos, copy one self-contained rules file.

- `AGENTS.md`: workflow and rules
- `mck-chart` skill: question to QA
- `dist/agents/mck-viz-rules.md`
- Cookbook pages with runnable snippets

The five steps behind every chart

1 Question. Write the question the chart answers in one sentence.

2 Chart. Pick the chart from the question type, not from habit.

3 Build. Grey first, then color only what the headline names.

4 Headline. Compute the finding, then write it as a sentence.

5 QA. Run the checklist and look at the rendered chart.

OVERVIEW

Contents

Fundamentals apply to every chart. Each chart family then gets the same treatment: anatomy, variants, best practice and how to build it.

Each chart page links to its cookbook page in `docs/cookbook/`, with tested code and click-by-click steps for every tool.

→ **Chart chooser** 4
Start here: from the question to the chart, linked to every chart page; the common traps

01 **Fundamentals** 6
Question first, how readers read, anatomy, headlines, color, type, clutter, honest scales, people data, accessibility

02 **Magnitude** 17
Bar and column charts: anatomy, variants, best practice, build it

03 **Part-to-whole** 22
Stacked bars, waterfalls, treemaps, donuts

04 **Change over time** 33
Line, stacked area, bubble timeline, small multiples

05 **Ranking** 44
Slope, bump, dot plot and dumbbell, heat table

06 **Distribution** 53
Strip, box plot, histogram

07 **Correlation** 59
Scatter and bubble, why not dual axes, heatmaps

08 **Relationships and flows** 67
Sankey, parallel coordinates, Venn

09 **Geo-spatial** 74
US choropleth and bubble maps, world maps, zoom-ins, choosing the geography

10 **Single number** 84
KPI and big number, progress to target

11 **Formats** 90
Slides, two charts on a slide, reports and print, web and mobile, dashboards, interactivity, animation, social

→ **Cheat sheet** 99
The whole guide on one page: anatomy, headline, color, type, checklist

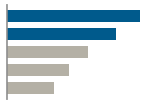
12 **Toolkit** 100
Downloads, set-up for each tool, the QA checklist, guidance for agents

Chart chooser

Pick the chart from the question, not from habit.

1. Write the question the chart answers in one sentence.
2. Find the question type and start with its first choice.
3. Check the traps on the next page.
4. Follow the chart's page to build it.

Every chart name links to its page in this guide. **Avoid** lists the charts that look right for the question but mislead or slow the reader.



Magnitude

How big? Which is bigger, by how much?

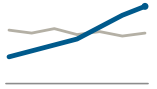
START WITH	**Sorted bar**
ALSO	Dot plot for values far from zero; paired bars
AVOID	Rainbow, 3D or cut-off bars



Part-to-whole

What is it made of? How does a total break down?

START WITH	**100% stacked bar**
ALSO	Waterfall between totals; treemap for many parts
AVOID	Pies over 5 slices



Change over time

How is it moving? When did it change?

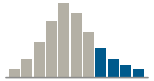
START WITH	**Line**, focus series in blue
ALSO	Stacked area; small multiples; bubble timeline
AVOID	Ten colored lines



Ranking

Who's on top? Who moved?

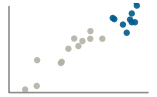
START WITH	**Sorted bar**; **slope** for change
ALSO	Dumbbell for us vs. them; heat table
AVOID	A color for every line



Distribution

How spread out? What's typical?

START WITH	**Strip plot** or **histogram**
ALSO	Box plot to compare many groups
AVOID	An average with no spread



Correlation

Do two measures move together?

START WITH	**Scatter** or bubble scatter
ALSO	Heatmap: two categories, one measure
AVOID	Dual axes



Relationships and flows

How do things connect or move?

START WITH	**Sankey**
ALSO	Parallel coordinates; Venn, 3 sets max
AVOID	A color per stage



Geo-spatial

Where? Does place matter?

START WITH	**Choropleth** for rates
ALSO	Bubble map for counts; world map; zoom-in
AVOID	Counts on a choropleth



Single number

What one figure should they remember?

START WITH	**Big number** with a comparison
ALSO	Progress bar or ring for a target
AVOID	Speedometers

CHART CHOOSER

Before you build: the common traps

The right chart type can still go wrong. If you are about to do one of these, use the fix instead.

The page numbers refer to this guide's PDF; on screen, each fix links to its page.

ABOUT TO	INSTEAD	
Cut the bar axis so small differences show	Bars from zero; a dot plot for a narrow range	p. 19
Give each bar its own color	Grey bars, the one that matters in blue	p. 19
Put two units on two axes	Two aligned panels, or index both to 100	p. 62
Draw ten lines in ten colors	All grey, one or two in blue; or small multiples	p. 35
Make a pie with eight slices	A sorted bar; donuts take five parts at most	p. 31
Stack eight segments in a bar	Five at most, the story segment blue on the baseline	p. 24
Color a waterfall green up, red down	Favorable teal, unfavorable orange; or grey with one blue	p. 27
Map raw counts by state	Map rates; show counts as bubbles	p. 80
Show a KPI as a speedometer	A big number with a comparison, or a bar to target	p. 87
Paste a 30-row table to make one point	Chart the point; keep the table for the appendix	p. 50
Put four charts on one slide	Two at most, each with its own insight headline	p. 92
Show a team of 7 people	Suppress or combine groups under 10	p. 15

Start with the question, not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

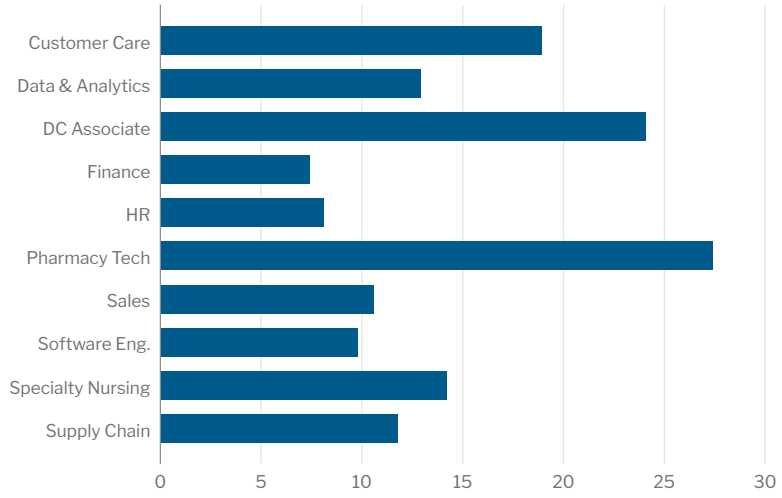
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

Attrition by job family

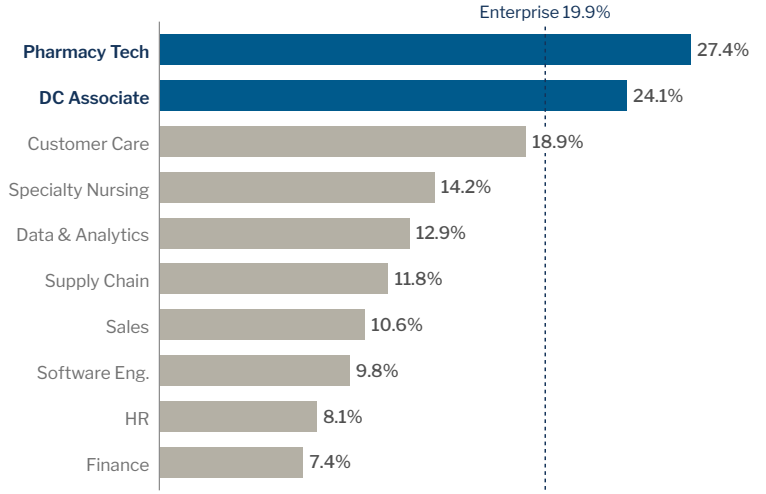
FY26



Shown
A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Explained
The headline states the finding, the order ranks the families, and color points to the two the headline names.

How readers read a chart

Readers move through a chart in three steps. Design for each one, in order.

1 Perceive

In the first second the eye takes in shape, length and color before reading a word. Make the thing that matters the thing that stands out: one color against grey.

2 Interpret

Next the reader looks for meaning: headline, labels, units, benchmarks. The headline should confirm what they just saw, or sharpen it.

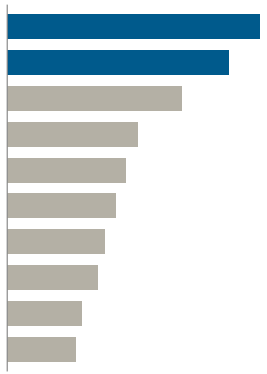
3 Act

Finally they ask "so what?". A callout that sizes the implication turns a finding into a decision.

STEP 1 · PERCEIVE

Two bars stand out

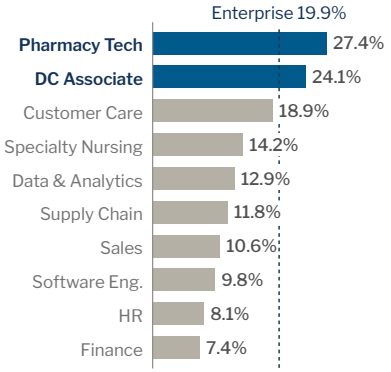
No words yet. Grey context and blue focus already tell the reader where to look.



STEP 2 · INTERPRET

Labels give it meaning

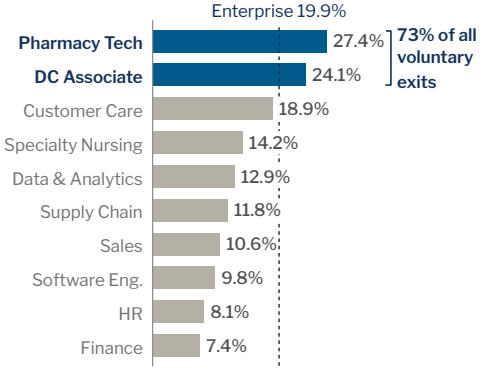
Names, values and a labelled benchmark show the two families sit well above 19.9%.



STEP 3 · ACT

The callout sizes the prize

Together they hold 58% of headcount but 73% of voluntary exits: a retention plan for two families moves the enterprise number.



FUNDAMENTALS

Anatomy of a chart

Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

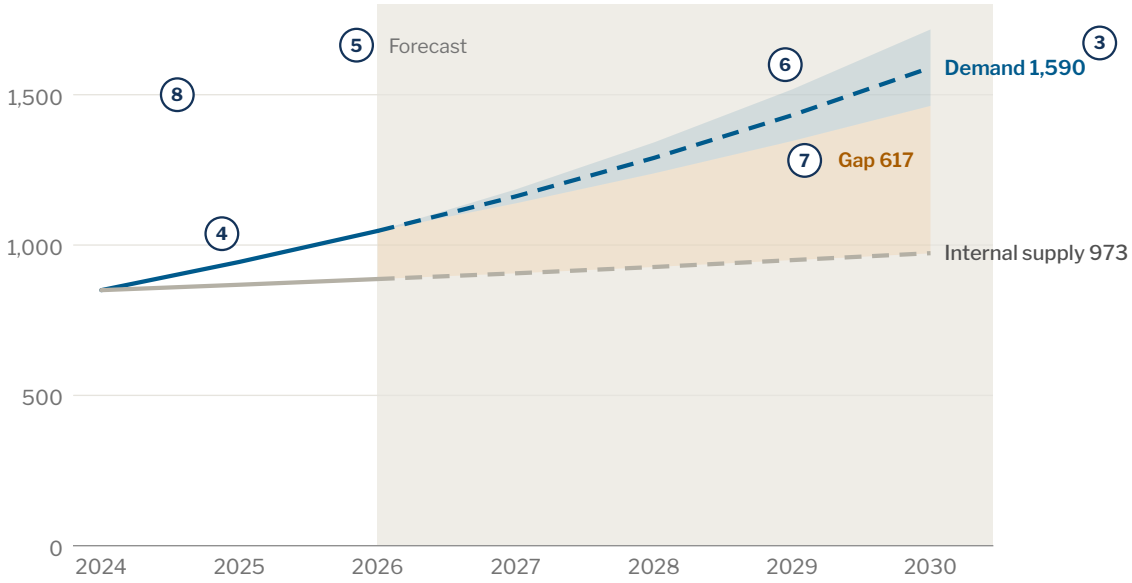
Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.

Internal supply covers only 61% of Data & AI demand by 2030, a gap of about 620 FTE (1)

Data & AI roles: forecast demand vs. internal supply, 2024–2030 (FTE) (2)



Forecast range on demand widens to ±8% by 2030. Internal supply = current staff plus planned reskilling, less expected attrition.

Source: Workday HCM and SWP model (synthetic data for illustration), as of 31 Mar 2026 (9)

Write the headline last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

The formula

Who or what + verb + number + so what.

Before you keep it

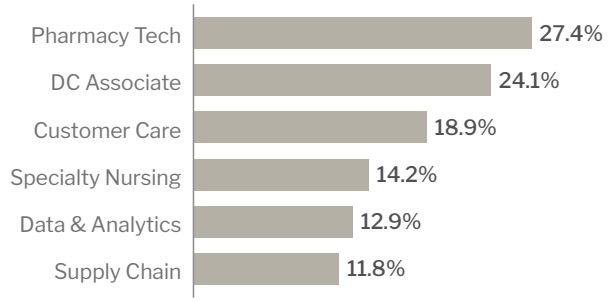
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

LABEL	Attrition by job family · names the topic; says nothing
VAGUE	Attrition varies across job families · true of every chart ever made
OPINION	Attrition is out of control in the frontline · a claim the chart can't support
INSIGHT	**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate**
INSIGHT + ACTION	**Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate**

Attrition varies across job families

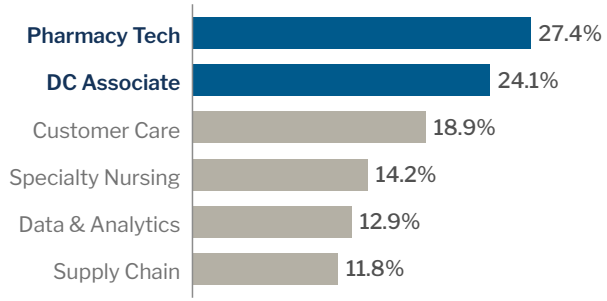
Voluntary attrition by job family, FY26 (%)



Headline and chart both neutral
Nothing tells the reader where to look.

Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)



Headline and highlight agree
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

Grey first, then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

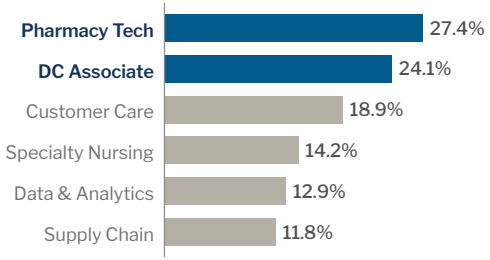
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

Orange means "worse". Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

Focus

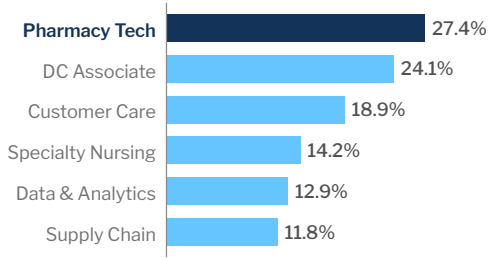
All grey; the items the headline names in blue.



PATTERN 2

Uniform + accent

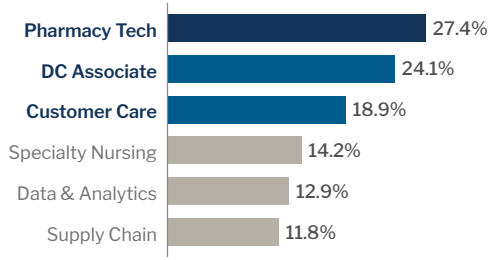
One series in light blue; the single exception in navy.



PATTERN 3

Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.



Color roles

Context · everything the reader should see but not focus on	`#B4B0A5`
Focus · what the headline names	`#005A8C`
Accent · the one exception that must pop (navy, not orange)	`#143359`
Light blue · the series when navy marks the exception	`#66C4FF`
Orange · only when it means worse: below target, shrinking, a shortfall	`#EF8200`
Total and benchmark · totals, targets, reference lines	`#143359`
Context light · de-emphasized fills in stacks and areas	`#DEDBD1`

Three palettes, three jobs

Pick the palette from what the color encodes.

Categories

Only when categories must be told apart. At most six, always in the same order, so "Full-time" is the same blue on every slide.

One measure, light to dark

For heat tables and choropleths of a single measure. Darker means more.

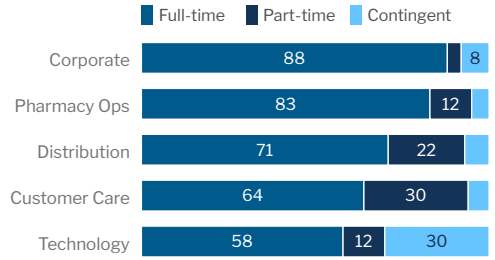
Gap to a benchmark

For values centered on zero or a target. Orange below, blue above, a neutral middle.

Text on a fill switches between white and navy, whichever contrasts more (`mv.text_on`).

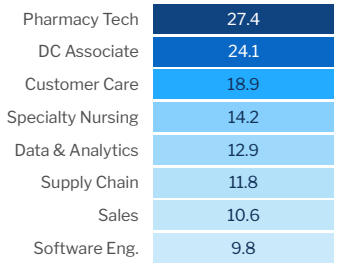
CATEGORICAL

Workforce mix by function (%)



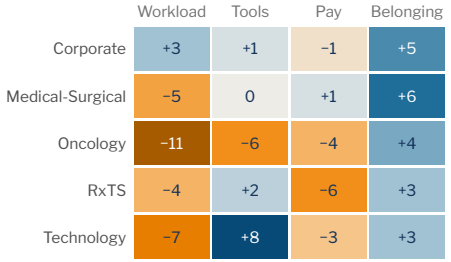
SEQUENTIAL

Voluntary attrition, FY26 (%)

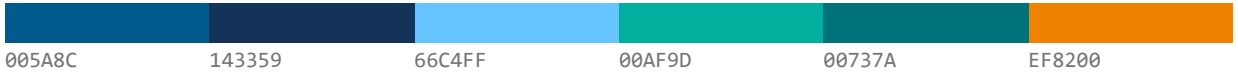


DIVERGING

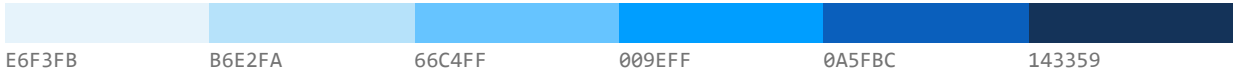
Engagement vs. healthcare benchmark (pts)



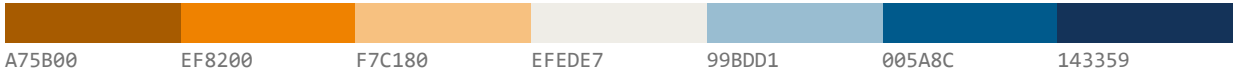
Categorical
Max 6, always this order



Sequential
One measure, light to dark



Diverging
Centered on 0 or a target



Type and proportion

One typeface, Libre Franklin, in a few fixed roles. Type should frame the data, not compete with it.

The headline is the largest text on the chart but the data is the largest *thing*. If the headline is the first thing you notice from across the room, it is too heavy.

- Nothing below 9 pt on slides, 11 px on screens
- Sentence case everywhere; no all-caps headlines
- Headline, subtitle, key, footnote and source share one left edge
- Slide titles Bold; a chart's own headline SemiBold
- Big numbers Bold, so they pop

Libre Franklin is open source (SIL OFL). Set-up for Office, the web and Power BI is in the Toolkit.

Role	Sample	Weight	Slide	Screen
Slide title	**Attrition is concentrated in two families**	Bold	24 pt	28 px
Chart headline	**Pharmacy Technicians leave at the highest rate**	Semibold	18 pt	20 px
Subtitle	Voluntary attrition, FY26 (%)	Regular	14 pt	15 px
Data labels	27.4%	Medium	12 pt	13 px
Axis and category labels	Pharmacy Technician	Regular	11 pt	12 px
Callouts	Hiring freeze, Q2	Regular	12 pt	13 px
Footnote and source	Source: Workday HCM, as of 31 Mar 2026	Light	9 pt	11 px
Big number	**75%**	Bold	54 pt	56 px

Proportion on a 16:9 slide

Margins of about 0.5 in. The chart takes at least 60% of the slide height. One message per chart, at most two charts per slide. With one chart, the slide title is its headline; with two, each chart gets its own SemiBold insight headline (see Formats).

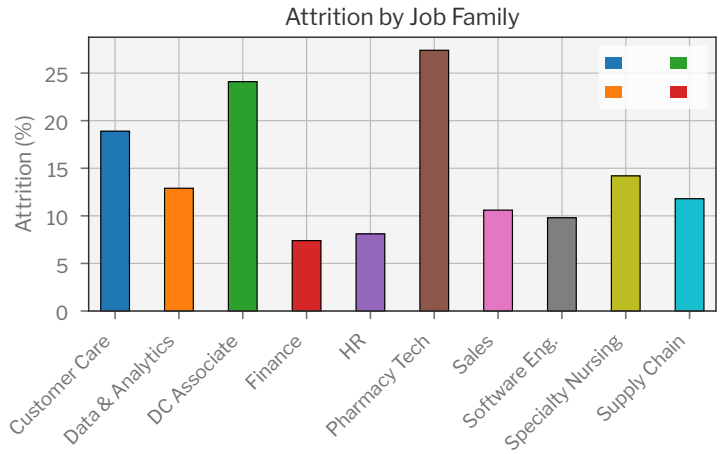
Proportion on a screen

The chart card sits on a quiet background with generous padding. Headline about 1.4× the subtitle. On phones, swap end labels for a top key and let wide exhibits scroll sideways rather than shrink.

Label directly, delete the rest

Every element must earn its place. If removing it loses nothing, remove it.

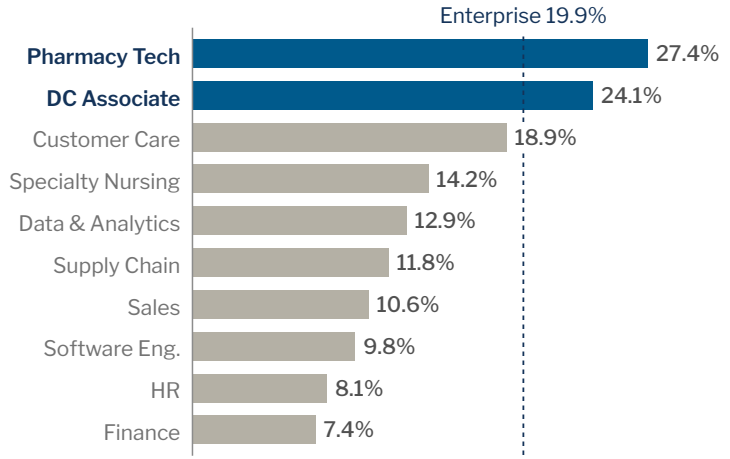
1. Replace the label-style title with an insight headline
2. Sort by value, not alphabet
3. Turn long-named columns into horizontal bars
4. Grey everything, then color the point
5. Label values in place; delete the value axis, gridlines and tick marks
6. Delete the legend, borders and background fill
7. Add the benchmark, labelled where it sits



Tool defaults
Rainbow fills, borders, rotated labels, gridlines, a legend that decodes nothing.

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)



Decluttered
The same data. Fewer elements, and every one of them carries information.

Honest scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

Bars start at zero. If the differences only show on a narrow range, change the chart, not the axis.

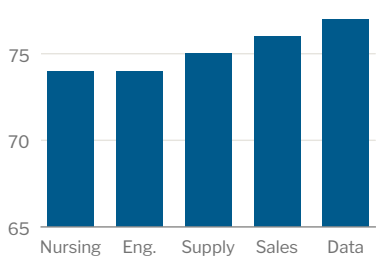
No dual axes. Two scales let you make any two lines cross wherever you like.

Bubble area, not diameter, is proportional to value.

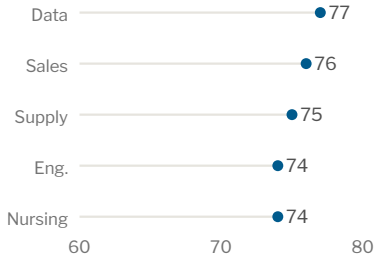
Pies take five slices at most, and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

Truncated bars

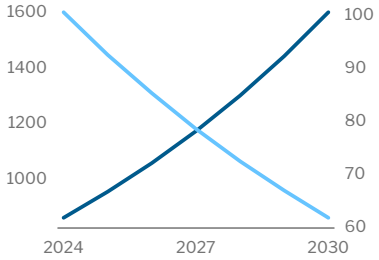


Try to avoid
An axis from 65 turns a 4% difference into a 33% one.

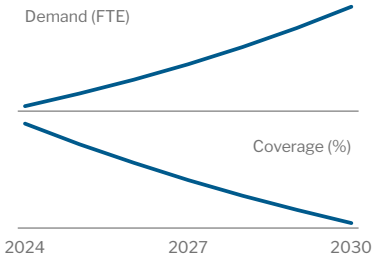


Recommended
A dot plot compares position, so the range can be narrow.

Dual axes

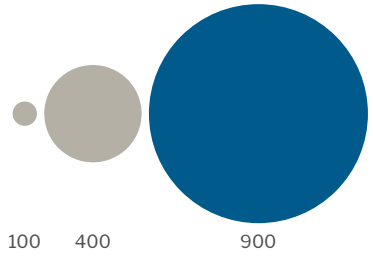


Try to avoid
The crossing point is an artifact of two chosen scales.

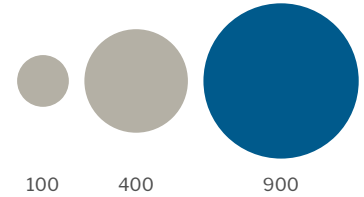


Recommended
Two aligned charts share the time axis.

Bubble size

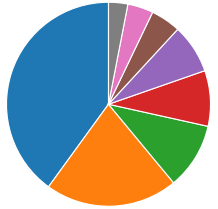


Try to avoid
Sized by diameter, 900 looks 81× bigger than 100.

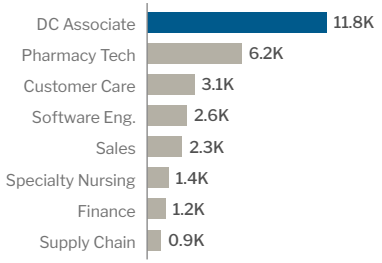


Recommended
Sized by area, 900 is 9× bigger, as it should be.

Too many slices



Try to avoid
Eight similar angles can't be ranked by eye.



Recommended
A sorted bar ranks them instantly.

People data is personal

Workforce charts describe colleagues, and carry extra obligations.

Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

Engagement by team, one distribution center

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Returns	7	52
Quality	4	88
Site leadership	9	81

Try to avoid
Reporting groups below the minimum, even in a table or tooltip.

Engagement by team, one distribution center

Small teams combined into one group that clears the threshold.

Team	Respondents	Engagement
Inbound	46	71
Outbound	38	68
Other teams (3 combined)	20	72

Recommended
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

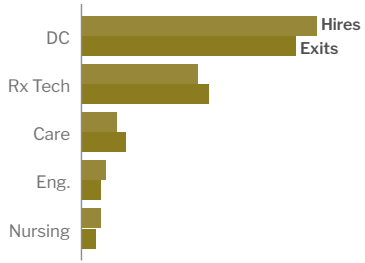
Accessible by default

About 1 in 12 men and 1 in 200 women have a color-vision deficiency. In a room of 25, assume someone does.

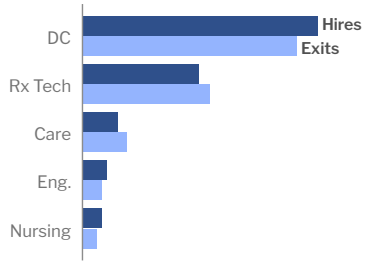
- Text contrast at least 4.5:1
- Never red vs. green; use blues and navy, with orange only for "worse"
- Never color alone: pair it with labels, position or shape
- Nothing below 9 pt on slides or 11 px on screens
- Alt text for every chart that leaves the team: the headline plus two or three key numbers

The red–green examples are simulated for deuteranopia, the most common form.

Red vs. green, as a color-blind reader sees it

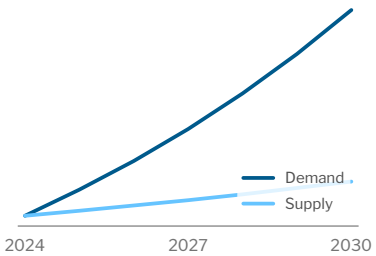


As seen with deuteranopia
Red and green hires vs. exits become one color.

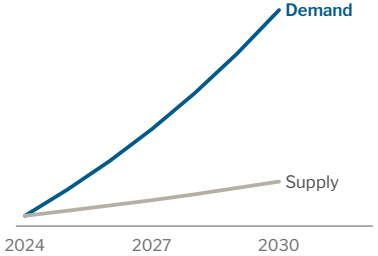


As seen with deuteranopia
Blue and light blue stay distinct.

Color as the only cue

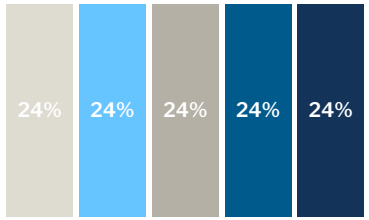


Try to avoid
The reader matches colors to a legend.

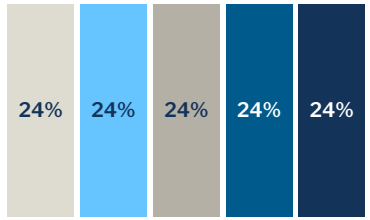


Recommended
Names at the line ends; color only reinforces.

Text on fills



Try to avoid
White on every fill fails on the light ones.



Recommended
White or navy, whichever contrasts more.

Alt text

Write it from the headline and the numbers that prove it.

> **Alt:** Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate. Pharmacy Technician 27.4%; Distribution Center Associate 24.1%; all other job families are below the enterprise rate.

In PowerPoint: right-click the chart, View Alt Text. In HTML: `aria-label` on the figure.

MAGNITUDE

Bar chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

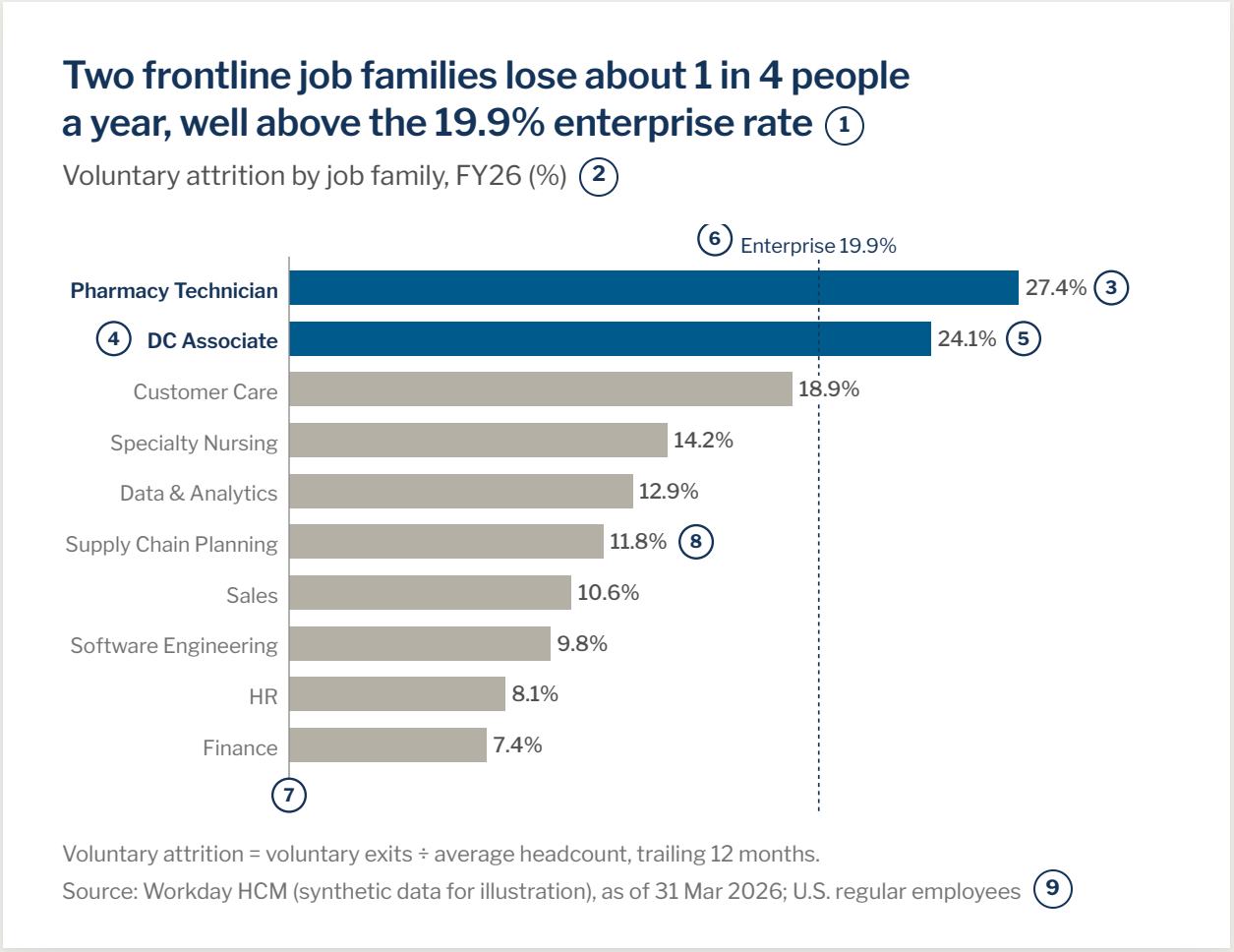
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

Bar chart variants

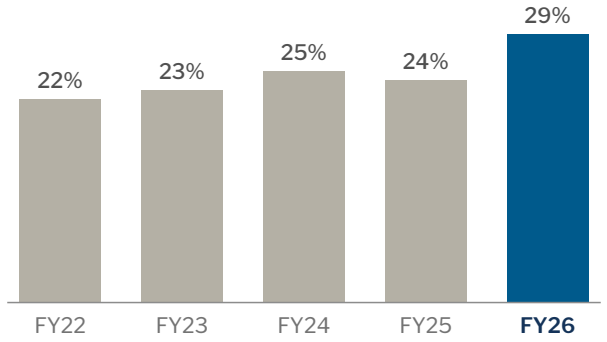
Same rules, different shapes. Pick the variant from the comparison you need to make.

When a bar is the wrong chart

- Values bunched far from zero (engagement scores of 66–77): use a **dot plot**
- Largest more than 20× the smallest: use **proportional bubbles** or split the chart
- Parts of a whole: use a **100% stacked bar** or **waterfall**
- Many periods: use a **line**
- Spread across people: use a **distribution**

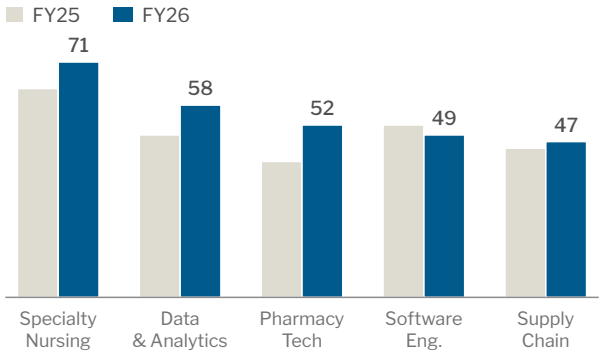
Column chart

A few periods in time order, or short category names. Here internal fill rose from 22% to 29%.



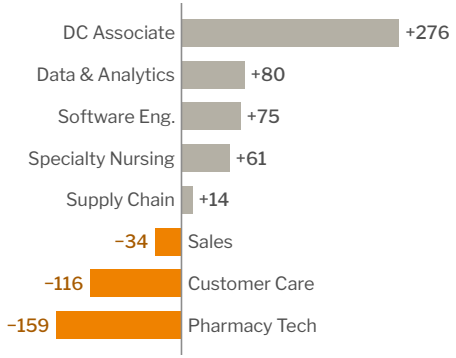
Grouped bars

Two or three series per category. Prior period in light grey, current in blue; key top-left.



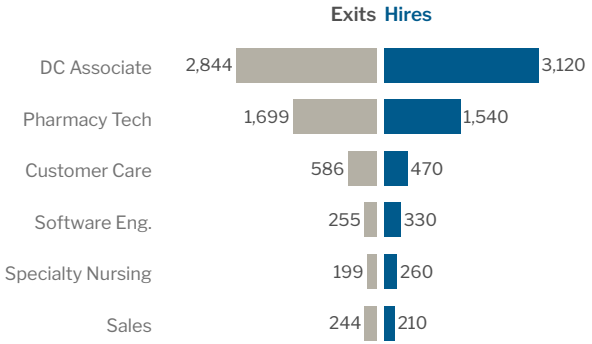
Diverging bars

Values on both sides of zero. Three job families shrank in FY26 despite hiring; negatives in orange.



Butterfly chart

Two measures back to back on the same items: FY26 exits and hires by job family.



MAGNITUDE

Bar chart best practice

Four rules that fix most bar charts.

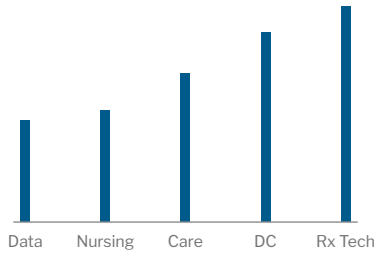
Size of bars. Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

Color of bars. A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

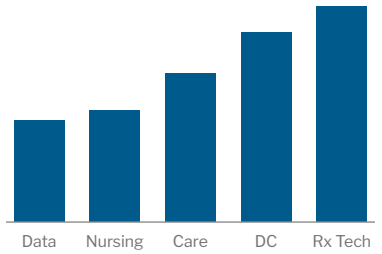
Direct labels. Gridlines and a value axis add noise. Label the bars and delete both.

Zero baseline. Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

Size of bars

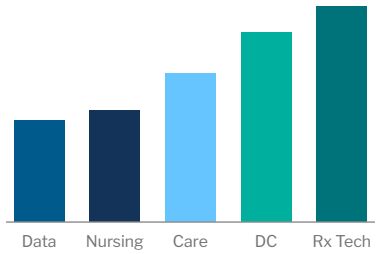


Try to avoid
Stick-thin bars; the gaps dominate.

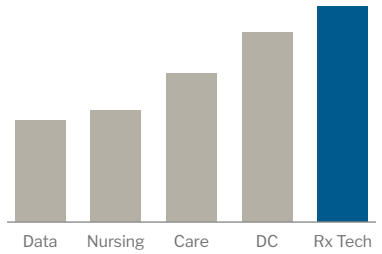


Recommended
Bar about twice the gap.

Color of bars

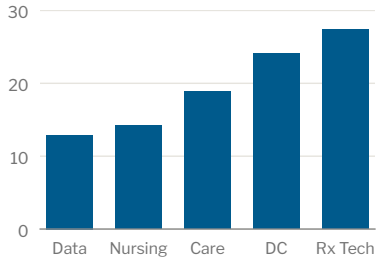


Try to avoid
A color per bar, encoding nothing.

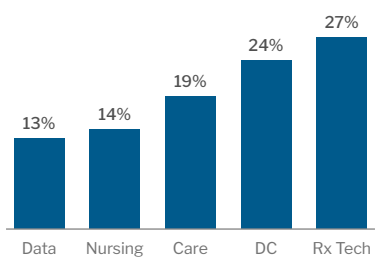


Recommended
Grey bars; blue for the one that matters.

Direct labels

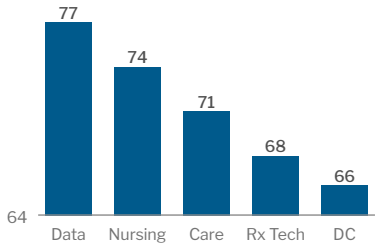


Try to avoid
Gridlines and an axis to read values from.

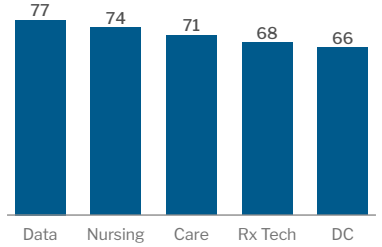


Recommended
Values on the bars; no axis, no grid.

Zero baseline



Try to avoid
Cut at 64, 77 looks 6.5× 66.



Recommended
From zero: 17% higher, as it is.

Bar chart best practice, continued

Four more, from reviewing workforce decks.

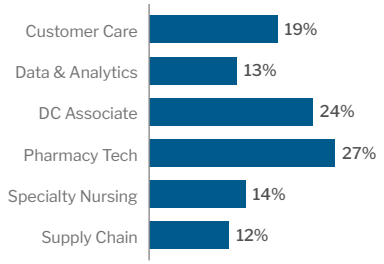
Sort with intent. Alphabetical order hides the ranking. Sort by value unless the categories have a natural order: tenure bands, job levels, survey scales.

Horizontal for long names. Rotated labels are slow to read. Job-family names need horizontal bars.

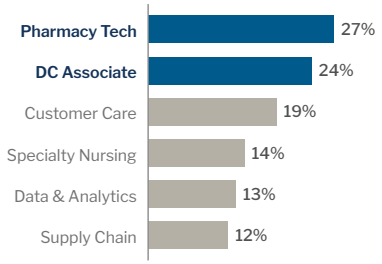
Label the benchmark in place. A reference line explained in a legend makes the reader search. Name it on the line.

Two highlights at most. Five colored bars in two colors is decoration, not emphasis.

Sort with intent

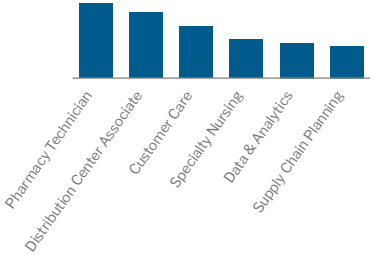


Try to avoid
Alphabetical: the ranking is hidden.

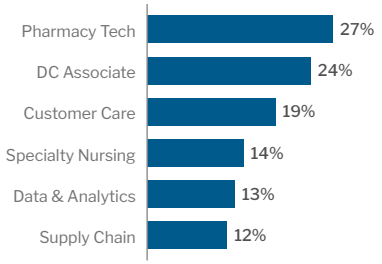


Recommended
By value, largest on top.

Horizontal for long names

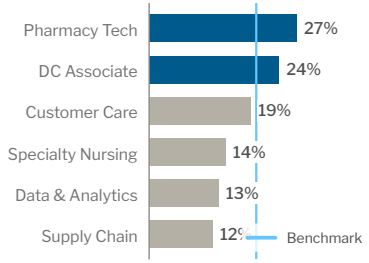


Try to avoid
Tilted names the reader has to decode.

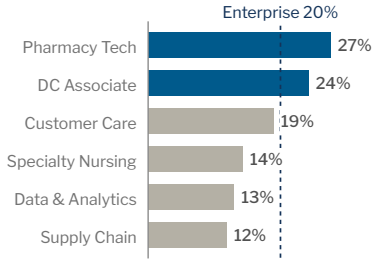


Recommended
Horizontal bars; names read naturally.

Label the benchmark

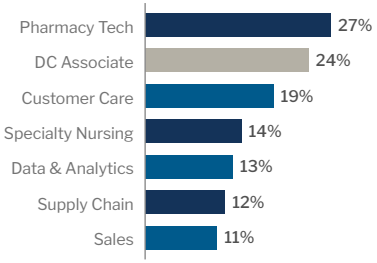


Try to avoid
A legend for one line, in a highlight color.

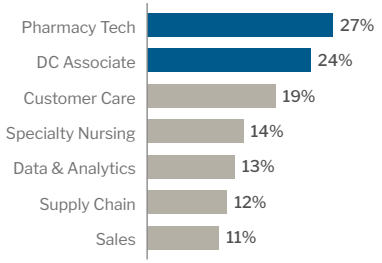


Recommended
Navy dashed line, named on the line.

Two highlights at most



Try to avoid
Everything highlighted means nothing is.



Recommended
The two the headline names.

MAGNITUDE

Build a bar chart

The same chart in every tool we use. Each recipe starts from the starter kit, so the colors and fonts are already right.

Full, tested code for each platform is on the cookbook page: docs/cookbook/bar.md.

Before you send it

- [ ] Starts at zero; bar about 2× the gap
- [ ] Sorted by value or a natural order
- [ ] Values labelled; no value axis or gridlines
- [ ] At most two highlights, matching the headline
- [ ] Benchmark labelled in place
- [ ] No group under 10 people

PowerPoint and Excel

Start from the starter deck or chart workbook in `dist/office/`. A clustered bar sorted ascending (largest on top), **Gap Width 50%**, bars Context grey and the focus bars Focus blue, labels Outside End; delete the value axis, gridlines, legend and chart title. The slide title is the headline.

Power BI

Import `mckesson-dataviz-theme.json` and sort the bars by the measure. Drive the highlight from a measure, not by clicking bars: a `Bar color` measure returns the focus blue when attrition is above the enterprise-rate measure and context grey otherwise, applied through **Colors › fx › Field value**. Data labels on; X axis and gridlines off (set-up).

Python

matplotlib with `mckviz` (Plotly: `mv.pl_highlight`), on the example data:

```
df = pd.read_csv("examples/data/attrition_by_job_family.csv").sort_values("vol_attrition_pct")
hot = df.job_family[df.vol_attrition_pct > df.enterprise_rate_pct]  # above enterprise
band = mv.TOKENS["chart"]["bar"]["band_fill"]  # bar about 2x the gap
bars = ax.barh(df.job_family, df.vol_attrition_pct, height=band,
               color=mv.highlight(df.job_family, focus=hot))
mv.label_bars(ax, bars, fmt="{:.1f}%"); mv.clean_axes(ax, keep=("left",), grid=None)
mv.titles(fig, headline, subtitle); mv.source(fig, source); mv.lint(fig)
```

HTML

Copy `templates/html/chart-card.html`: headline, subtitle and source are HTML, colors come from CSS variables such as `var(--mck-chart-focus)`, and the chart is drawn after the font loads.

R

`source("dist/r/theme_mck.R")`, then `geom_col(aes(fill = job_family), width = 0.67)` + `scale_fill_manual(values = mck_highlight(levels(df$job_family), focus = focus), guide = "none")` + `theme_mck()`.

Coding agents

"Make a McKesson bar chart of voluntary attrition by job family, FY26, highlighting the families above the enterprise rate." The `mck-chart` skill runs the workflow and returns the chart, a chart brief and alt text.

PART-TO-WHOLE

Stacked bar chart

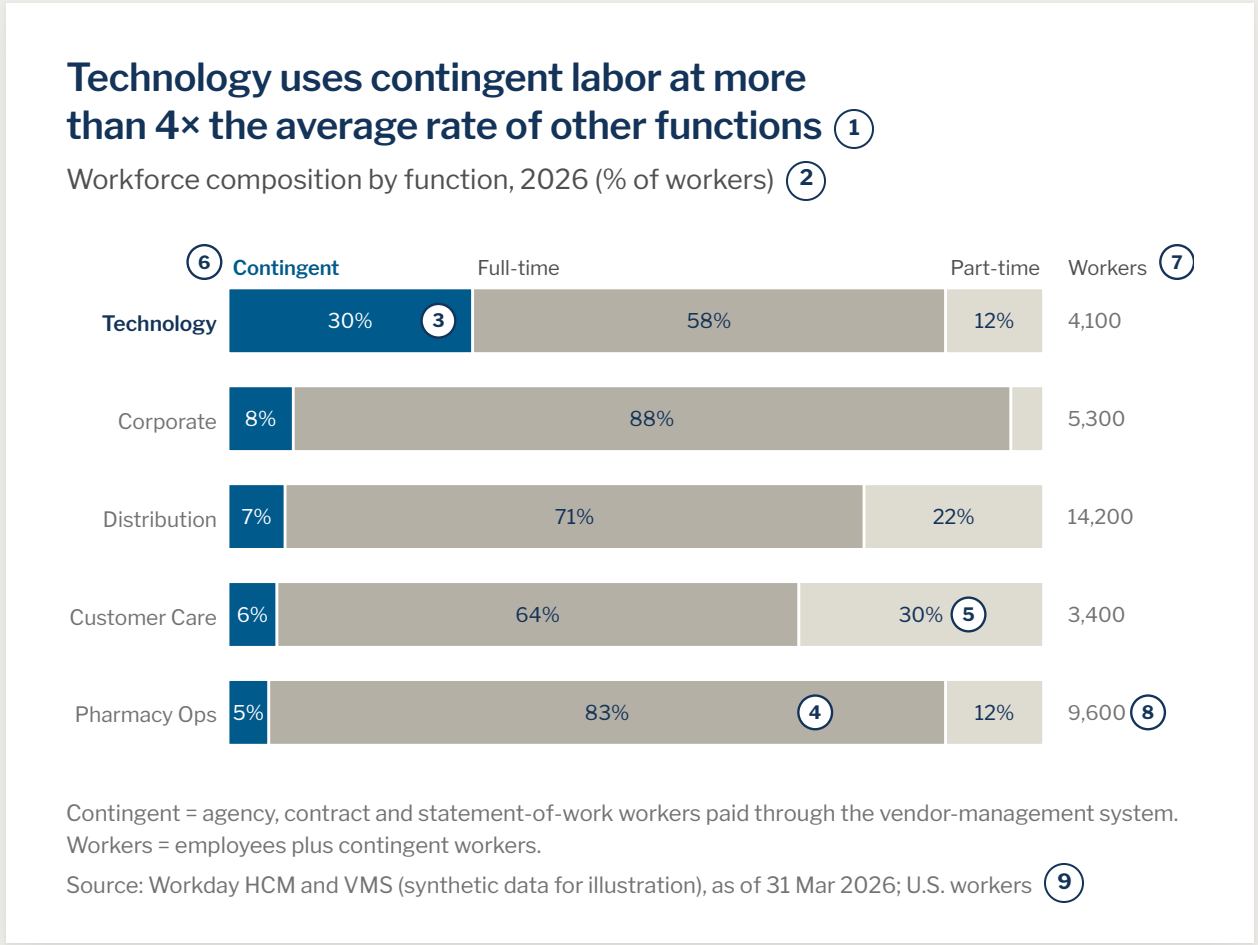
Part-to-whole charts show how a whole breaks into its parts, and how each part relates to the total.

A 100% stacked bar compares the mix of several wholes at once. Color only the segment the headline is about, and put it on the baseline so every bar's value starts level.

McKesson spec

- Five segments at most; group the rest as "Other"
- The story segment in blue, on the baseline
- Values inside segments that fit; names over the top bar
- Show each bar's base: 100% hides size

1. **Insight headline**
   Names the part and how far it stands out.
2. **Subtitle**
   Metric, unit, scope, period.
3. **The part that matters**
   Blue, on the baseline, so every bar compares from zero.
4. **Grey context**
   Two greys, split by white separators.
5. **Values in the segments**
   Where they fit; white on blue, navy on grey.
6. **Names, not a key**
   Each segment named once, over the top bar.
7. **The base**
   A 100% bar hides how big each whole is; give the count.
8. **Sorted by the story**
   Bars ordered by the blue segment, not alphabetically.
9. **Footnote and source**
   Define the parts.



PART-TO-WHOLE

Stacked bar variants

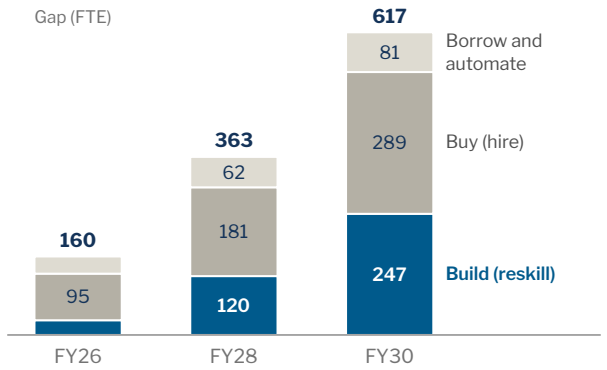
Same rules: one part in color, the rest grey, labels in place.

When a stack is the wrong chart

- One whole with up to five parts: a **donut** or a single bar
- How a total changes step by step: a **waterfall**
- Many parts or a hierarchy: a **treemap**
- Comparing the parts themselves across groups: **grouped bars** or **small multiples**; only the baseline segment of a stack compares cleanly

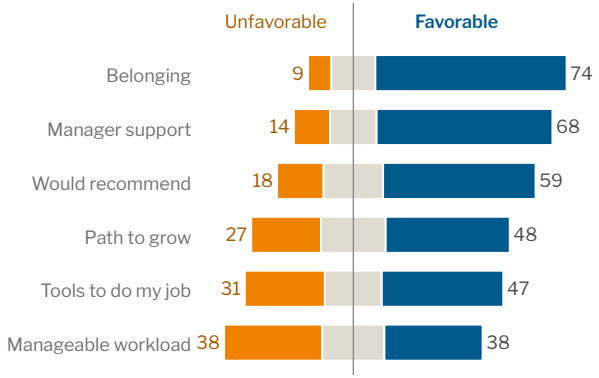
Stacked columns with totals

When the total matters too. The FY30 Data & AI gap of 617 FTE: reskilling (on the baseline) closes 40%.



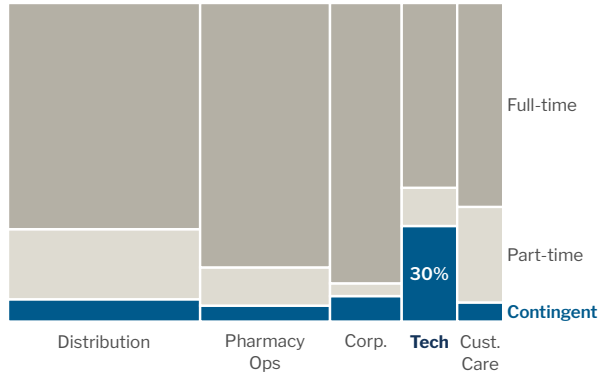
Diverging stacked bars

Survey scales. Unfavorable in orange left of zero (it means worse), favorable in blue, neutral split across the middle.



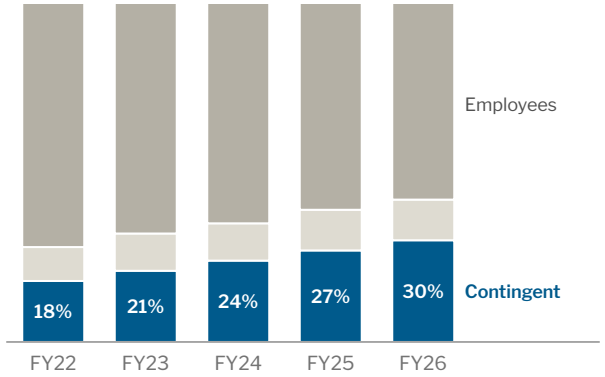
Marimekko

Width and height both carry data: column width is each function's workforce, height its mix. Powerful, but it needs explaining.



Mix over time

A few periods as 100% columns, the tracked segment on the baseline so its change reads directly.



PART-TO-WHOLE

Stacked bar best practice

Stacks are easy to make and easy to misread. Four rules keep them honest.

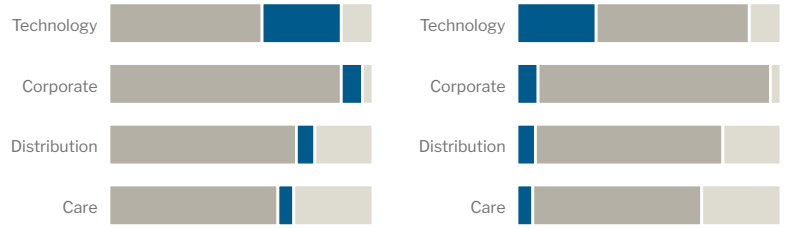
The part that matters on the baseline. Only segments that share a starting line can be compared by eye.

Five segments at most. More slices turn a bar into a barcode. Keep the parts the story needs and group the rest.

Label in place. Values inside the segments; name each segment once over the top bar. If a key is unavoidable, top-left, in stack order.

One order in every bar. Never re-sort segments within each bar.

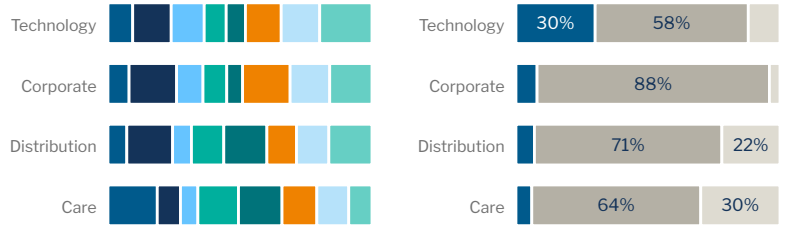
The part that matters on the baseline



Try to avoid
A floating segment can't be compared across bars.

Recommended
On the baseline, every bar starts level.

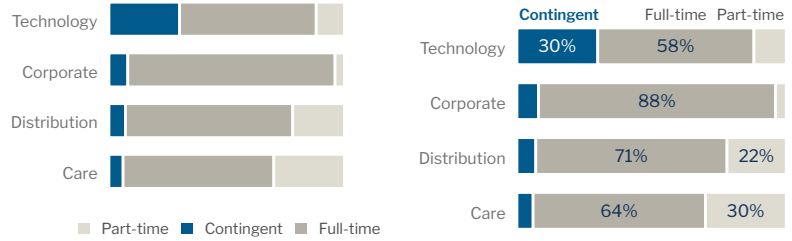
Five segments at most



Try to avoid
Eight colored slices nobody can follow.

Recommended
The part in blue, the rest in grey.

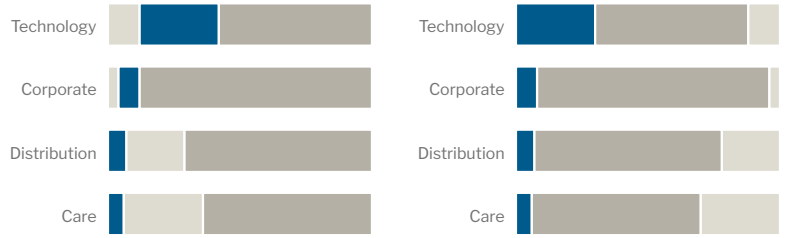
Label in place



Try to avoid
A bottom legend in another order; no values.

Recommended
Values inside; names over the top bar.

One order in every bar



Try to avoid
Segments re-sorted inside each bar.

Recommended
The same order everywhere.

PART-TO-WHOLE

Waterfall chart

A waterfall shows how a starting total becomes an ending total, one driver at a time.

It is a stacked column chart with an invisible spacer series that floats each driver at the running total. Use it for labor cost bridges, headcount bridges and budget-to-actual walks.

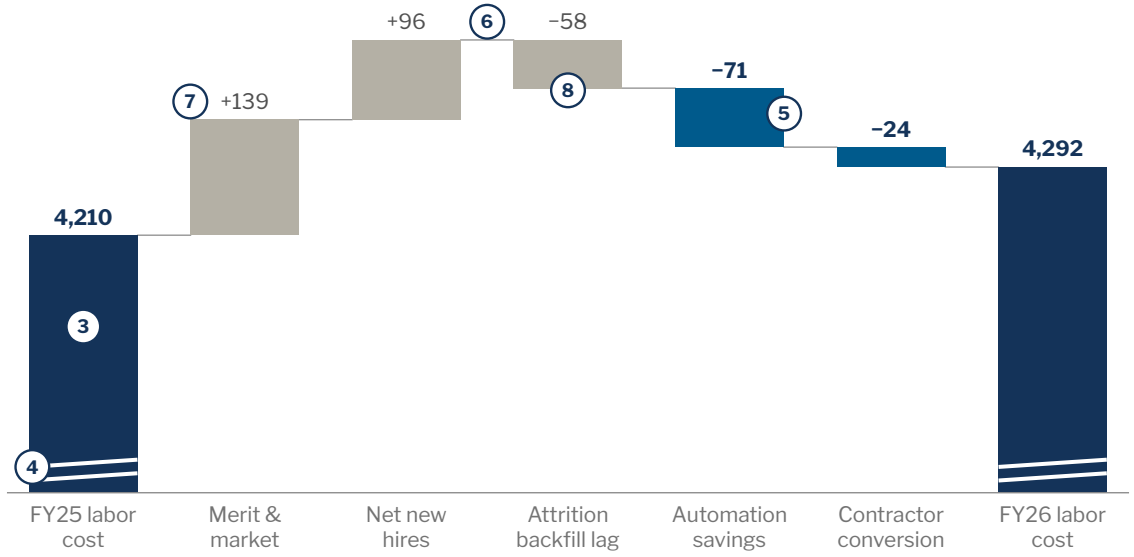
McKesson spec

- Totals in navy; drivers grey, the ones the headline names in blue
- Or by meaning: favorable teal, unfavorable orange
- Thin light connectors between bars
- Signed labels on drivers (true minus, explicit +); bold totals
- If the axis starts above zero, mark the break and say so

1. **Insight headline**
   Sizes what the named drivers did to the total.
2. **Subtitle**
   From and to, with the unit.
3. **Totals in navy**
   Start, end and any subtotal.
4. **Marked break**
   The axis starts at $3,900M; white slashes say so.
5. **The drivers that matter**
   In blue; label bold.
6. **Thin connectors**
   Light grey, so the bars stay dominant.
7. **Signed labels**
   +139, −58: direction without color.
8. **Grey context**
   Every other driver, in logical order.
9. **Footnote**
   States the axis break.

Automation and contractor conversion offset ~40% of FY26 labor cost growth (1)

Labor cost bridge, FY25 to FY26 ($M) (2)



Value axis starts at $3,900M (break marks on the totals) so the drivers are readable.

Source: Finance labor cost model (synthetic data for illustration), FY26 (9)

PART-TO-WHOLE

Waterfall variants

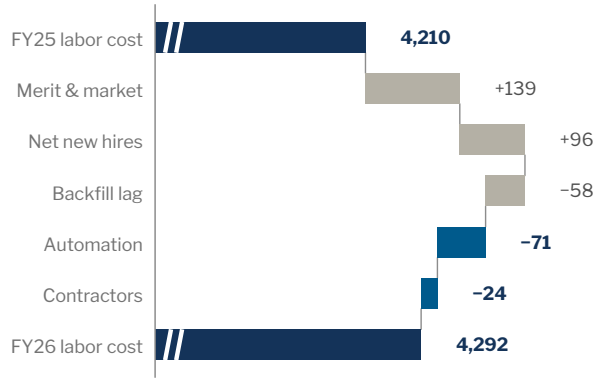
The same build, turned or recolored for the job.

Two color patterns. When good vs. bad is the message, color every driver by meaning: favorable teal, unfavorable orange. When the headline names one or two drivers, grey the rest and make those blue. Totals stay navy either way.

Excel's built-in Waterfall chart is vertical only. For horizontal waterfalls or per-bar colors, build a stacked bar with a transparent spacer series (see the cookbook).

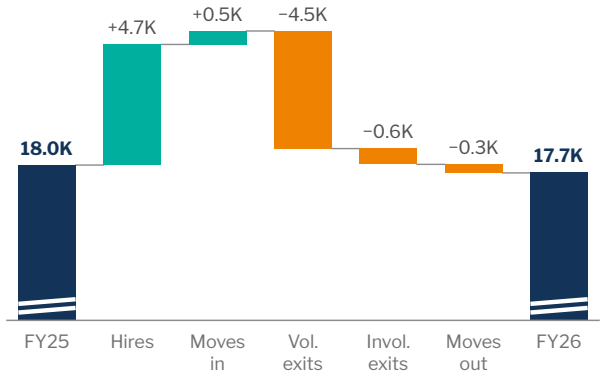
Horizontal waterfall

Long driver names, narrow layouts and phones.



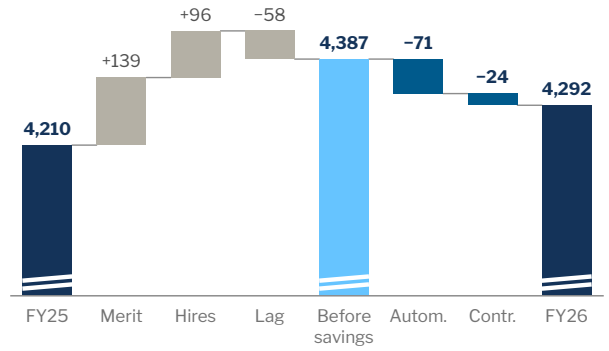
Colored by meaning

Frontline headcount bridge: voluntary exits equal 97% of external hires.



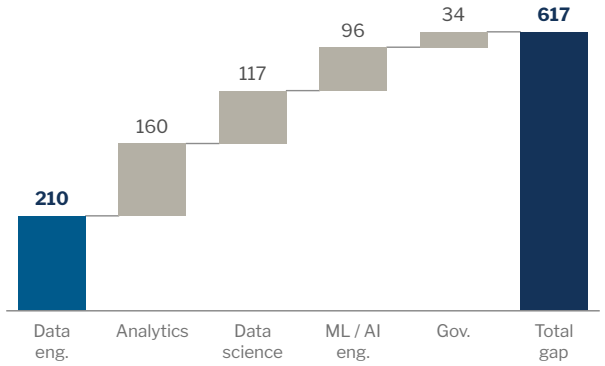
With a subtotal

A light-blue subtotal separates cost growth from the savings that offset it.



Distribution waterfall

Contributions to one total instead of steps in time: data engineering is 34% of the FY30 gap.



PART-TO-WHOLE

Waterfall best practice

The bars carry the story; everything else stays quiet.

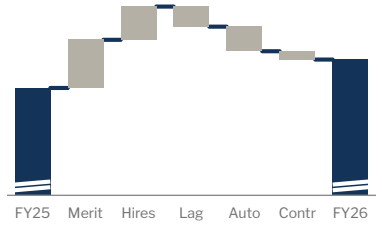
Thin connectors. Heavy dark connectors compete with the bars. Use hairlines in light grey.

Color by meaning, in the palette. Favorable teal, unfavorable orange, judged from the reader's view: in a cost bridge, growth is the bad news. Never red and green: they fail for color-blind readers, and green-up makes cost growth look good.

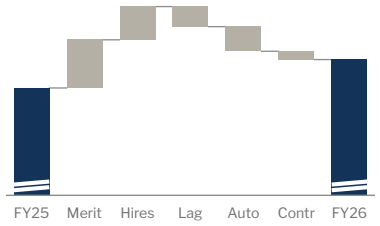
Mark a broken axis. Totals often dwarf the drivers. Starting the axis above zero is allowed here, and only here, if the break is drawn and footnoted.

Group the small steps. Keep the drivers the story needs; fold the rest together.

Connector lines



Try to avoid
Heavy, dark connectors.

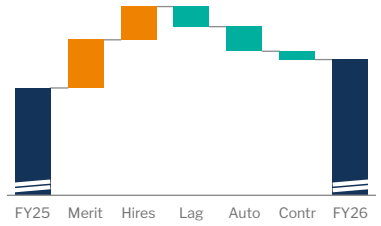


Recommended
Thin, light grey.

Coloring the bars

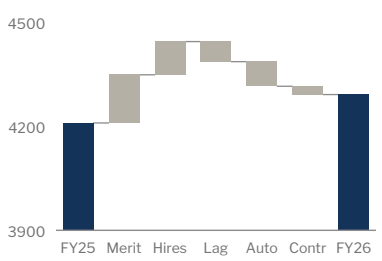


Try to avoid
Green up, red down: cost growth looks good.

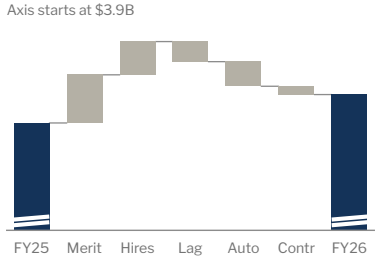


Recommended
Cost growth orange, savings teal.

A broken axis

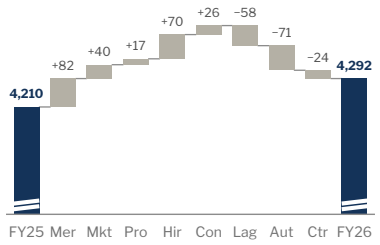


Try to avoid
Cut totals with no warning.

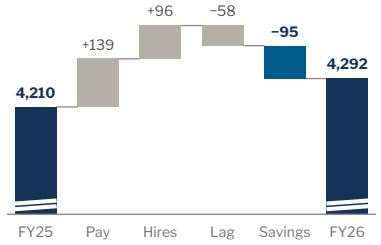


Recommended
Break marks and a note.

Group the small steps



Try to avoid
Ten narrow drivers to decode.



Recommended
Five steps, savings combined.

PART-TO-WHOLE

Treemap

Nested rectangles whose area is proportional to value. A treemap shows many parts, or a hierarchy, in one compact block.

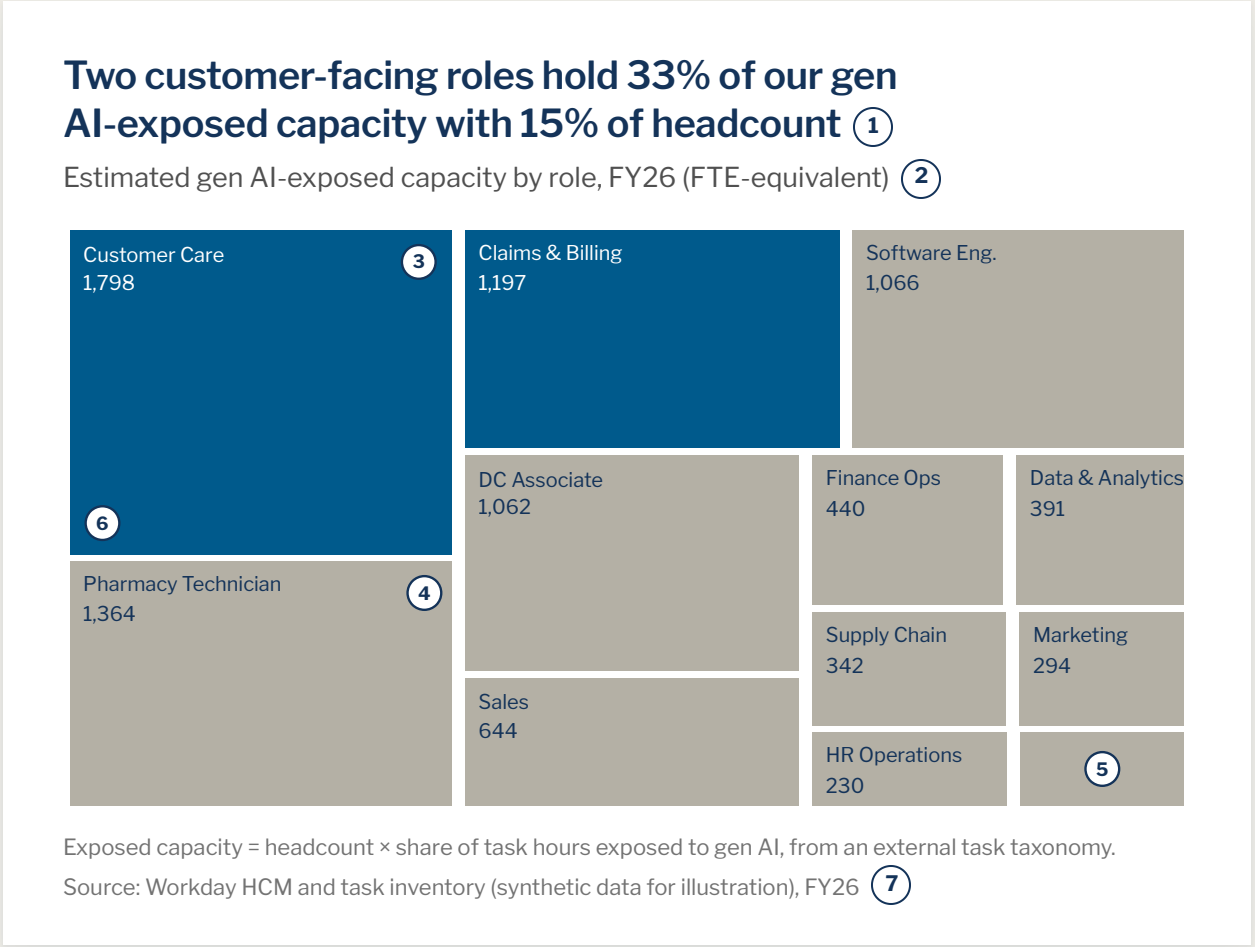
Use it when there are too many parts for a donut or a stack (ten to thirty roles, sites or skills) and the point is which few dominate.

Don't use it when readers need to compare similar sizes precisely, or there are only a handful of parts: a sorted bar is clearer.

McKesson spec

- Squarified layout, largest top-left
- Thin, even white gutters
- Grey tiles; the ones the headline names in blue
- Name and value inside each tile that fits

1. **Insight headline**
   Share of the whole held by the named few.
2. **Subtitle**
   What area measures, with the unit.
3. **Focus in blue**
   The two roles the headline names.
4. **Grey context**
   Every other role, the same grey.
5. **Smallest last**
   Bottom-right. Tiles too small for a label get none; list them in a footnote if they matter.
6. **Largest first**
   Top-left, flowing to the smallest; labels top-left inside each tile.
7. **Footnote and source**
   Define the measure.



PART-TO-WHOLE

Treemap best practice

A treemap works when it reads as one shape that flows from large to small.

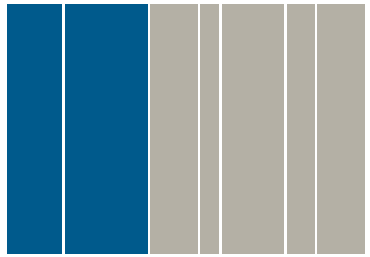
Largest to smallest. Don't arrange tiles alphabetically or by an org grouping that breaks the flow.

Labels inside. No label bands or boxes over the tiles; they distort the areas. If a tile can't hold its label, leave it unlabelled or label it outside.

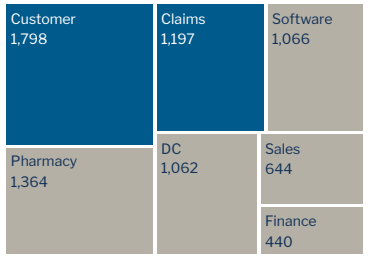
Even padding. Uneven gaps make similar tiles look different.

Few parts? Use a bar. Areas are harder to compare than lengths.

Arrange by size

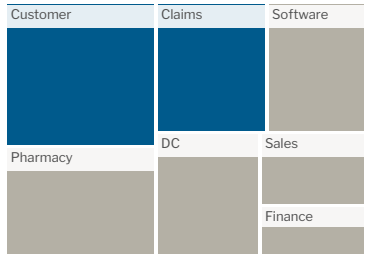


Try to avoid
Alphabetical strips.

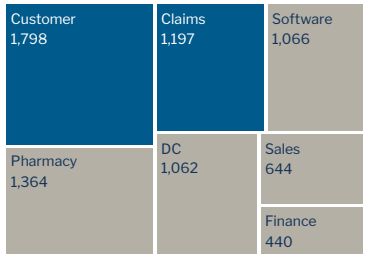


Recommended
Largest top-left.

Labelling



Try to avoid
Label bands over the tiles.



Recommended
Inside, top-left.

Padding

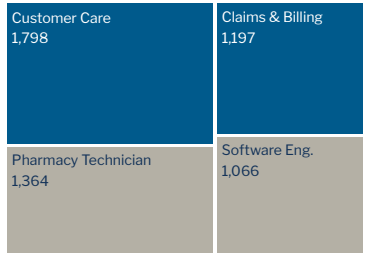


Try to avoid
Uneven gaps.

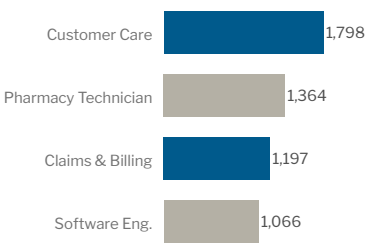


Recommended
Thin, even gutters.

Few parts



Try to avoid
Four similar tiles.



Recommended
A sorted bar ranks them.

PART-TO-WHOLE

Donut and pie

A donut shows one whole split into a few parts. Prefer it to a pie: readers follow the arc length around the ring, and the center holds the total.

Use it for one whole and at most five parts. To compare several wholes, use 100% stacked bars; for more parts, a bar or treemap.

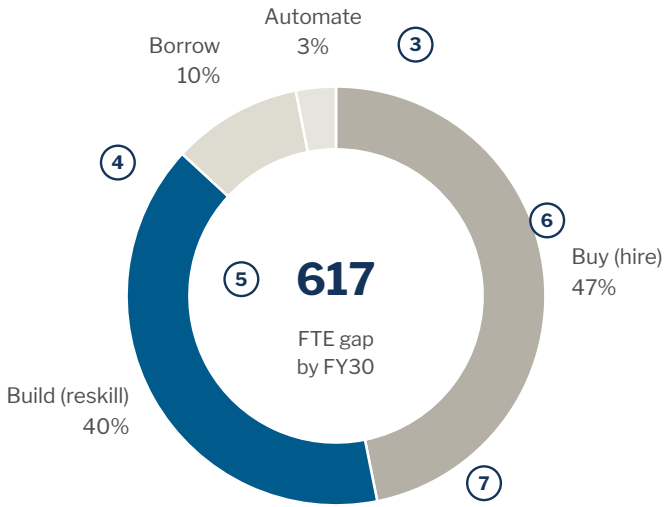
McKesson spec

- Five segments at most; "Other" last
- Largest first, clockwise from 12 o'clock
- Hole about 60% of the diameter; white separators
- Labels and values outside; no legend

1. **Insight headline**
   The share, and what it means for the plan.
2. **Subtitle**
   The whole, the unit, the year.
3. **Start at 12**
   Largest first, then clockwise.
4. **The part that matters**
   In blue; the rest grey.
5. **Total in the center**
   Bold, large.
6. **Labels outside**
   Name and share, beside each segment.
7. **White separators**
   Thin gaps keep segments distinct.
8. **Footnote and source**
   Define the parts.

Reskilling can close 40% of the FY30 Data & AI gap; hiring still has to cover nearly half (1)

Planned sources to close the Data & AI capability gap, FY30 (% of 617 FTE) (2)



Base scenario. Build = reskilling current employees; buy = external hires; borrow = contractors and partners; automate = roles removed by automation.

Source: Strategic workforce planning model (synthetic data for illustration), Mar 2026 (8)

PART-TO-WHOLE

Donut best practice

People judge length faster than angle or area, which is why the donut beats the pie.

Donut over pie. The ring turns angles into arc lengths, and the hole holds the total.

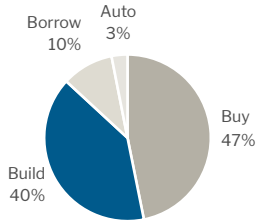
Labels outside. Squeezed labels on the ring are hard to read; put names and values outside.

Few segments. Group the smallest into "Other".

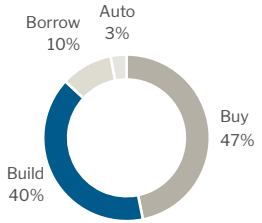
Largest first, clockwise from 12. Don't mix small and large at random.

Progress rings and dials for a single percentage are in Single number.

Donut over pie

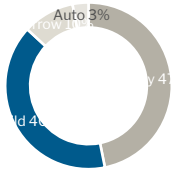


Try to avoid
Angles and areas.

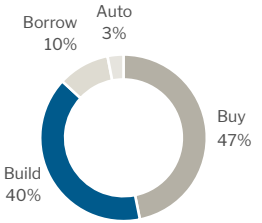


Recommended
Arc lengths on a ring.

Labelling



Try to avoid
Labels squeezed onto the ring.

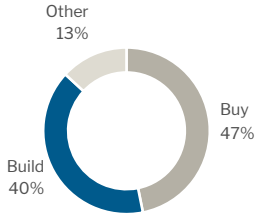


Recommended
Outside, with the share.

Many small segments

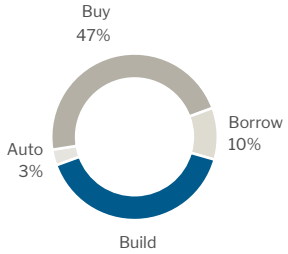


Try to avoid
Nine slices, nine colors.

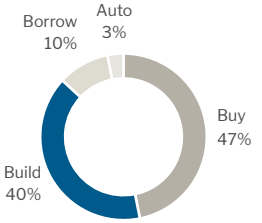


Recommended
The story, plus "Other".

Order of segments



Try to avoid
No order, odd start.



Recommended
Largest first from 12.

PART-TO-WHOLE

Build part-to-whole charts

Every tool can make these natively. The settings below are the ones that matter; tested code and click-by-click steps are on the cookbook pages.

stacked-bar.md · waterfall.md · treemap.md · donut-pie.md

Before you send it

- [ ] Parts sum to the whole (100%, or the end total)
- [ ] Five parts at most; "Other" last
- [ ] One part in color, matching the headline
- [ ] Waterfall axis break marked and footnoted
- [ ] No group under 10 people, in any part

PowerPoint and Excel	**Stacked:** Stacked or 100% Stacked Bar, Gap Width 50%, Overlap 100%, white 1.5 pt borders; move the story series first or last in Select Data. **Waterfall:** Insert › Waterfall, then right-click each total › *Set as Total*; drivers grey, totals navy. **Treemap:** Insert › Hierarchy › Treemap; labels *Overlapping*, not Banner. **Donut:** Doughnut, angle of first slice 0°, hole 62%, data sorted descending.
Power BI	Native 100% stacked bar, Waterfall, Treemap and Donut visuals all take the theme. Set the story color per series or with a field-value color measure; legend top-left or off; data labels on. The native waterfall can't color single bars or break the axis: use a stacked column with a transparent spacer measure when you need that.
Python	`ax.barh(..., left=running_total, edgecolor="white")` for stacks; the waterfall, squarified treemap and donut recipes in the cookbook use only matplotlib and `mckviz`. Plotly has native `go.Waterfall`, `go.Treemap` and `go.Pie(hole=0.6)`.
HTML	Plotly with `MCK_PLOTLY_TEMPLATE`: `barmode: "stack"`, `waterfall` and `treemap` traces. Read colors from `var(--mck-chart-focus)` and `var(--mck-chart-context)`.
R	`geom_col(position = "fill")` for 100% stacks; `treemapify::geom_treemap()` for treemaps; waterfalls with `geom_rect()`. Colors from `mck_highlight()`.

```
colors = mv.highlight(parts, focus="Contingent")          # one part blue, rest grey
mv.lint(fig)    # flags > 6 colors, legends, and bars that don't start at zero
```

CHANGE OVER TIME

Line chart

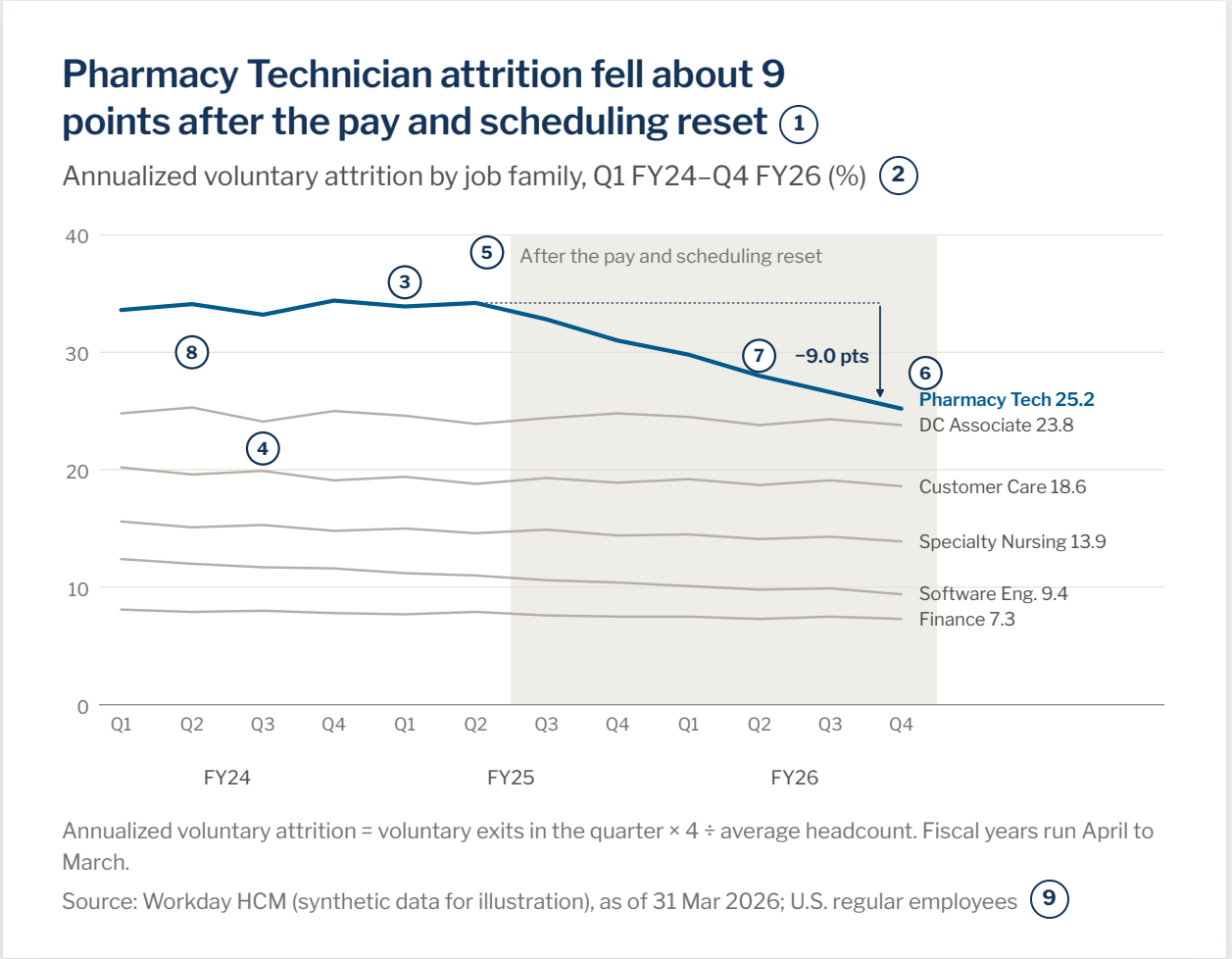
Change-over-time charts show how something moves, and when it changed, at any time scale from weeks to decades.

A line connects values at equal time intervals, so the eye reads the trend rather than the individual amounts. Use it for monthly or quarterly attrition, hiring velocity, or a forecast.

McKesson spec

- The series the headline names in blue; others grey
- Data lines about twice the gridline weight
- Names at the line ends, not in a legend
- Shade the period that explains the change

1. **Insight headline**
   What changed, by how much, and why.
2. **Subtitle**
   Metric, unit, scope and period.
3. **Focus line**
   Blue and heavier than the rest.
4. **Context lines**
   Grey and thinner: there to compare against.
5. **The period that matters**
   Lightly shaded and named.
6. **Direct labels**
   Names and latest values at the line ends, nudged apart.
7. **The change, sized**
   A short callout carries the headline number.
8. **Light scaffolding**
   A few thin gridlines; the unit lives in the subtitle.
9. **Footnote and source**
   Define the metric.



CHANGE OVER TIME

Line chart variants

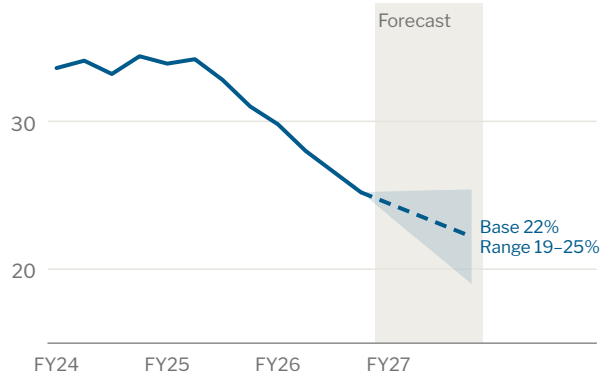
The same rules hold: one line in blue, names at the ends, light scaffolding.

When a line is the wrong chart

- Only two to five periods, and the amounts matter: **columns**
- How a total splits over time: **stacked area**
- Discrete events of different sizes: a **bubble timeline**
- Many series that tangle: **small multiples**
- Two measures in different units: two aligned charts, never a **dual axis**

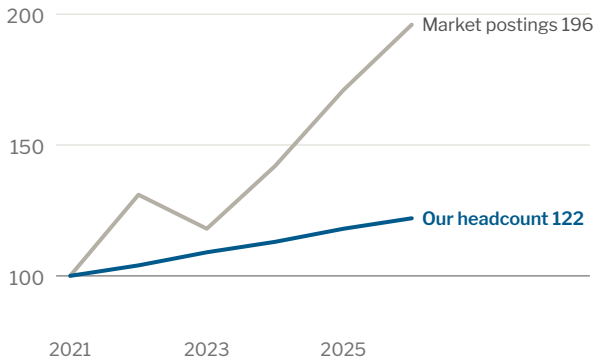
Forecast with a range

Dashed beyond the last actual, inside a band that widens with time. Label it as an estimate.



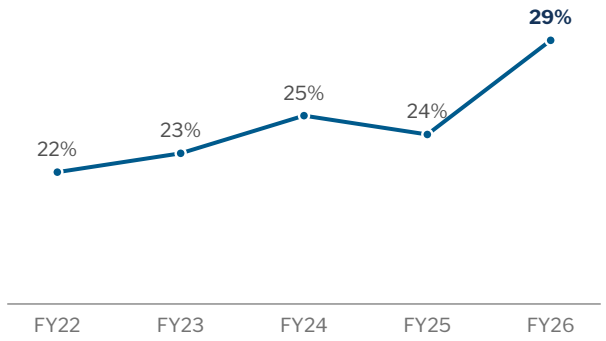
Indexed to 100

Compares growth when the units or sizes differ: market Data & AI postings reached 196, our headcount 122.



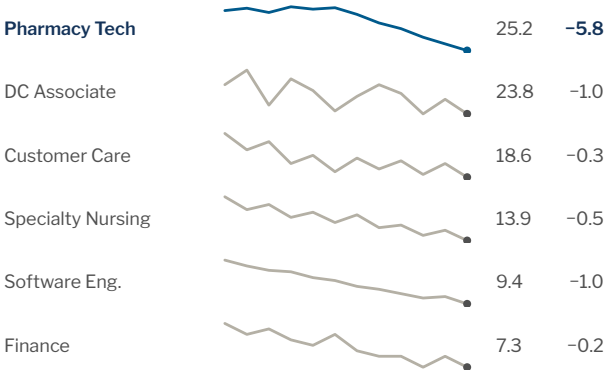
Few points, every value labelled

With five annual values, label each one and drop the axis.



Sparklines

One small line per row in a table or dashboard: shape, latest value, change. Worse in orange.



CHANGE OVER TIME

Line chart best practice

Four rules that make trends readable.

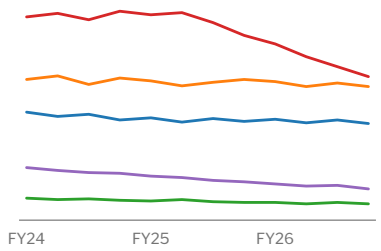
Highlight one line. A color per series doesn't show the trend; it makes the reader decode. Grey the background, color the message.

Lines for trends. Stacked columns hide how each part changes. When the trend is the point, draw lines.

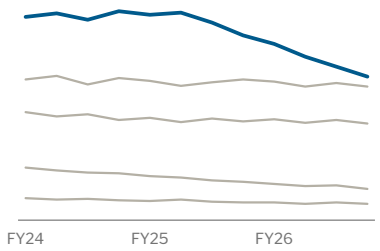
No fill without a zero. Shading under a line turns it into an area chart, which implies a zero baseline.

Names at the line ends. A legend above the chart breaks the reading order.

Highlights

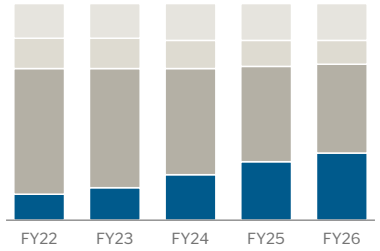


Try to avoid
Every line its own color.

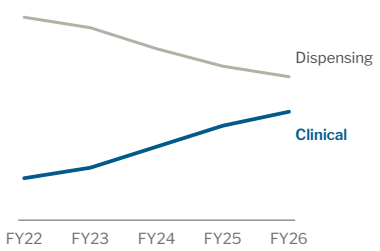


Recommended
Grey context, one blue line.

Showing trends

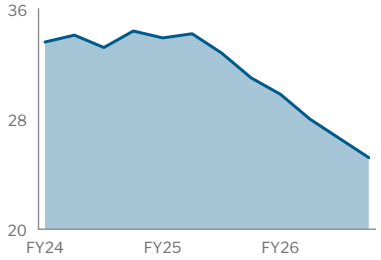


Try to avoid
Stacks hide each part's change.

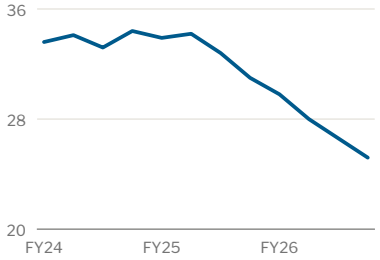


Recommended
Lines show the crossover.

Shading lines

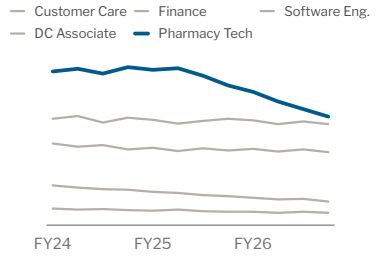


Try to avoid
A filled area on a 20–36 axis.

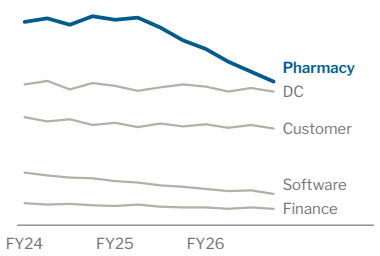


Recommended
A plain line.

Labelling lines



Try to avoid
A legend to decode.



Recommended
Names where the lines end.

CHANGE OVER TIME

Line chart best practice, continued

Four more, from HR and workforce data in particular.

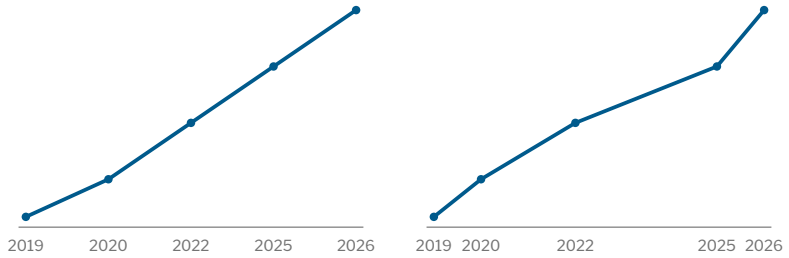
Honest time spacing. Missing years still take up space. Use a real date axis, not evenly spaced labels.

Show gaps as gaps. A system change or survey break is not a straight line. Leave the gap and say why.

Weight with intent. Data lines about twice the gridlines; gridlines light, or gone.

Too many lines? Split them. More than four or five crossing lines become spaghetti. Use small multiples.

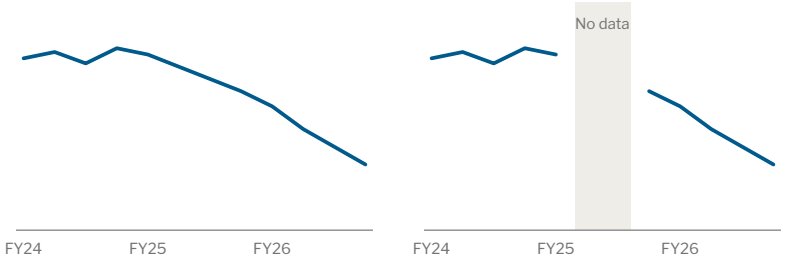
Even time intervals



Try to avoid
Missing years squeezed out.

Recommended
True spacing; the slope is honest.

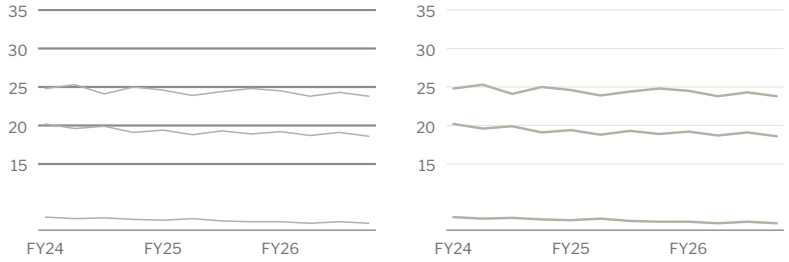
Missing data



Try to avoid
Drawing across the gap.

Recommended
A visible, labelled gap.

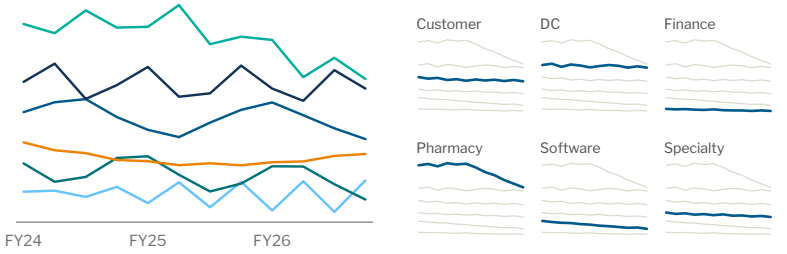
Line weight



Try to avoid
Gridlines outweigh the data.

Recommended
Data lines twice the grid.

Too many lines



Try to avoid
Six crossing colors.

Recommended
Small multiples.

CHANGE OVER TIME

Stacked area chart

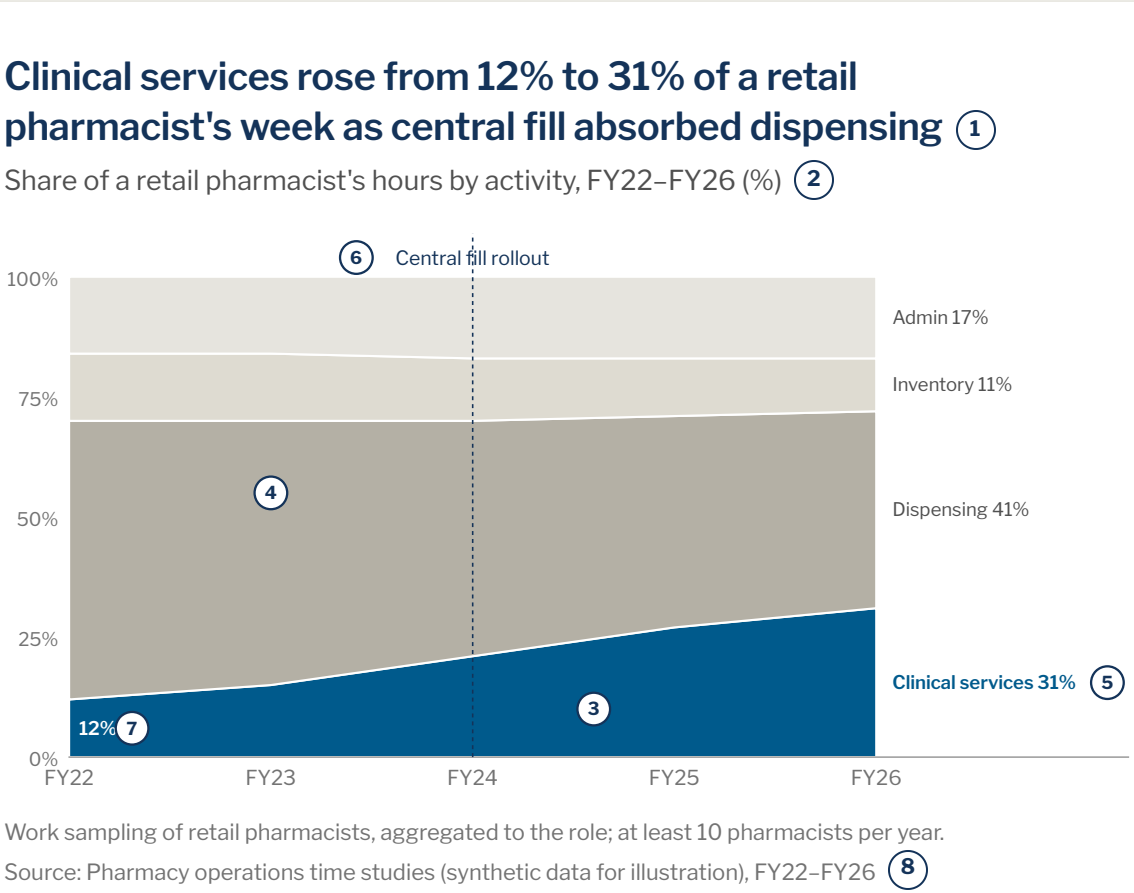
Layers stacked on each other show how a total splits over time, and how each part's share shifts.

Use it for a few parts that change gradually: how a role's hours divide between activities, or how the workforce mix moves. The layer on the baseline is the only one that reads precisely, so put the story there.

McKesson spec

- Four or five layers at most
- The story layer on the baseline, in blue; the rest ordered greys
- Labels inside or at the right edge, not in a legend
- Mark the event that explains the shift

1. **Insight headline**
   Start, end and the reason.
2. **Subtitle**
   Whose time, what unit, which years.
3. **Story on the baseline**
   Blue, so its growth reads directly.
4. **Grey context**
   Ordered from dark to light.
5. **Labels at the end**
   Name and latest share beside each layer.
6. **The event, marked**
   A thin dashed line and a short name.
7. **Where it started**
   The first value, so the change is explicit.
8. **Footnote and source**
   How time was measured.



CHANGE OVER TIME

Stacked area best practice

Areas are heavy marks. Keep them few, ordered and calm.

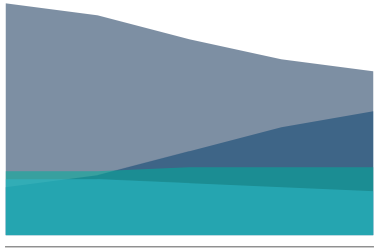
Stack, don't overlap. Areas drawn over each other hide what's behind. Stack them, or use lines.

Order the colors. Random colors make neighbors clash. Story layer blue on the baseline; the rest in ordered greys.

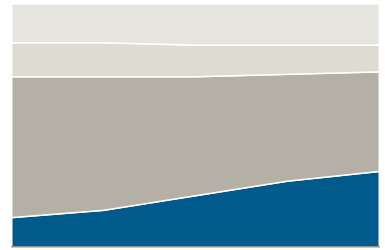
Volatile on top. A jumpy layer at the bottom makes every layer above it jump. Keep steady layers low.

Few layers. Four or five at most; fold the rest into "Other".

Overlapping areas

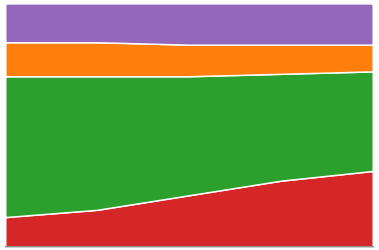


Try to avoid
Translucent layers over each other.

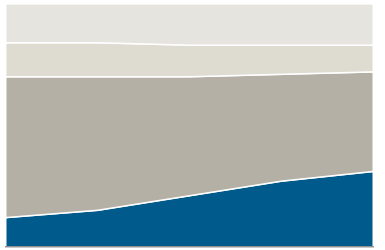


Recommended
Stacked layers.

Color order



Try to avoid
Unrelated bright colors.



Recommended
Blue story, ordered greys.

High variability

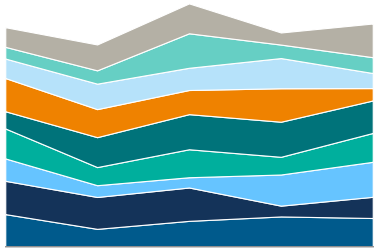


Try to avoid
The volatile layer at the bottom.



Recommended
Steady layers low, volatile on top.

Number of layers



Try to avoid
Nine thin layers.



Recommended
Three: story, comparison, rest.

CHANGE OVER TIME

Bubble timeline

Events placed on a time axis, with bubble area showing size. It combines a timeline with a proportional-area chart.

Use it for discrete events of different sizes: announced layoffs in the talent market, reorganizations, hiring waves. Each category gets its own row.

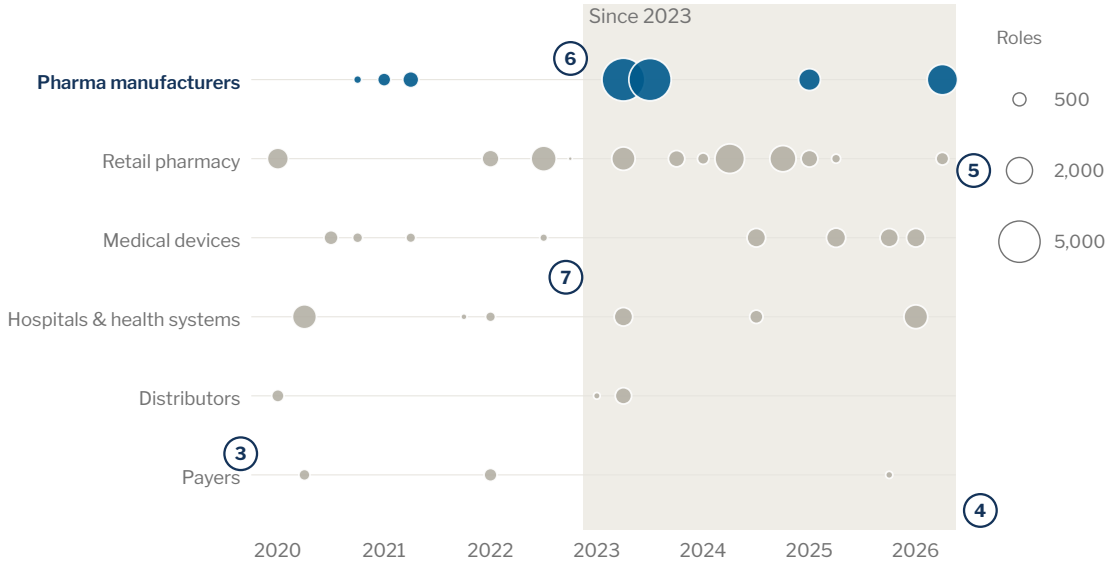
McKesson spec

- Bubble area, not diameter, proportional to value
- One grey for all rows; the headline's row in blue
- Largest bubbles drawn first, with white outlines
- A size key with two or three reference circles

1. **Insight headline**
   Who stands out, by how much.
2. **Subtitle**
   Says what bubble area means.
3. **One row per category**
   Sorted by total, largest on top.
4. **Time along the bottom**
   Even intervals, years labelled.
5. **Size key**
   Reference circles; area ∝ value.
6. **Focus row**
   Blue; white outlines keep overlaps readable.
7. **The period that matters**
   Shaded and named.
8. **Footnote and source**
   What counts as an event.

Pharma manufacturers have announced 14,690 job cuts since 2023, 1.7× the next healthcare sub-sector (1)

Announced workforce reductions by healthcare sub-sector, 2020–Q2 2026 (bubble area = roles affected) (2)



Announcements in public WARN notices and company statements; one bubble per sub-sector per quarter.

Source: External labor-market feed (synthetic data for illustration), as of Jun 2026 (8)

CHANGE OVER TIME

Bubble timeline best practice

The rows already separate the categories; let size do the rest.

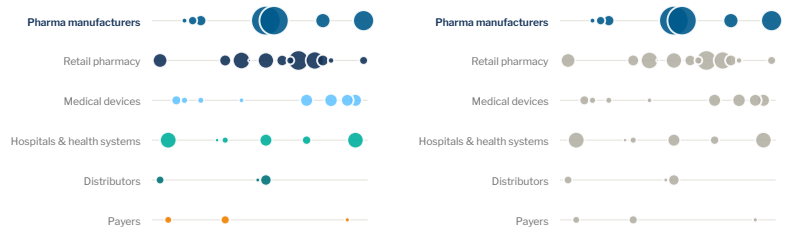
One color. Coloring each row repeats what the rows already say. Use grey, and blue for the headline's row.

Don't hide small bubbles. Draw the largest first, with white outlines.

Sort the rows. By total, or by a meaningful order, not alphabetically.

Show the scale. Without a size key, readers can only say "bigger".

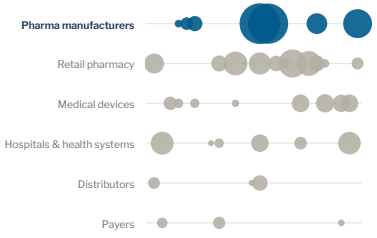
Coloring bubbles



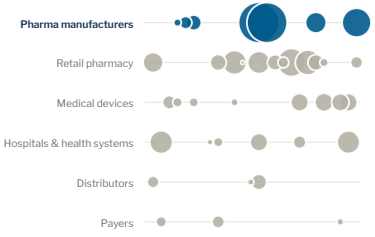
Try to avoid
A color per row.

Recommended
Grey, one row blue.

Overlapping bubbles

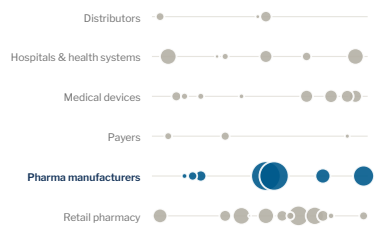


Try to avoid
Big bubbles on top.

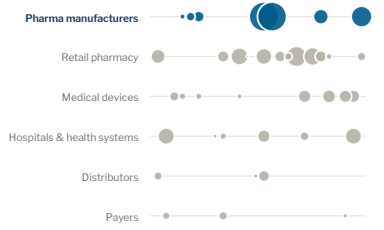


Recommended
Big behind, white outlines.

Row order

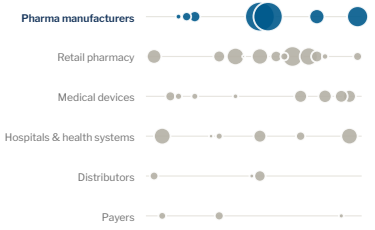


Try to avoid
Alphabetical rows.

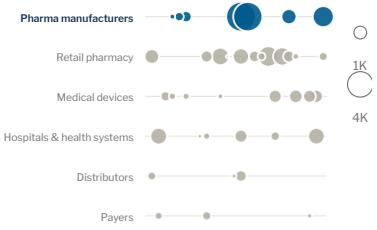


Recommended
Sorted by total.

Size key



Try to avoid
No way to read size.



Recommended
Two reference circles.

CHANGE OVER TIME

Small multiples

The same small chart repeated for each group, on one shared scale. The eye compares shapes across panels instead of untangling crossing lines.

Use it when six or more groups each need their own trend: business units, regions, job families. In each panel, the panel's own line is blue and the others sit faintly behind it.

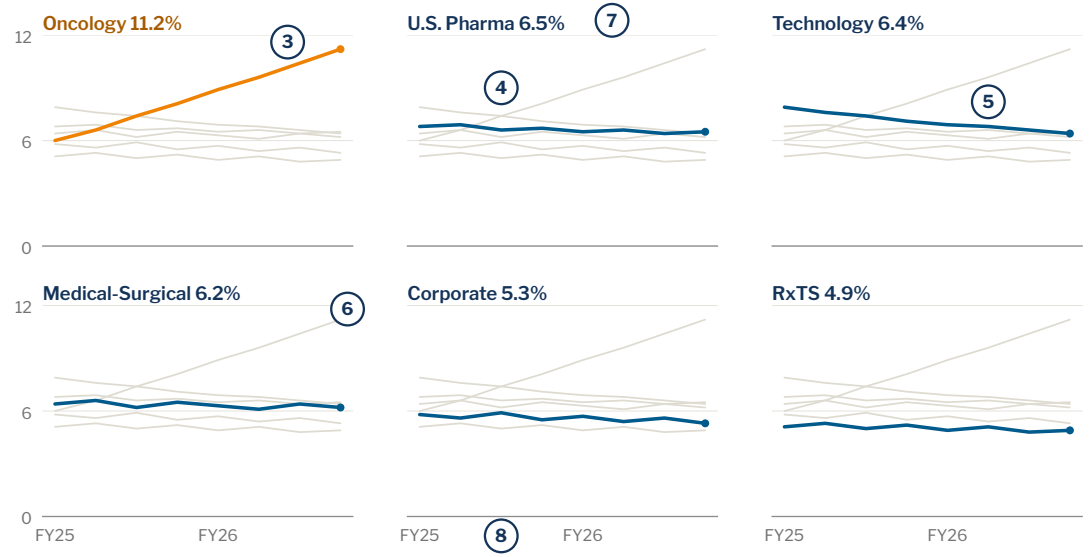
McKesson spec

- One shared scale, stated once
- Panels ordered by value or a natural order
- Panel title = name + latest value
- Axis labels only on the outer edge

1. **Insight headline**
   The one panel that differs.
2. **Subtitle**
   Metric, unit, period, once for all panels.
3. **The exception**
   In orange, because it is getting worse.
4. **Its own line in blue**
   The other units faint behind it, for context.
5. **Ordered by value**
   Highest latest value first.
6. **Shared scale**
   0–12% in every panel.
7. **Title carries the number**
   Name and latest value.
8. **Outer axes only**
   Labels on the left column and bottom row.
9. **Footnote and source**
   Define the metric.

Oncology regretted attrition nearly doubled in two years while every other business unit stayed within 2 points (1)

Regretted attrition by business unit, Q1 FY25–Q4 FY26 (% of high performers, annualized) (2)



Regretted attrition = voluntary exits of employees rated "exceeds" in the last review, annualized.

Source: Workday HCM (synthetic data for illustration), as of 31 Mar 2026 (9)

CHANGE OVER TIME

Small multiples best practice

Small multiples only work if every panel is read the same way.

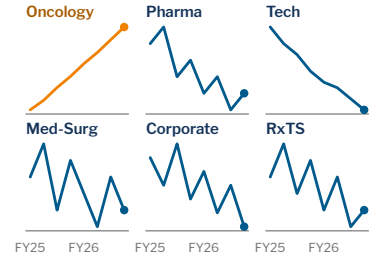
One scale. Independent scales make a 0.5-point wobble look like the 5-point rise.

Context in every panel. The other units, faint behind each line, show whether a panel is unusual.

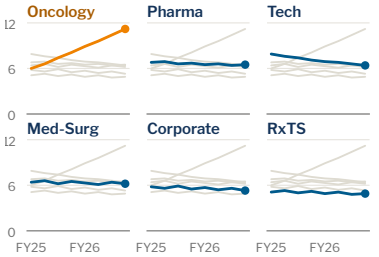
Order with meaning. By value, size or geography, not the alphabet.

Label the edges only. Repeated axes are clutter the eye reads six times.

Shared scale

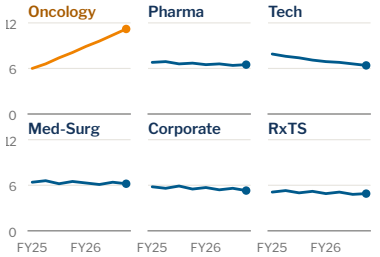


Try to avoid
Every panel zooms itself.

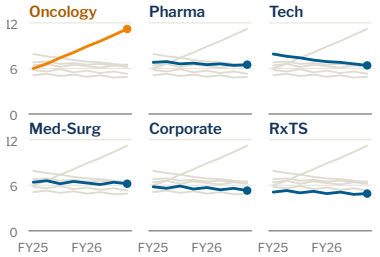


Recommended
One scale; the rise stands out.

Context

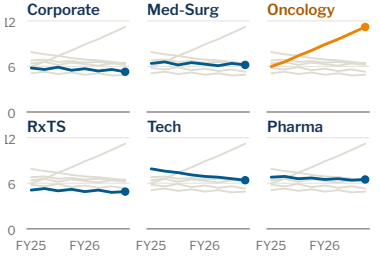


Try to avoid
Lines alone.

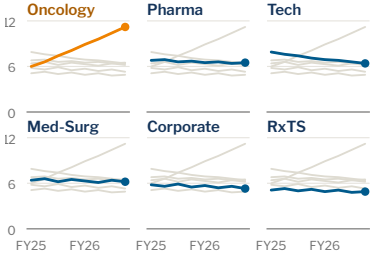


Recommended
The others faint behind.

Panel order

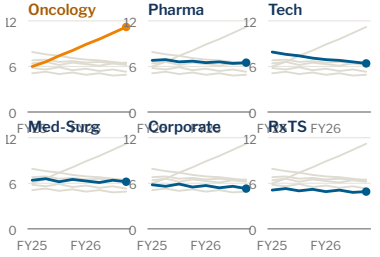


Try to avoid
Alphabetical.

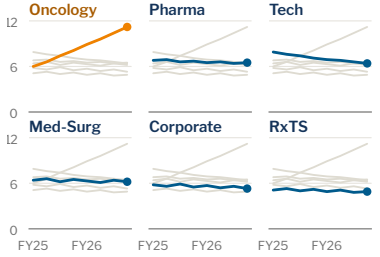


Recommended
By latest value.

Axes



Try to avoid
Axes on every panel.



Recommended
Outer edge only.

CHANGE OVER TIME

Build change-over-time charts

Native, editable charts in every tool. Tested code and click-by-click steps are on the cookbook pages.

line.md · stacked-area.md · bubble-timeline.md · small-multiples.md

Before you send it

- [ ] Time left to right, at true intervals
- [ ] One line or layer in blue, matching the headline
- [ ] Names at line ends; no legend
- [ ] Forecasts dashed, labelled and with a range
- [ ] Small multiples share one scale

PowerPoint and Excel	**Line:** Insert › Line; format the focus series 2.25 pt Focus blue, the rest 1.25 pt grey; delete the legend and add data labels to the last point only (Series Name + Value). Use a *date* axis so missing periods keep their space. **Stacked area:** Insert › 100% Stacked Area; story series first in Select Data; white 1 pt borders. **Small multiples:** build one chart, fix the axis bounds (Minimum/Maximum, not Auto), then copy it per group.
Power BI	Line chart with the focus series colored by measure; turn off the legend and use *Series labels*. Forecast bands as low/high measures shown with *Error bars* › *Shaded area*. Small multiples are built in: add the group to *Small multiples* and keep *Y-axis* › *Shared*.
Python	matplotlib with `mckviz`; `mv.label_line_ends(ax)` places and nudges end labels; `mv.shade_period(ax, start, end, label)` marks the period.

```
for fam, s in df.items():                # one column per job family, one row per quarter
    ax.plot(x, s, label=fam, color=C.focus if fam == focus else C.context,
            lw=2.4 if fam == focus else 1.4)
mv.shade_period(ax, 6, 11, "After the reset")
mv.label_line_ends(ax, fmt="{:.1f}")      # names at the ends, nudged apart
```

HTML	Plotly `scatter` traces with `mode: "lines"`, an annotation per line end, and a `shape` for the shaded period; `facet_col` (Plotly Express) for small multiples with shared axes.
R	`geom_line()` with `mck_highlight()` colors and `geom_text()` end labels; `geom_area()` for stacks; `facet_wrap(~ unit)` keeps a shared scale by default.

RANKING

Slope chart

Ranking charts show order: who comes first, who moved, and how far apart the items really are.

A slope chart joins each item's rank (or value) at two points in time. The eye reads the angle: steep lines are the movers, and crossings show who overtook whom. Use it for skills in job postings, sites or job families before and after a change.

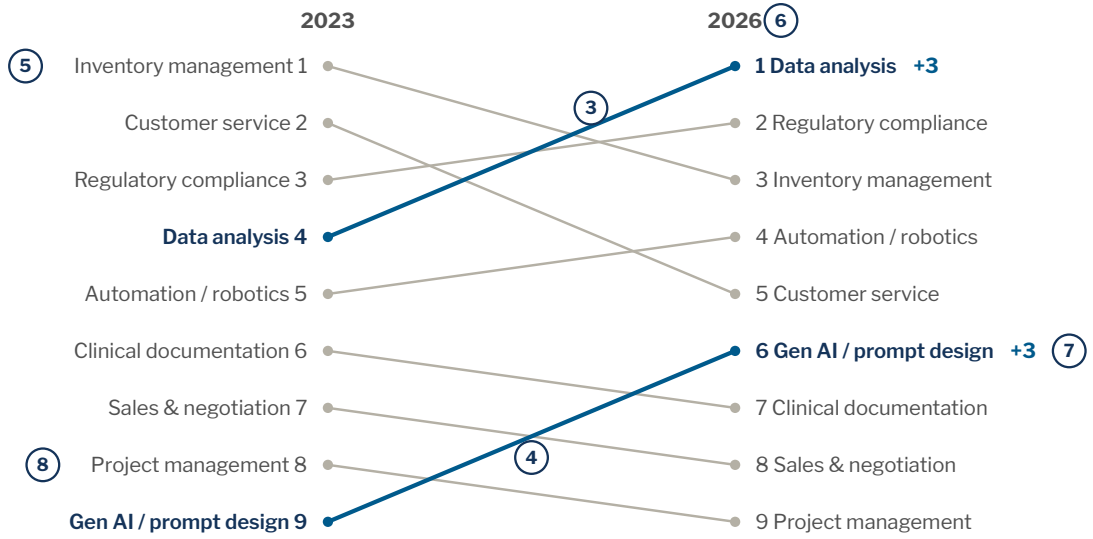
McKesson spec

- Five to ten items; one or two movers in blue
- Names at both ends, in rank order; no legend
- Rank 1 at the top; no axis or gridlines
- Footnote what the rank is based on

1. **Insight headline**
   Names the movers and how far they moved.
2. **Subtitle**
   What is ranked, where, which two years, and what 1 means.
3. **Movers in blue**
   The two the headline names, drawn last so they sit on top.
4. **Grey context**
   Every other item, thinner.
5. **Names at both ends**
   Left in the starting order, right in the ending order.
6. **Period headers**
   Bold, above each column.
7. **The move, sized**
   A signed change beside the focus names.
8. **Rank 1 on top**
   Ranks evenly spaced; the numbers sit beside the names.
9. **Footnote and source**
   What the rank measures.

Data analysis and gen AI are the fastest-rising skills in sector job postings, up three places each since 2023 (1)

Rank of skills in U.S. healthcare-distribution job postings, 2023 vs. 2026 (1 = most requested) (2)



Rank by share of job postings that mention the skill; the 9 most-requested skills in 2023.
Source: External job-postings feed (synthetic data for illustration), U.S. healthcare distribution, 2023 and 2026 (9)

RANKING

Slope and bump variants

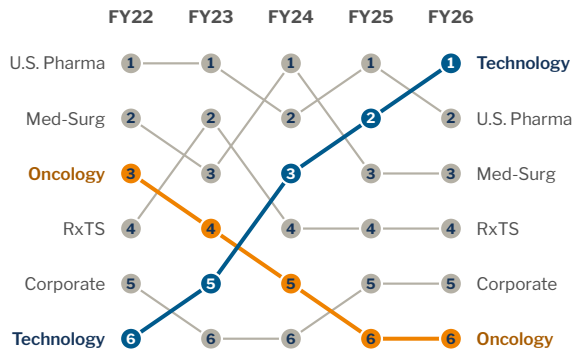
The same rules: grey context, one or two movers in color, names where the lines end.

When a slope is the wrong chart

- The gaps matter more than the order: a **sorted bar** or a **dot plot**
- More than about six periods, or the shape between them matters: a **line chart**
- More than about ten items: show the **top N**, or use **small multiples**
- One period only: a **sorted bar**

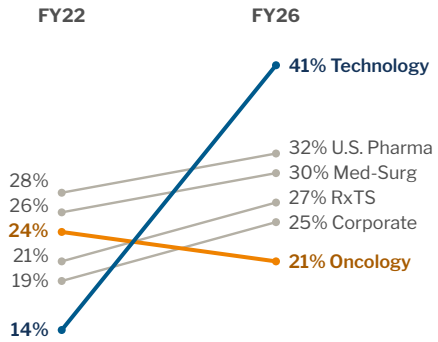
Bump chart

Rank order over five years. Technology climbed from last to first in internal fill; Oncology slid from third to last. A fall to last is worse, so it is orange.



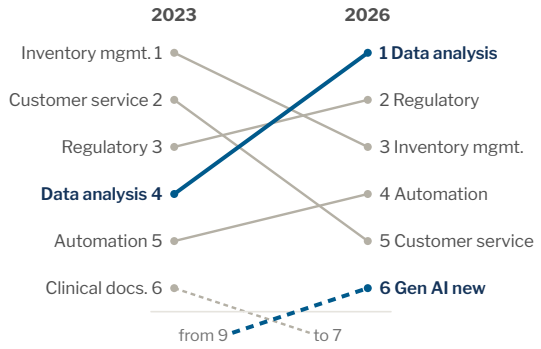
Value slope

Values on one shared scale, when the size of the change matters: Technology went from 14% to 41%, Oncology from 24% to 21%.



Top N, with entries and exits

“Top 6” in the subtitle, and a stub for what enters or leaves: gen AI enters from ninth; clinical documentation drops to seventh.



Ranked list with movement

For dashboards and long lists: the current order, with the change in rank in a column of its own.

2026			vs. 2023
1	**Data analysis**	▲	**+3**
2	Regulatory compliance	▲	+1
3	Inventory management	▼	−2
4	Automation / robotics	▲	+1
5	Customer service	▼	−3
6	**Gen AI / prompt design**	▲	**+3**
7	Clinical documentation	▼	−1
8	Sales & negotiation	▼	−1
9	Project management	▼	−1

Slope and bump best practice

Crossing lines are the story, so make them easy to follow.

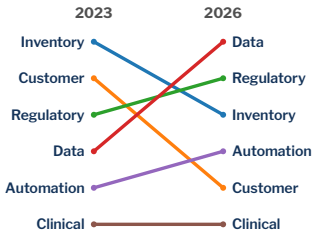
Highlight the movers. A color per line makes every crossing compete. Grey everything, then color the one or two the headline names.

Names at both ends. A legend sends the reader back and forth. Label each line where it starts and where it ends.

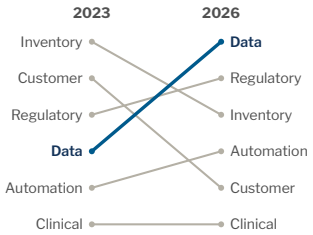
Rank 1 on top. A normal value axis puts first place at the bottom. Reverse it and drop the axis.

Ranks hide gaps. A swap of places looks like an even trade. When the size of the change matters, plot values.

Highlights

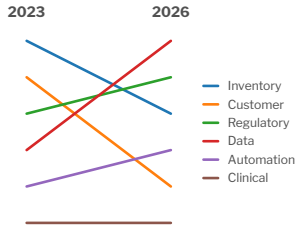


Try to avoid
A color for every line.

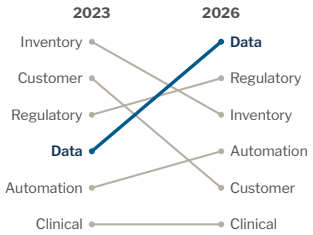


Recommended
Grey context, the mover in blue.

Labelling

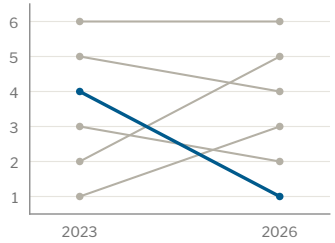


Try to avoid
A legend to decode.

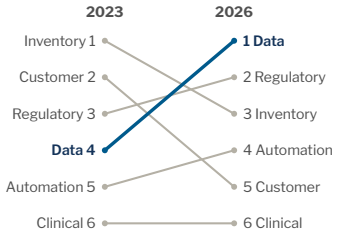


Recommended
Names at both ends.

Rank order

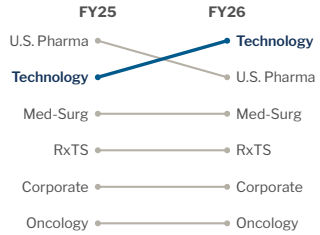


Try to avoid
Rank 1 at the bottom of an axis.

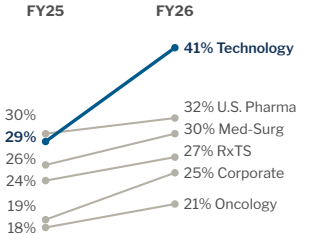


Recommended
Rank 1 on top, numbers by the names.

Ranks hide gaps



Try to avoid
A swap that looks like an even trade.



Recommended
Values show who really moved.

RANKING

Dot plot and dumbbell

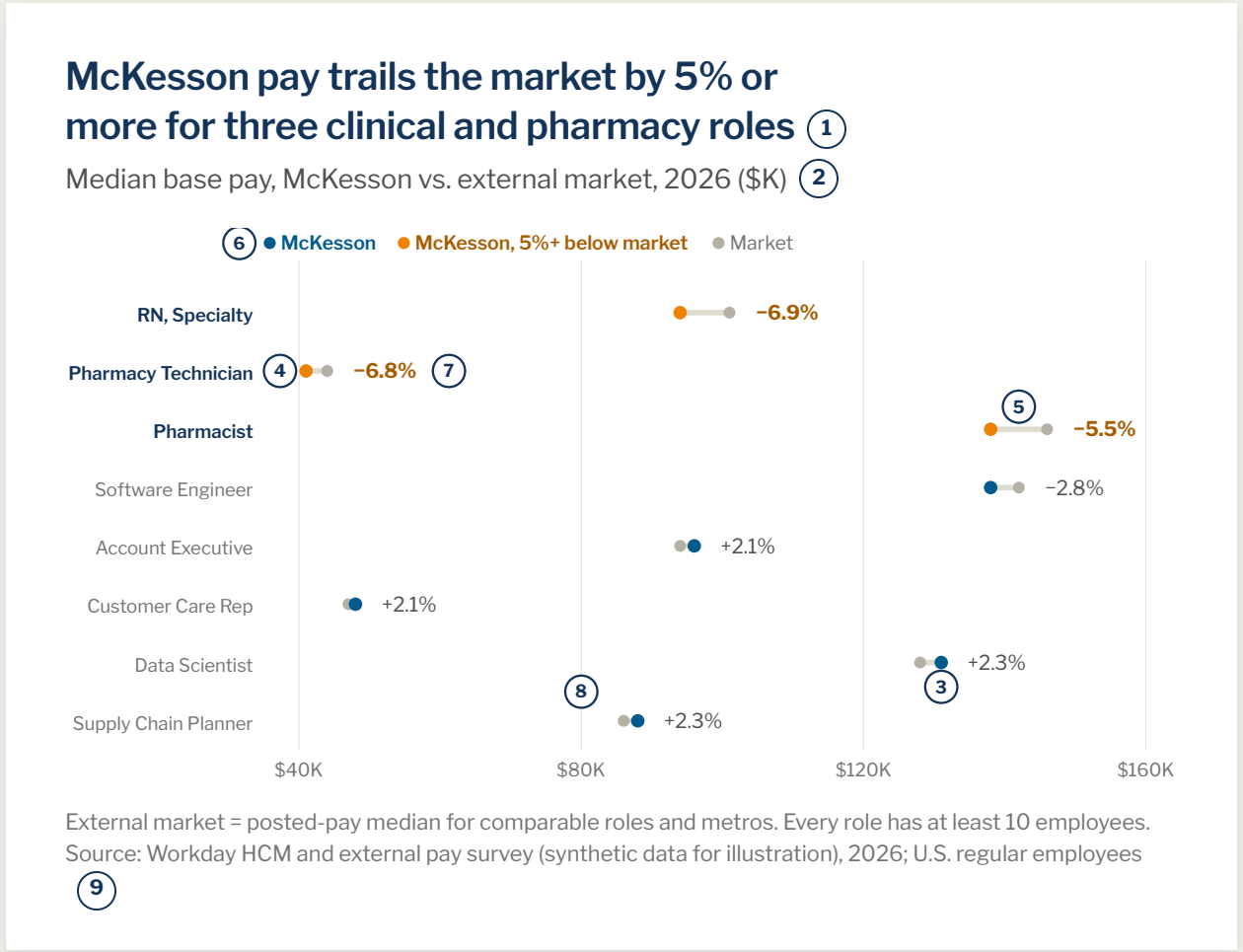
A dot encodes position, not length, so the axis can start where the data starts. A dumbbell joins two dots per row, and the eye reads the gap.

Use a dot plot for scores far from zero (engagement, compa-ratio, proficiency) and a dumbbell for us vs. market, before vs. after, or plan vs. actual.

McKesson spec

- Comparator grey, focus blue; orange only for worse
- Light connector behind the dots
- Signed gap at the row end; sorted by the gap
- Round, non-zero range with light gridlines

1. **Insight headline**
   How many rows fall short, and by how much.
2. **Subtitle**
   Both measures, the year and the unit.
3. **Grey comparator, blue focus**
   Market grey; McKesson blue.
4. **Worse in orange**
   The roles 5% or more below market.
5. **A light connector**
   The gap is what the eye reads.
6. **Every dot named**
   A one-line key above the rows: blue, orange and grey.
7. **Signed gap**
   At the row end, with a true minus; bold for the exceptions.
8. **Position, not length**
   A round, non-zero range and light gridlines.
9. **Footnote and source**
   Define the comparator.



RANKING

Dot plot variants

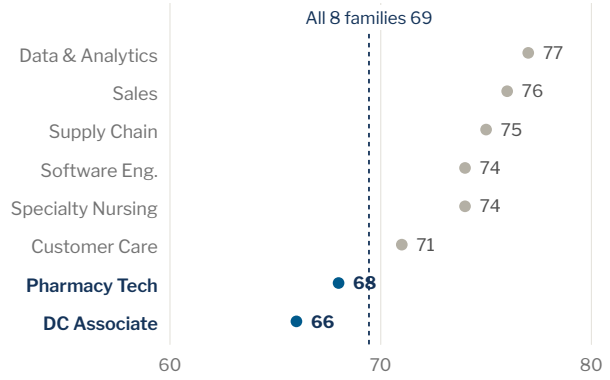
Dots carry one, two or three values per row, on any scale that suits the data.

When a dot plot is the wrong chart

- The size from zero matters (headcount, cost): a **bar chart**
- More than two or three points in time: a **line chart**
- A change in rank: a **slope or bump chart**
- The spread across people: a **distribution**
- Gaps smaller than a dot: plot the **difference** instead

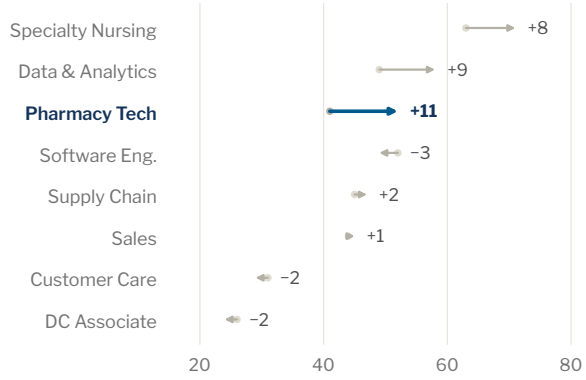
Dot plot with a benchmark

Engagement runs 66–77, so bars would need a cut axis. The two lowest sit below the 69 average for all 8 families.



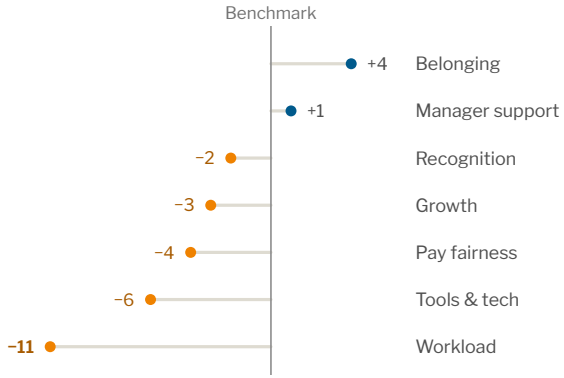
Before and after

Arrows from FY25 to FY26 make direction explicit: pharmacy technician roles take 11 more days to fill.



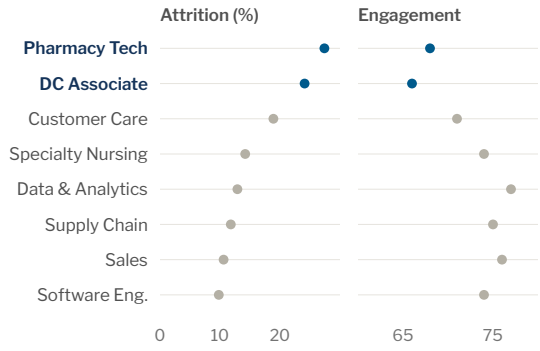
Gaps to a benchmark

Centered on zero, below in orange: Oncology's workload score is −11 points from the survey benchmark.



Two measures, one row order

Side-by-side panels, each on its own scale, keep the rows aligned so the pattern reads across.



RANKING

Dot plot best practice

Dots are quiet marks. These four rules keep the comparison obvious.

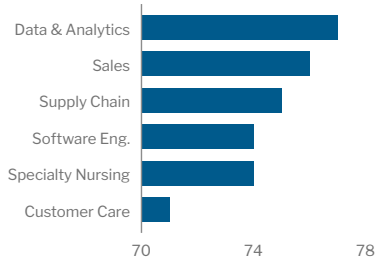
Dots, not cut bars. When values sit far from zero, switch to dots. Never truncate a bar.

Connect the pair. A light connector shows the gap; name each dot type once, in a one-line key above the rows.

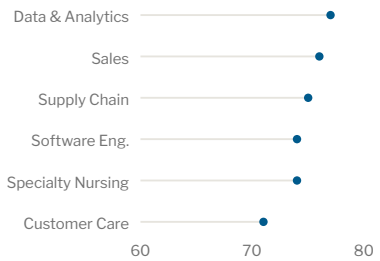
Grey comparator. Two strong colors leave the reader asking which one is us.

Sort by the gap. Alphabetical order hides the story. Put the largest shortfall first and write the gap at the row end.

Values far from zero

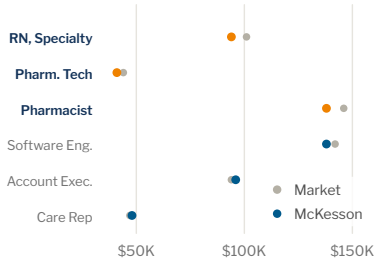


Try to avoid
Bars on an axis cut at 70.

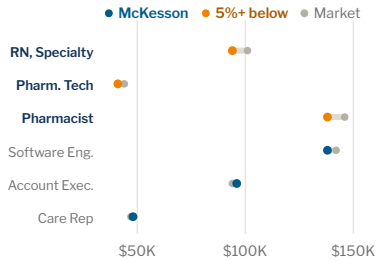


Recommended
Dots on an honest range.

Connect the pair

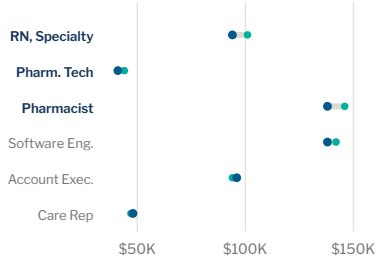


Try to avoid
Loose dots and a legend.

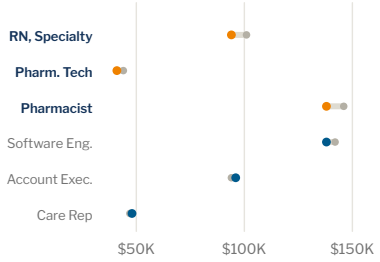


Recommended
A connector; types named once.

Colors

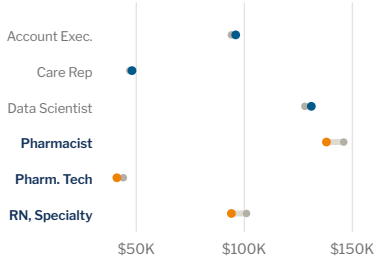


Try to avoid
Two equally strong colors.

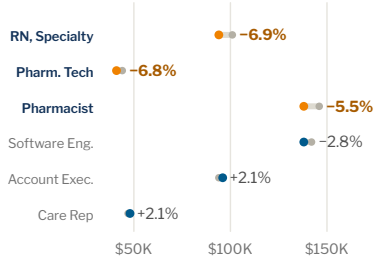


Recommended
Grey market, blue McKesson.

Row order



Try to avoid
Alphabetical rows.



Recommended
Largest shortfall first, gap labelled.

Heat table

A table for looking things up, with one column shaded so the pattern shows before anyone reads a number.

Use it for scorecards and quarterly workforce packs: several measures in different units, which readers will copy into their own work. If there is one message about one measure, make the chart instead.

McKesson spec

- Two shaded columns at most, blue by default
- Horizontal hairlines only; no boxes or zebra stripes
- Twelve rows and seven data columns at most
- No cell under 10 people

1. **Insight headline**
   Says what the table shows.
2. **Subtitle**
   Scope, period, and how totals are built.
3. **Focus rows**
   Bold navy across the whole row.
4. **One shaded column**
   Blue classes, light to dark; text flips to white on dark fills.
5. **Inline bars**
   Size at a glance, from a common edge.
6. **Delta column**
   Signed; only the worse direction in orange.
7. **Units in the headers**
   Bare, right-aligned numbers, same decimals per column.
8. **Weighted total, last**
   Under a navy rule; says what it covers.
9. **Footnote and source**
   The shading key and definitions.

The two largest frontline job families combine the highest attrition with the lowest engagement (1)

Workforce scorecard by job family, FY26; totals weighted by headcount (2)

Job family	Headcount (5)	Voluntary attrition (%) (4)	Engagement index	Median days to fill (7)	Change vs. FY25 (days) (6)
Pharmacy Tech (3)	**6,200**	**27.4**	**68**	**52**	**+11**
DC Associate	**11,800**	**24.1**	**66**	**24**	**−2**
Customer Care	3,100	18.9	71	29	−2
Specialty Nursing	1,400	14.2	74	71	+8
Data & Analytics	850	12.9	77	58	+9
Supply Chain	900	11.8	75	47	+2
Sales	2,300	10.6	76	45	+1
Software Eng.	2,600	9.8	74	49	−3
All 8 job families (8)	**29,150**	**20.7**	**69**	**38**	**+2**

Attrition shaded in 5-point classes from under 5% to 25% and above. Orange change = slower to fill than FY25. HR and Finance are not in the scorecard; the enterprise rate across all 10 job families is 19.9%. Source: Workday HCM, Workday Recruiting and engagement survey (synthetic data for illustration), FY26; U.S. regular employees (9)

RANKING

Heat table best practice

Formatting should point at the message, not decorate every cell.

Shade one or two columns. A color scale on every column is a Christmas tree: nothing stands out.

No traffic lights. Red, amber and green fail for color-blind readers and aren't McKesson colors. Use blue classes; orange only when darker means worse.

Hairlines only. Boxes and zebra stripes add ink between the numbers. A header rule and light row lines are enough.

Weight the total. An average of the rates (19.5) is not the rate for the group (23.3, weighted by headcount).

How much to shade

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	27.4	68	52
DC Associate	24.1	66	24
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58

Try to avoid
Every column on its own scale.

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	**27.4**	**68**	**52**
DC Associate	**24.1**	**66**	**24**
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58

Recommended
One column; focus rows bold.

Color scale

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	27.4	68	52
DC Associate	24.1	66	24
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58

Try to avoid
Red, amber, green.

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	**27.4**	**68**	**52**
DC Associate	**24.1**	**66**	**24**
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58

Recommended
Blue classes, light to dark.

Rules

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	27.4	68	52
DC Associate	24.1	66	24
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58

Try to avoid
Boxes and zebra stripes.

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	**27.4**	**68**	**52**
DC Associate	**24.1**	**66**	**24**
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58

Recommended
Header rule, hairlines.

Total row

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	**27.4**	**68**	**52**
DC Associate	**24.1**	**66**	**24**
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58
Average	**19.5**		

Try to avoid
An average of averages.

	Attrition (%)	Engage-ment	Days to fill
Pharm. Tech	**27.4**	**68**	**52**
DC Associate	**24.1**	**66**	**24**
Customer Care	18.9	71	29
Spec. Nursing	14.2	74	71
Data & Analytics	12.9	77	58
All 5 families	**23.3**		

Recommended
Weighted by headcount.

RANKING

Build ranking charts

Native, editable charts and tables in every tool. Tested code and click-by-click steps are on the cookbook pages.

slope-bump.md · dot-plot-dumbbell.md · tables.md

Before you send it

- [ ] Rank 1 on top; names at both ends
- [ ] One or two highlights, matching the headline
- [ ] Dots, not cut bars, for values far from zero
- [ ] Units in table headers; two shaded columns at most
- [ ] Totals weighted by headcount
- [ ] Nothing ranked or shown under 10 people

PowerPoint and Excel	**Slope and bump:** Insert › Line with Markers; vertical axis *Values in reverse order*, bounds 0.5 to n + 0.5, labels none; context series grey 1 pt, focus 2.25 pt blue; Series Name labels on the first and last points. **Dumbbell:** XY scatter with custom horizontal error bars (no cap, 3 pt, light sand) as the connector. **Heat table:** clear the table style; bottom borders only; one conditional-formatting rule per blue class on the shaded column; data bars with *Show Bar Only* in a helper column.
Power BI	Bump: a Line chart on a `RANKX` measure with *Y-axis › Invert range* and series labels on. Dumbbell: the Deneb visual, or a line chart with markers and error bars. Heat table: a Table visual with *Cell elements › Background color › fx › Rules* on one measure. Blank small groups: `IF([Headcount] < 10, BLANK(), [Rank])`.
Python	matplotlib with `mckviz`; colors from `mv.highlight`, signed gaps from `mv.signed`, cell text from `mv.text_on(fill)`.
HTML	Plotly `lines+markers+text` with `autorange: "reversed"` for bumps; one connector trace with `None` breaks for dumbbells. Tables as real `<table class="mck-table">` markup with `is-focus`, `is-total` and `.bar`.
R	`geom_line()` + `geom_point()` with `scale_y_reverse()`; `geom_segment()` under two `geom_point()` layers for dumbbells; `gt` with `data_color(method = "bin")` for heat tables.

```
colors = dict(zip(ranks, mv.highlight(ranks, focus=risers)))
for name, (r0, r1) in ranks.items():          # rank 1 at the top
    ax.plot([0, 1], [r0, r1], color=colors[name])
    ax.text(-0.05, r0, f"{name}  {r0}", ha="right", va="center")
    ax.text(1.05, r1, f"{r1}  {name}", ha="left", va="center")
ax.set_ylim(len(ranks) + 0.5, 0.5); ax.axis("off")
```

DISTRIBUTION

Strip and box plot

Distribution charts show how a measure spreads across people, teams or sites: what is typical, how wide the range is, and where the tails sit.

Average compa-ratio is 0.97, yet 25% of pharmacy technicians are below 0.90. A strip plot shows every person as a dot; the box marks the middle half.

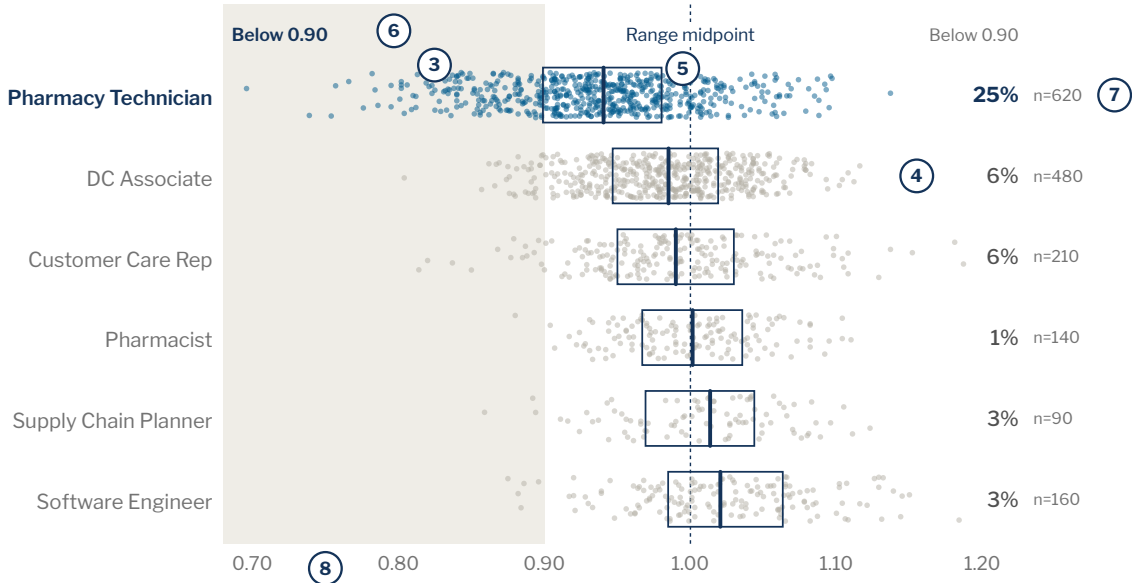
McKesson spec

- Two to eight groups, each n ≥ 10, n shown
- Grey translucent dots; the focus group blue
- Zone shaded; reference line labelled
- Anonymous dots: no names, IDs or values

1. **Insight headline**
   The share in the tail, and how it compares.
2. **Subtitle**
   Defines the measure, scope and date.
3. **Focus group in blue**
   Each dot one employee; overlaps read as density.
4. **Grey context**
   Every other group, the same grey.
5. **Box on top**
   Middle 50% of people; the thick line is the median.
6. **Zone and reference**
   Shaded below 0.90; the midpoint dashed and named.
7. **Share and n**
   The number the headline uses, at the row end.
8. **Keep the value axis**
   Its ticks are how positions are read; no gridlines.
9. **Footnote and source**
   Explain the dots and the box.

One in four pharmacy technicians is paid below 0.90 of range midpoint, about 4× the share in any other job family (1)

Compa-ratio (base pay ÷ salary-range midpoint) by job family, U.S. regular employees, Mar 2026 (2)



Each dot is one employee. Box = middle 50% of employees; thick line = median.

Source: Workday HCM (synthetic data for illustration), as of 31 Mar 2026; U.S. regular employees (9)

DISTRIBUTION

Histogram

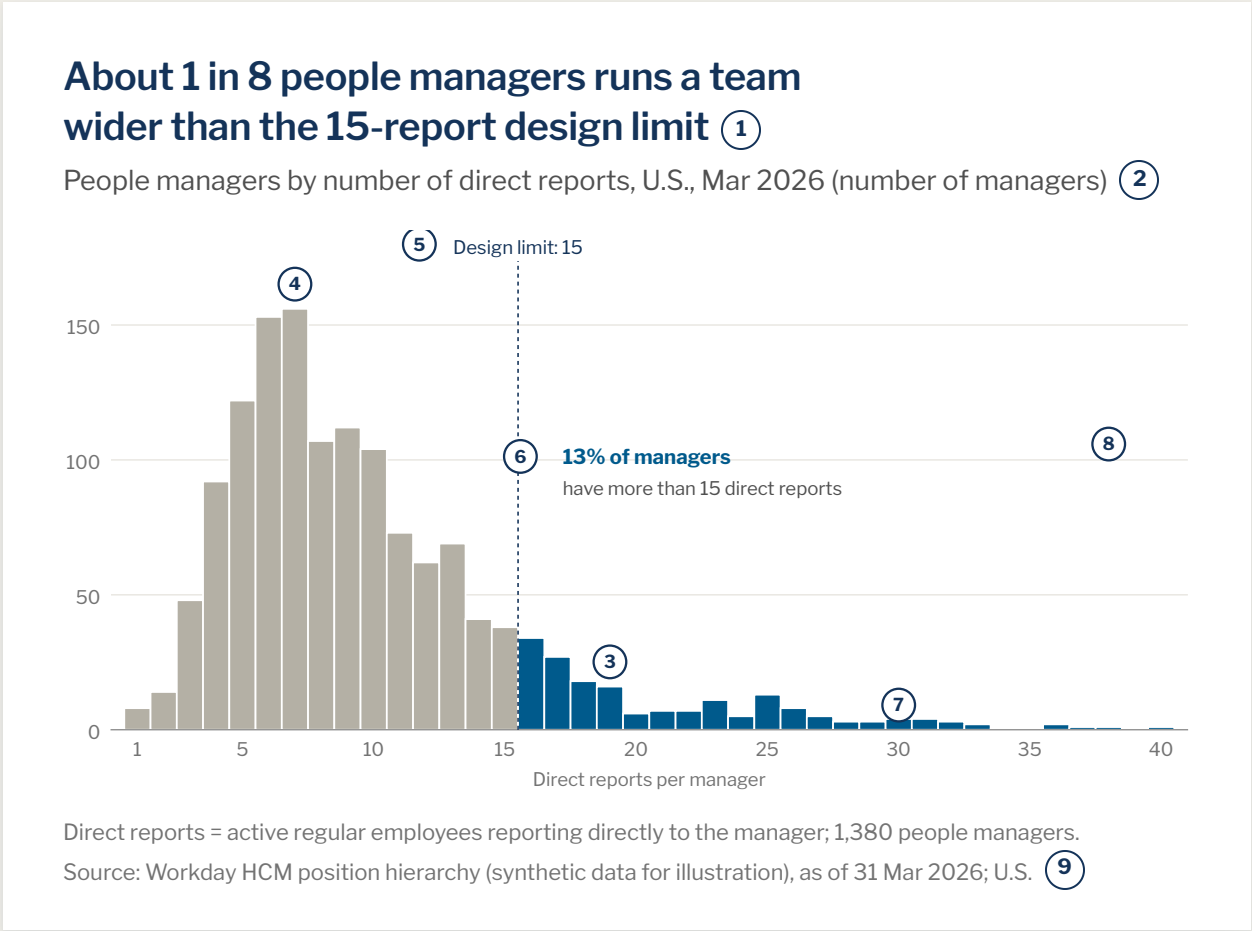
Bars that count how many people fall in each equal-width bin. A histogram shows the shape of one group: skew, a long tail, two peaks.

Use it for one population in detail: span of control across all people managers, days to fill across all requisitions. Here the median manager has 8 direct reports, but the tail runs to 40.

McKesson spec

- Equal bins that touch, with a thin white seam
- A bin edge exactly on the policy line
- One bin per value for counts; 15–40 bins otherwise
- Bins past the threshold in blue, the rest grey

1. **Insight headline**
   How many people are past the line.
2. **Subtitle**
   Population, date, and what the bars count.
3. **Bins that matter**
   Past the limit, in blue.
4. **Grey context**
   The rest of the distribution.
5. **The limit on a bin edge**
   Dashed navy, named in place.
6. **The share, written out**
   Computed with the same rule as the colors.
7. **One bin per value**
   Whole numbers; bins touch.
8. **Light count gridlines**
   No axis line; the baseline stays.
9. **Footnote and source**
   Define who counts.



DISTRIBUTION

Distribution variants

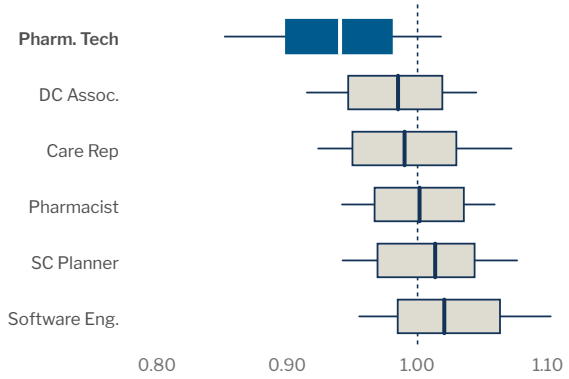
Pick the form by how many groups you compare and how many people are in them.

When a distribution is the wrong chart

- One value per group (median pay by family): a **bar** or **dot plot**
- How the medians shifted between two dates: a **dumbbell**
- Two measures per person: a **scatter**, or a binned **heatmap** for thousands
- Groups under 10 people: merge them or leave them out

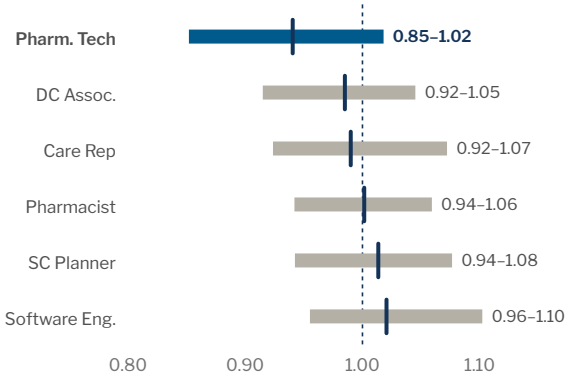
Box plot

For many groups or tens of thousands of people. Whiskers at the 10th and 90th percentiles, footnoted.



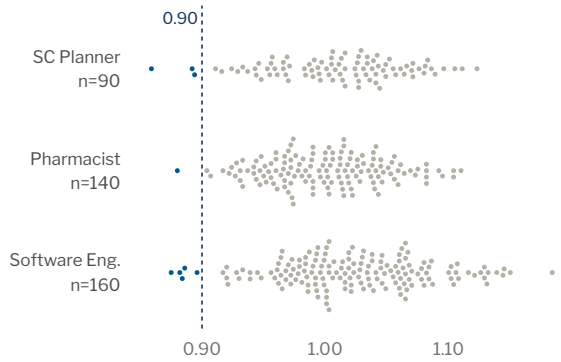
Range bars

A simpler box for executive decks: the middle 80% as a bar, the median as a tick. Pharmacy technicians start at 0.85.



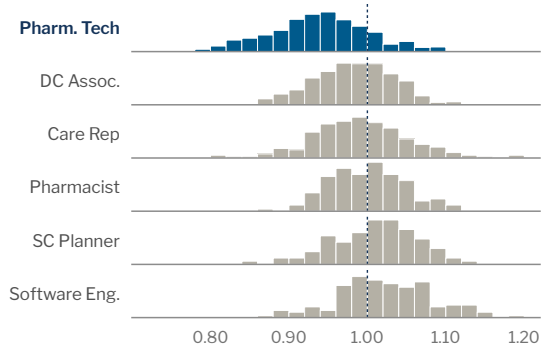
Beeswarm

Dots that never overlap, so every person is countable. Works up to about 300 per row; above that, jitter.



Small-multiple histograms

Three to six groups whose shapes differ. Same bins, same scale, shown as a share of each group.



Distribution best practice

The spread is the message; protect it, and protect the people in it.

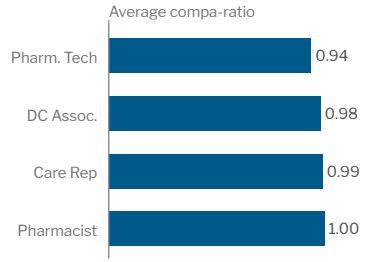
Show the spread, not just the average. Group averages look alike; the tails below 0.90 do not.

A bin edge on the threshold. Wide bins that straddle 0.90 color the wrong people: 17% instead of 25%.

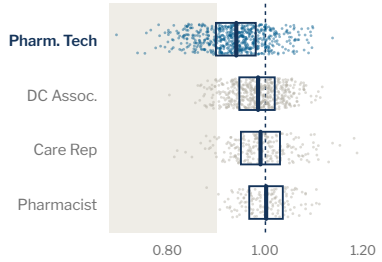
One scale for every panel. Rescaled panels make every group look the same shape.

No group under 10. A dot in a team of four is a findable person. Merge small groups and show n.

Spread, not average

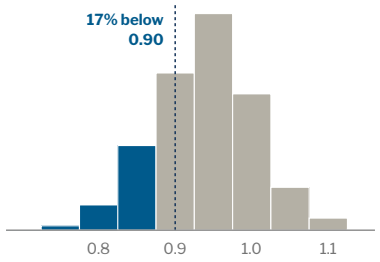


Try to avoid
Averages that hide the tail.

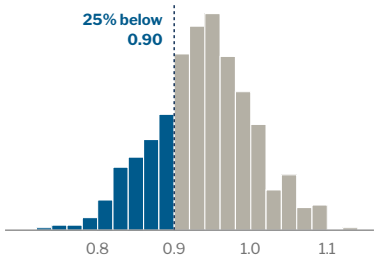


Recommended
Every person, with the box.

Bin edges

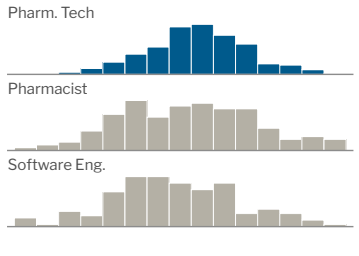


Try to avoid
A bin straddling 0.90.

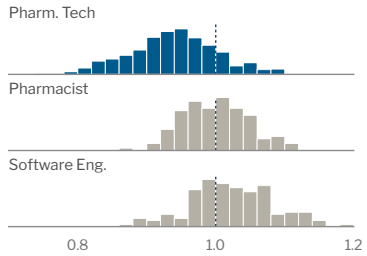


Recommended
An edge exactly on 0.90.

Shared scale

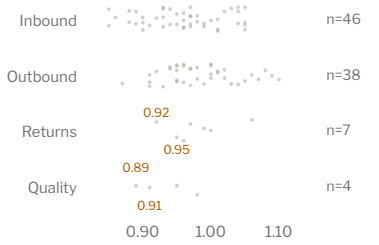


Try to avoid
Each panel on its own range.

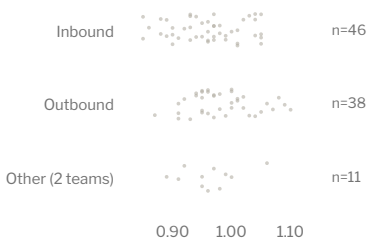


Recommended
Same bins, same scale.

Minimum group size



Try to avoid
Teams of 7 and 4, values labelled.



Recommended
Small teams merged; n shown.

Distribution best practice, continued

Four more, for strip and box plots in particular.

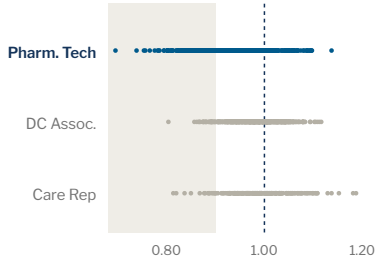
Jitter and fade. Opaque dots on one line turn into a bar. Spread them across the row and make them translucent.

Explain the box. Few readers know 1.5 × IQR whiskers. Use the 10th–90th percentile and say so.

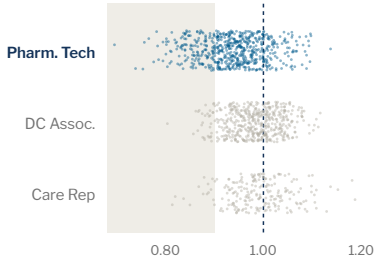
Don't clip the tails. A tight axis silently drops people (64 here). Widen it, or cap it and footnote how many.

Sort by median. Put the focus group next to the zone, not where the alphabet puts it.

Overlapping dots

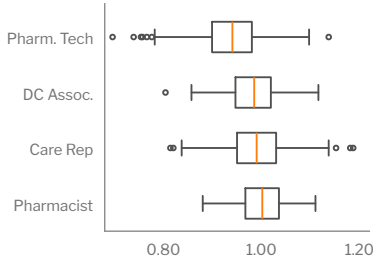


Try to avoid
Opaque dots on one line.

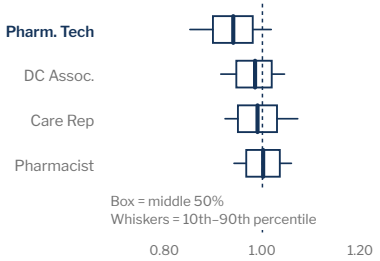


Recommended
Jittered and translucent.

Box and whiskers

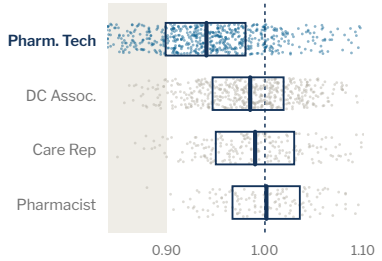


Try to avoid
Software defaults, unexplained.

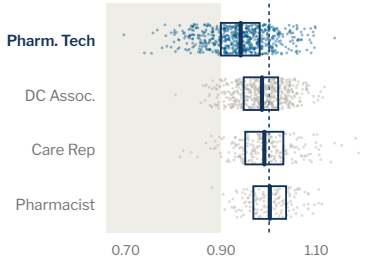


Recommended
10th–90th whiskers, explained.

Axis range

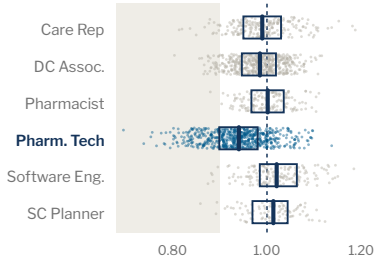


Try to avoid
Tails cut off silently.

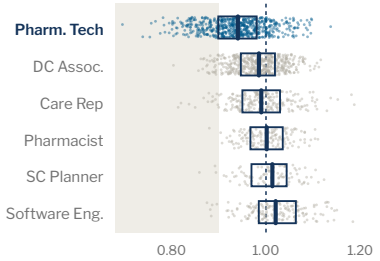


Recommended
Room for everyone.

Group order



Try to avoid
Alphabetical.



Recommended
By median, focus by the zone.

DISTRIBUTION

Build distribution charts

Set the bins and the jitter yourself, not the software. Tested code and click-by-click steps are on the cookbook page.

distribution.md

Before you send it

- [ ] Form matches the number of groups
- [ ] Equal bins; an edge on the threshold
- [ ] n for every group; none under 10
- [ ] No individual's value in labels or tooltips
- [ ] One x scale across rows and panels
- [ ] Box footnoted; nothing clipped

PowerPoint and Excel	**Histogram:** make your own bin column and count with `COUNTIFS`, then a clustered column with Gap Width 0% and white 0.75 pt borders; bins past the threshold in Focus blue. **Box plot:** Insert › Statistic Chart › Box and Whisker, with outlier points and mean markers off. **Strip:** XY scatter with y = group index + `RAND()` jitter, then Paste Values so it stays fixed.
Power BI	Histogram: *New group › Bin type: Size of bins*, a clustered column chart with inner padding 0, colors by *fx › Rules*, and the threshold as an X-axis constant line. No native box plot: use the certified Box and Whisker visual or a scatter with a jitter column. Turn tooltips off for individual points.
Python	matplotlib with `mckviz`: fixed-seed jitter, translucent dots, the box drawn on top.
HTML	Plotly: pre-bin with `np.histogram` and draw `go.Bar` so you control the edges; `go.Box` with `boxpoints: "all"` for strips, and `hoveron: "boxes"` so no one can hover a single person's pay.
R	`geom_jitter()` under `geom_boxplot(fill = NA, outlier.shape = NA)`; `geom_histogram(breaks = ...)` with explicit edges; `facet_wrap()` keeps one scale.

```
rng = np.random.default_rng(1)                # fixed jitter: reproducible
for y, fam in enumerate(order):               # groups with n >= 10 only
    x = df.compa_ratio[df.job_family == fam]
    ax.scatter(x, y + rng.uniform(-0.28, 0.28, len(x)), s=7, alpha=0.45,
               color=C.focus if fam == focus else C.context, lw=0)
ax.axvspan(0.68, 0.90, color=C.surface_subtle, zorder=0)
```

CORRELATION

Bubble scatterplot

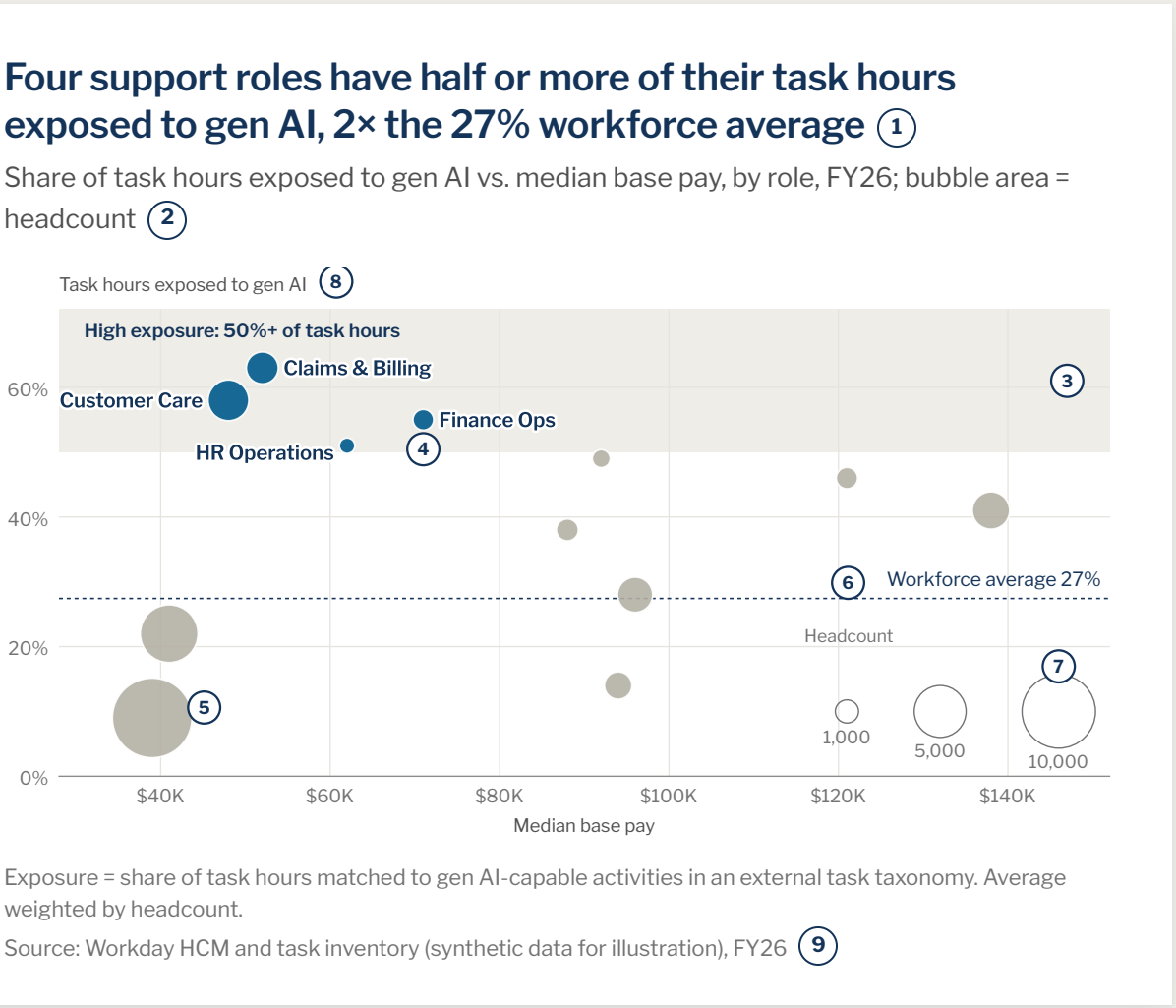
Correlation charts show how two or more measures relate to each other: does X go with Y, and who sits where it matters?

Each entity is one mark. Position carries two measures; bubble area can carry a third, such as headcount. Use it for exposure vs. pay by role, engagement vs. span of control by team, or time to fill vs. market tightness by metro.

McKesson spec

- Grey bubbles; the ones the headline names in blue
- Bubble area, not diameter, ∝ value; a size key
- Label five points at most; shade the zone that matters
- Say "is associated with", never "causes"

1. **Insight headline**
   Who is in the zone, and by how much.
2. **Subtitle**
   Both measures, the period, and what bubble area means.
3. **The zone that matters**
   Shaded behind the marks and named in place.
4. **Focus in blue, labelled**
   Only the bubbles the headline names.
5. **Grey context**
   Unlabelled; big bubbles drawn first, white outlines.
6. **Reference line**
   Navy, dashed, its value computed and labelled.
7. **Size key**
   Three round values on the same scale.
8. **Horizontal axis titles**
   Units on the axes; the Y title top-left.
9. **Footnote and source**
   Define the measure and the average.



CORRELATION

Scatterplot variants

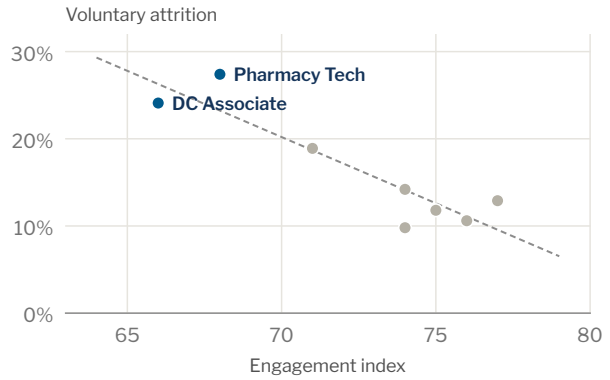
The same rules hold: grey context, blue for the story, labels only where the headline points.

When a scatter is the wrong chart

- One measure, ranked: a **bar**
- One measure over time: a **line**
- The X measure is a category: a **dot plot**
- Thousands of overlapping people: a **binned heatmap**
- Four or more measures per entity: **parallel coordinates**

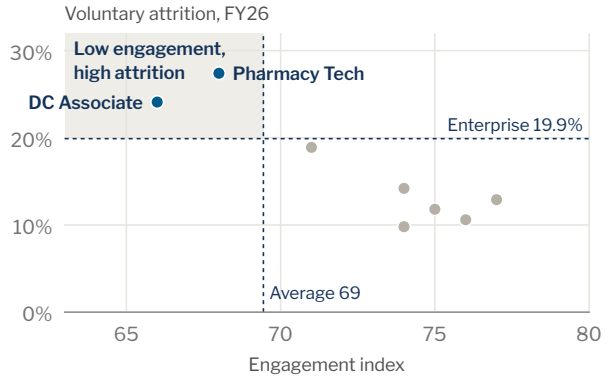
Plain scatter

No size measure, equal dots. Lower engagement goes with higher attrition across job families (r = −0.91).



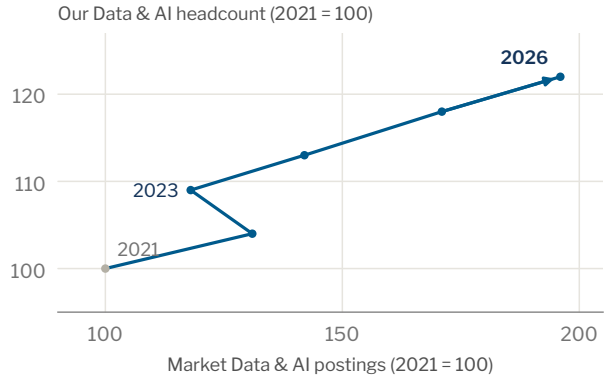
Quadrant

Reference lines at meaningful values (the enterprise rate, the average); shade only the quadrant the headline names. Two families sit in it.



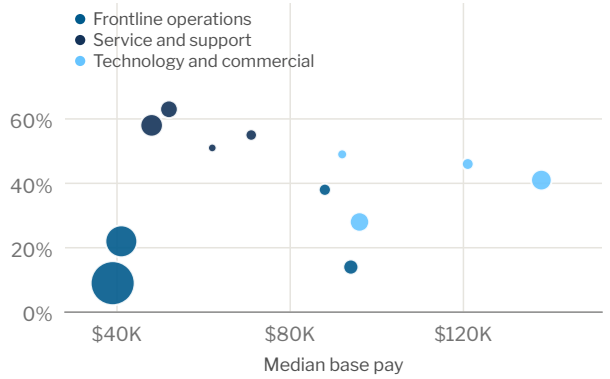
Connected scatter

One entity over time, points joined in order. Market postings reached 196 while our headcount reached 122.



Color for a real group

Color only when it encodes a variable, with a key: four groups at most, in palette order.



CORRELATION

Scatterplot best practice

A scatter invites the eye to roam. These four rules give it somewhere to land.

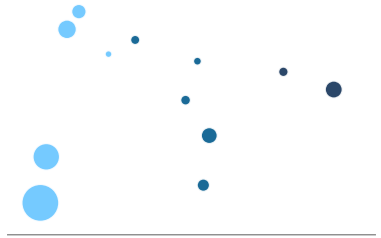
Color carries a variable, or the story. Coloring by x-range repeats the position and encodes nothing.

Two trend lines at most. One line per highlighted group; the rest stay grey, with no line.

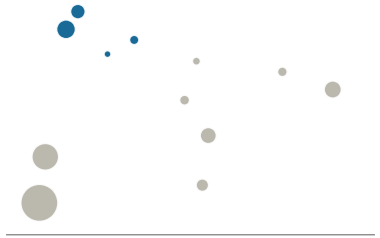
Label what the headline names. Twelve labels bury the point. Put the rest in tooltips.

Big bubbles behind. Draw the largest first, with white outlines, so small ones stay visible.

Using color

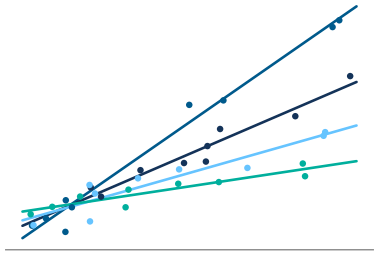


Try to avoid
Colors by pay range.

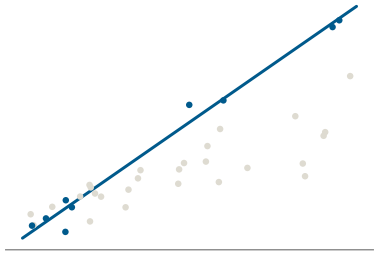


Recommended
Grey, the story in blue.

Trend lines



Try to avoid
Four crossing lines.



Recommended
One group, one line.

Labelling points



Try to avoid
Every bubble named.



Recommended
Only the focus.

Overlapping bubbles



Try to avoid
The focus hidden underneath.



Recommended
Big first, white outlines.

CORRELATION

Dual axis charts

Two measures in different units, one axis each. Both ranges are arbitrary, so where the lines cross or track is the designer's choice, not the data's.

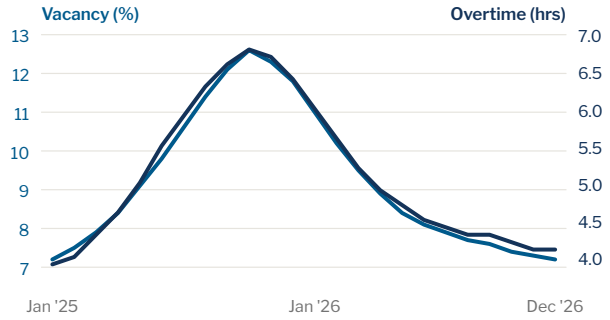
Pick the alternative from the question

- Did B move when A moved? **Two aligned panels** on one time axis
- Which grew faster? **Index both** to 100 at a stated base
- How did B respond to A? A **connected scatter**

If a template forces one: columns for the quantity, a line for the rate, both axes from zero, no horizontal gridlines, each axis title colored to its series.

Did overtime move with vacancies? Dual axis

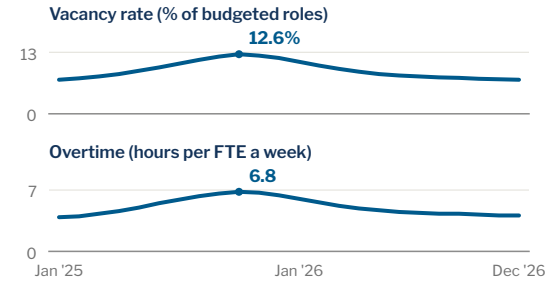
Ranges picked so the lines lie on top of each other: "overtime tracks vacancies exactly".



Try to avoid
The fit is an artifact of two chosen ranges.

Did overtime move with vacancies? Two aligned panels

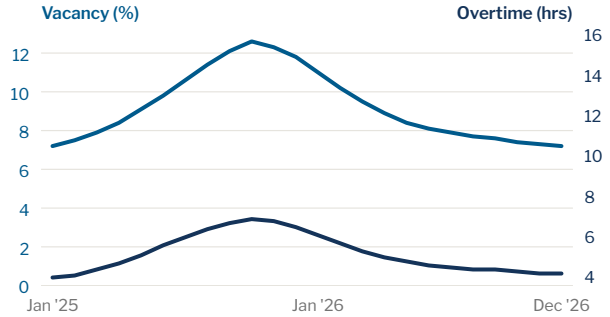
Each measure on its own scale from zero; the shared time axis makes timing exact. Both peaked in Oct 2025.



Recommended
When the question is timing.

Which grew faster? Dual axis

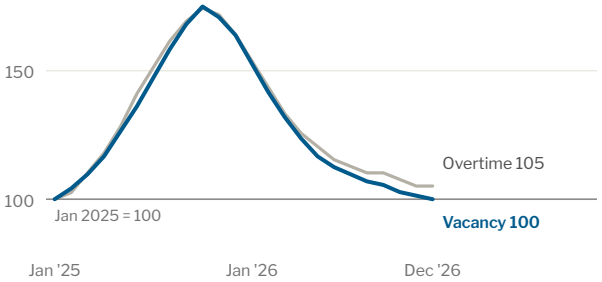
The same data, another right-hand range: now overtime looks flat while vacancies soar.



Try to avoid
Two scales, two contradictory stories.

Which grew faster? Indexed to 100

One honest axis: at the Oct 2025 peak, vacancies were up 75% and overtime 74%; both have since fallen back near 100.



Recommended
When the question is relative change.

CORRELATION

Diverging heatmap

A heatmap shows one measure across two sets of categories as color, so the eye finds patterns in 30 to 150 cells at once: a weak column, a strong row, one hot cell.

Use a diverging scale when the data has a meaningful middle: a benchmark, a target, change vs. last year. Use a sequential scale when the reference sits at one end, such as zero.

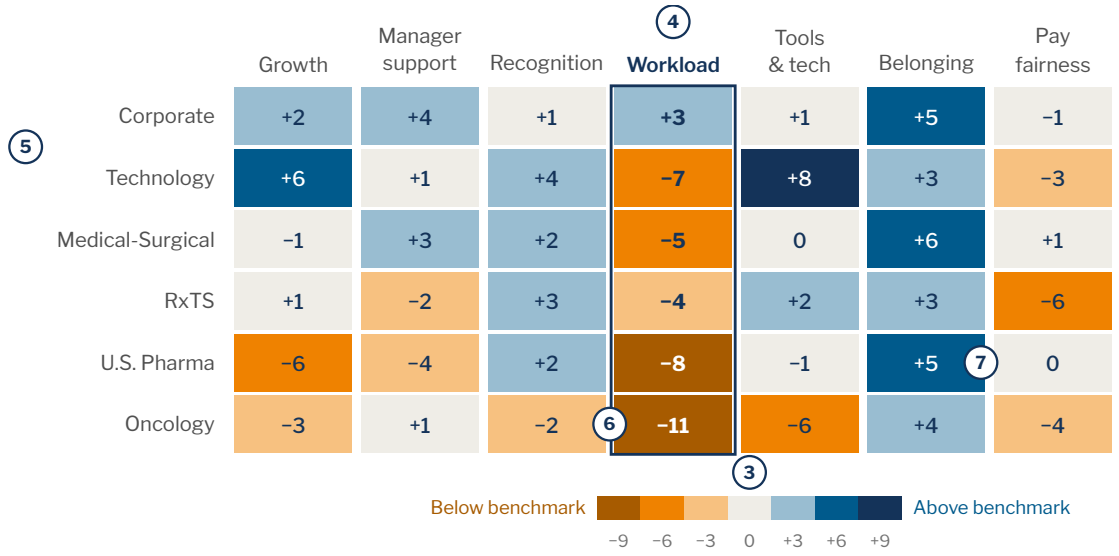
McKesson spec

- Seven even classes; the middle one Light Sand
- Orange below the reference (worse), blue above
- Rows ranked by value, never alphabetical
- Outline one column or row; don't add a hue

1. **Insight headline**
   The pattern, counted from the data.
2. **Subtitle**
   Measure, reference, scope, unit.
3. **Neutral middle**
   Symmetric classes around zero, stepped key with end labels.
4. **Focus column**
   Outlined in navy, header bold.
5. **Ranked rows**
   By average gap, so the pattern emerges.
6. **Orange means worse**
   Below the benchmark, darker the further below.
7. **Values in the cells**
   White on dark fills, navy on light.
8. **Footnote and source**
   Define the gap; note any suppression.

Workload trails the benchmark in five of six business units, Oncology most of all (1)

Engagement driver score vs. external healthcare benchmark, by business unit, 2026 (points) (2)



Gap = business-unit favorable score minus the external healthcare benchmark. Rows ranked by average gap; cells under 10 respondents would show grey, marked n<10.

Source: Employee engagement survey and external benchmark (synthetic data for illustration), 2026 (8)

CORRELATION

Heatmap variants

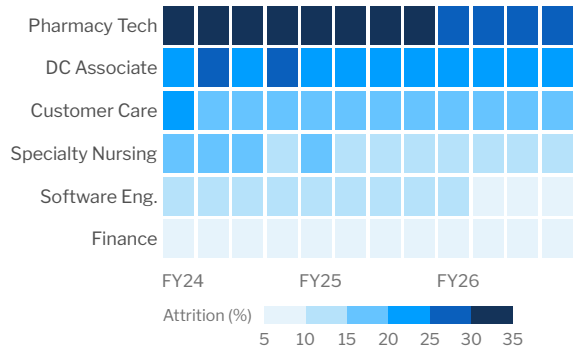
Same build: white gutters, even classes, ranked rows. Blue is the default ramp for any single measure.

When a heatmap is the wrong chart

- A few values to compare precisely: a **bar** or **dot plot**
- A trend for one to four series: a **line**
- Columns in different units: a **table**, or one scale per column
- Places: a **map**

Sequential, over time

Reference at one end: darkest = highest. Pharmacy Tech is darkest in every quarter.



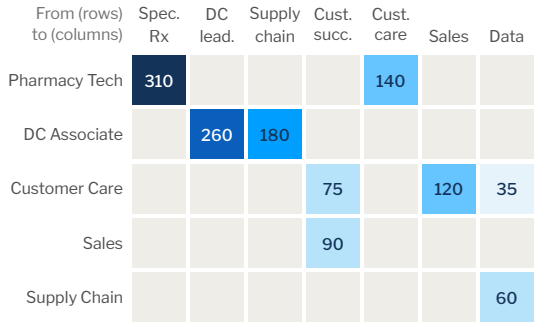
Heat table

Values in every cell, color as a second cue. Oncology reached 11.2% in Q4 FY26.

	Q1	Q2	Q3	Q4
Oncology	**8.9**	**9.6**	**10.4**	**11.2**
U.S. Pharma	6.5	6.6	6.4	6.5
Technology	6.9	6.8	6.6	6.4
Medical-Surgical	6.3	6.1	6.4	6.2
Corporate	5.7	5.4	5.6	5.3
RxTS	4.9	5.1	4.8	4.9

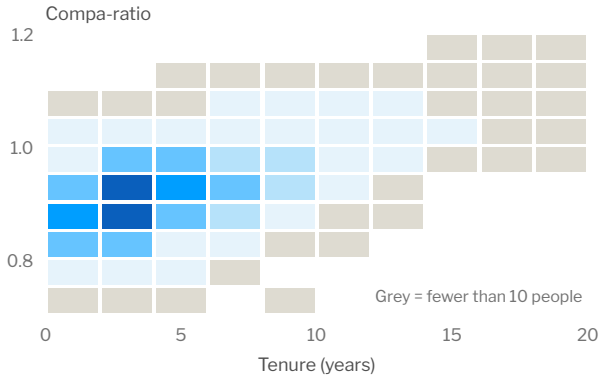
Origin × destination

Many-to-many moves read better as a matrix than a tangled Sankey. Largest: Pharmacy Technician to Specialty Pharmacy Ops, 310 moves.



Binned density

Thousands of people on two measures: count them into a grid instead of overplotting dots.



CORRELATION

Heatmap best practice

Color is read less precisely than length, so the scale has to be honest and the order has to help.

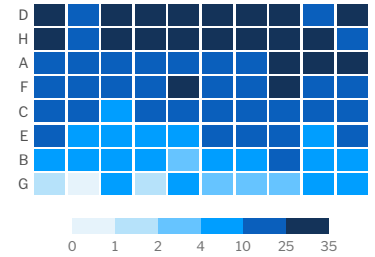
Even steps. Hand-picked breaks (1, 2, 4, 10, 25) make small differences look large. Use five to seven equal classes.

Rank the rows. Alphabetical order scatters the pattern; ranking by average reveals it.

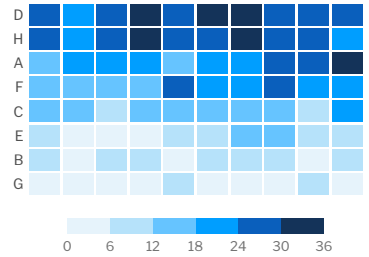
Keep the zero step. A diverging scale without a neutral class forces every cell above or below.

No cell under 10 people. Grey it, mark it n<10 and footnote it, in tooltips too.

Color steps

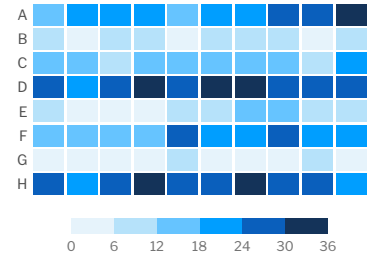


Try to avoid
Uneven breaks.

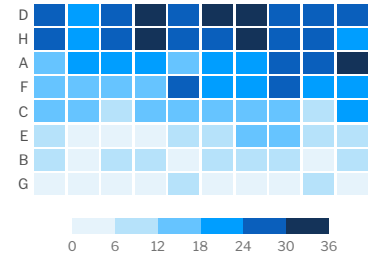


Recommended
Six equal classes.

Sorting

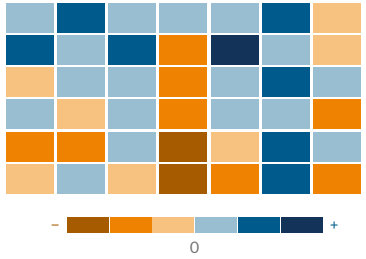


Try to avoid
Alphabetical rows.

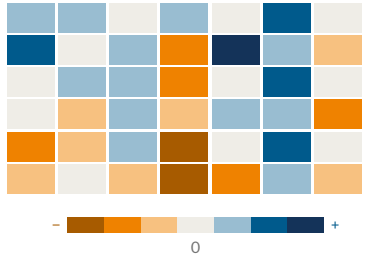


Recommended
Ranked by average.

Zero on a diverging scale



Try to avoid
No neutral class.



Recommended
A Light Sand middle.

Small groups

	Growth	Mgr	Load	Tools
Inbound (46)	+4	+2	−3	+1
Outbound (38)	−2	+1	−6	0
Returns (6)	−9	−11	−4	−8
Quality (21)	+3	+5	+1	+2

Try to avoid
A six-person team scored.

	Growth	Mgr	Load	Tools
Inbound (46)	+4	+2	−3	+1
Outbound (38)	−2	+1	−6	0
Returns (6)	n<10	n<10	n<10	n<10
Quality (21)	+3	+5	+1	+2

Recommended
Greyed and marked n<10.

Build correlation charts

Native, editable charts in every tool. Tested code and click-by-click steps are on the cookbook pages.

scatter-bubble.md · dual-axis.md · heatmap.md

Before you send it

- [ ] Bubble area ∝ value, with a size key
- [ ] Only the headline's points labelled
- [ ] Association, not cause, in the headline
- [ ] No dual axis: panels or an index
- [ ] Diverging midpoint = the real reference
- [ ] No point or cell under 10 people

PowerPoint and Excel	**Bubble:** Insert › Insert Scatter (X, Y) or Bubble Chart › Bubble; split focus and context into two series; *Size represents: Area of bubbles*; label the focus series with *Value From Cells*. **Instead of a dual axis:** an index helper column `=B2/B$2*100`, or two line charts with identical width, aligned left. **Heatmap:** Conditional Formatting › Color Scales › More Rules › 3-Color Scale with fixed Number bounds (−10.5, 0, 10.5), or one rule per class; white cell borders as gutters.
Power BI	**Scatter chart** with the entity in Values; color by measure through *Markers › Colors › fx*; averages and thresholds as Analytics-pane constant lines. **Matrix** with *Cell elements › Background color › fx* (Gradient with a middle color, or Rules per class). Index measures replace a dual axis. Blank small groups: `IF([Headcount] < 10, BLANK(), …)`.

Python	matplotlib with `mckviz`; stepped classes from `BoundaryNorm`.
	`ax.scatter(df.pay, df.exposure, s=df.headcount * K,  # area ∝ value`
	`  c=mv.highlight(df.role, focus=focus), edgecolors="white")`
	`norm = BoundaryNorm(np.arange(-10.5, 10.6, 3), 7)  # 7 even classes, 0 in the middle`
	`ax.imshow(m, cmap=ListedColormap(mv.DIVERGING), norm=norm)`

HTML	Plotly `scatter` with `sizemode: "area"` and tooltips for the unlabelled points; `heatmap` with a stepped colorscale and `xgap`/`ygap` 2; `make_subplots(shared_xaxes=True)` for aligned panels. Avoid `secondary_y`.
R	`geom_point(aes(size = headcount))` + `scale_size_area()`; `geom_tile()` + `scale_fill_mck_div(limits = c(-10.5, 10.5))`; `facet_wrap(~ measure, ncol = 1, scales = "free_y")` for aligned panels.

Sankey diagram

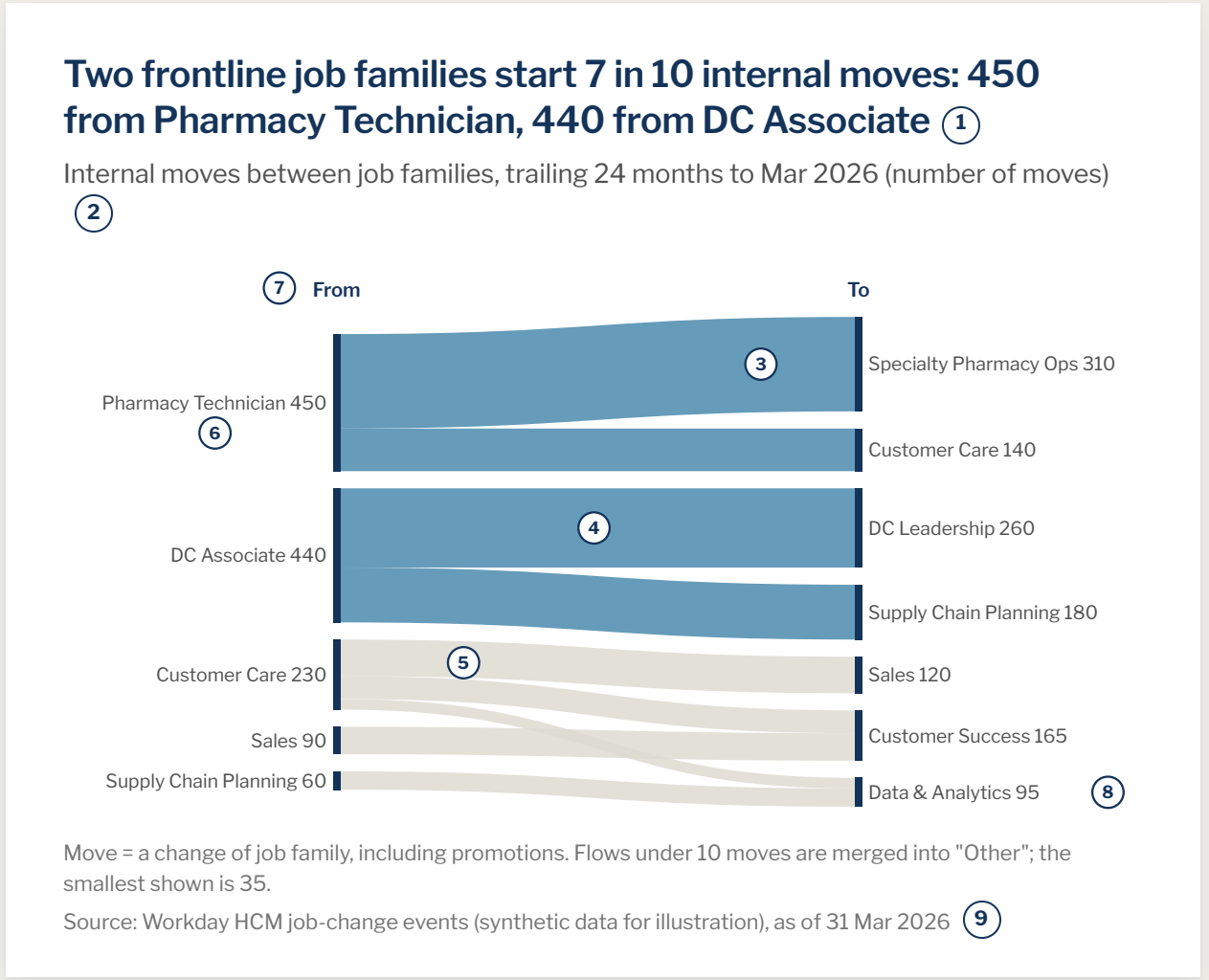
Relationship charts show how groups connect: how people move from one state to another, how they profile across many measures, or where groups overlap.

Each transfer is a band whose width is proportional to the people in it. Use it for internal mobility or a reskilling pipeline, when a few big flows matter most.

McKesson spec

- Focus flows in blue; the rest light grey
- Thin navy node bars; name and total beside each
- Two to four stages, eight nodes a column at most
- No flow under 10 people: merge into "Other"

1. **Insight headline**
   Where the volume comes from, with the numbers.
2. **Subtitle**
   What flows, over what period, in what unit.
3. **Width = people**
   Every band on one shared scale.
4. **Focus flows in blue**
   The paths the headline names, drawn on top.
5. **Light grey context**
   Visible, but quiet.
6. **Nodes**
   Thin navy bars, name and total, no boxes.
7. **Stage headers**
   Bold, above each column.
8. **Ordered to untangle**
   Largest sources first; destinations follow their source.
9. **Footnote and source**
   Define a move; state the minimum flow.



Sankey best practice

Color, shape and the n-size rule decide whether a Sankey can be read.

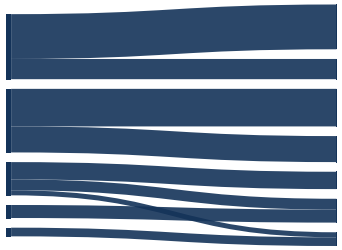
Color the story. One dark color makes crossing paths impossible to follow. Blue for the focus, light grey for the rest.

Color transfer type, not stage. The stages are already obvious. Give each type of transfer one color at every step; orange only for the loss.

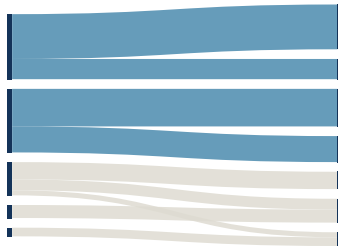
Keep it organic. Curved, semi-transparent bands and thin node bars. No boxed values, no angular arrows.

No flow under 10 people. A three-person move identifies people. Merge small flows into "Other".

Coloring flows

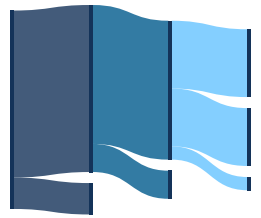


Try to avoid
One dark color.

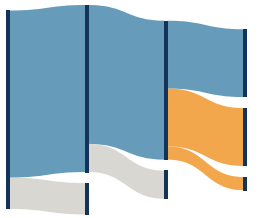


Recommended
The story in blue.

Color by transfer type

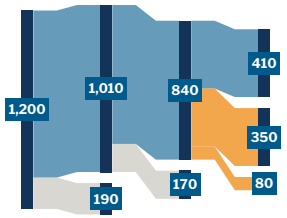


Try to avoid
A color per stage.

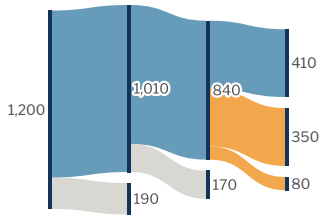


Recommended
Progressed, dropped, not placed.

Styling

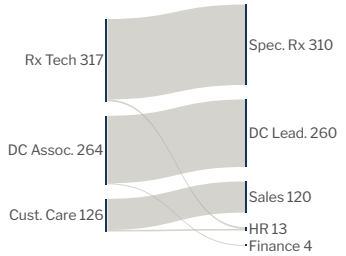


Try to avoid
Boxes and angles.

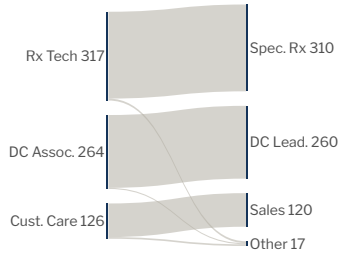


Recommended
Curves, plain numbers.

Small flows



Try to avoid
Moves of 4, 6 and 7 people.



Recommended
Merged into "Other".

Parallel coordinates

One vertical axis per measure, one line per group. It shows a profile: the shape a group makes across four to eight measures at once.

The axes are different measures, not points in time, so their order is a design choice. Put the story first, and orient every axis so up means the same thing.

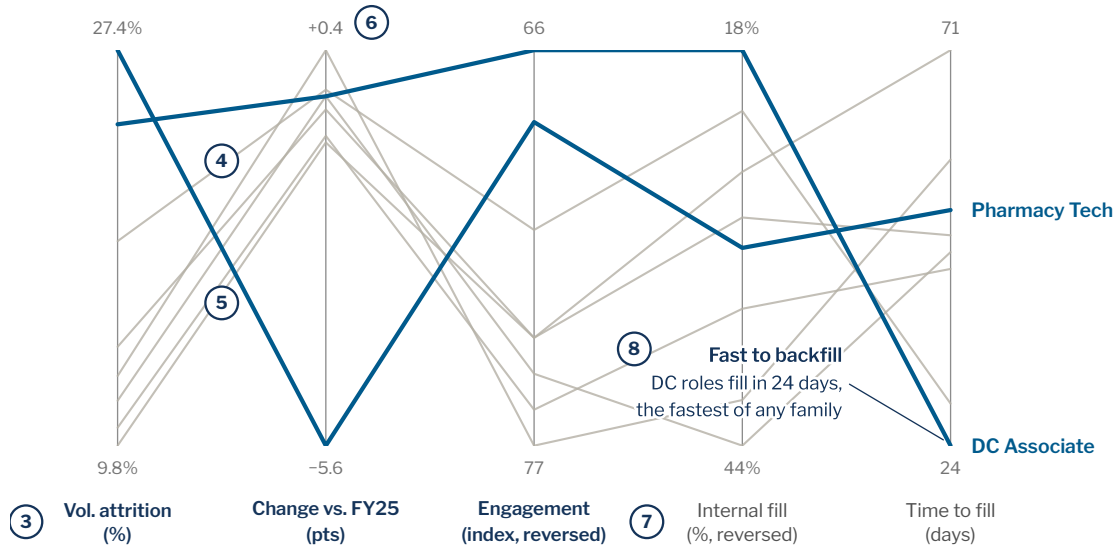
McKesson spec

- One or two groups in blue; every other line grey
- Each axis scaled min–max, real min and max printed
- Reversed axes labelled; "up = worse" in the subtitle
- Every line a group of at least 10 people

1. **Insight headline**
   What the profile shows, measure by measure.
2. **Subtitle**
   Groups, measures, scaling and direction.
3. **Story axes first**
   Bold names; related measures side by side.
4. **Focus lines in blue**
   Heavier, on top, named at the right end.
5. **Grey context**
   The other families, thin and unnamed.
6. **Real range at the ends**
   Max on top, min below, with units.
7. **Reversed, and said so**
   So up always means worse.
8. **One callout**
   The exception worth a sentence.
9. **Footnote and source**
   Scaling, reversals, group sizes.

Pharmacy Technicians and DC Associates combine the highest attrition with the lowest engagement of any job family (1)

Profile of eight job families on five measures, FY26 (each axis scaled min–max; up = worse) (2)



Engagement and internal-fill axes are reversed so that up always means worse. Every line is a job family of 850+ people.

Source: Workday HCM, ATS and engagement survey (synthetic data for illustration), FY26 (9)

Parallel coordinates best practice

The chart works only when one group stands out from a quiet background.

Use it instead of a dense table. A grid of 40 numbers hides the profile; lines show it at a glance.

One group in color. A shade of blue per line is a tangle with no message.

Scale each axis. On one raw scale, the measure with the biggest numbers flattens the rest.

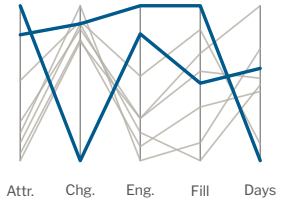
Point every axis the same way. Mixed directions make a consistently bad group zigzag.

Interactive versions can brush, reorder and hide axes; that beats cropping a static chart to two or three axes.

Instead of a table

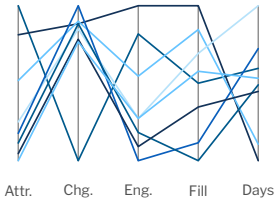
	Attr.	Chg	Eng.	Int.	Days
Pharmacy	27.4	-5.6	68	31	52
DC Associ	24.1	-0.3	66	18	24
Customer	18.9	-0.2	71	22	29
Specialty	14.2	-0.5	74	26	71
Data & An	12.9	0.4	77	41	58
Supply Ch	11.8	-0.3	75	44	47
Sales	10.6	-0.9	76	35	45
Software	9.8	-1.0	74	29	49

Try to avoid
A grid of numbers.

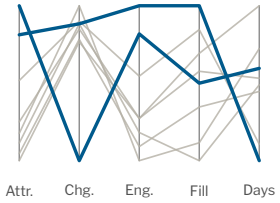


Recommended
The profile, highlighted.

Highlighting lines

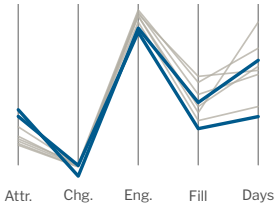


Try to avoid
A blue per line.

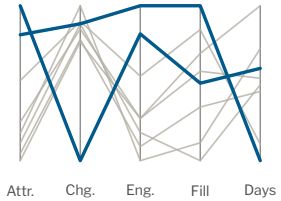


Recommended
Grey, one group blue.

Scaling the axes

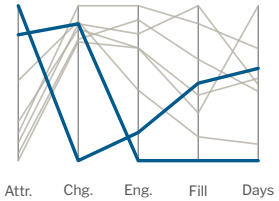


Try to avoid
One raw scale.

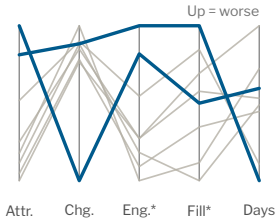


Recommended
Min–max per axis.

Axis direction



Try to avoid
Up means better on some axes.



Recommended
Reversed axes marked *.

RELATIONSHIPS AND FLOWS

Venn diagram

Overlapping circles show who sits in more than one group at once: people with several verified skills, mentors who also lead networks.

Use it for two or three sets when the overlap itself is the point. The circles don't encode size; the counts do. For four or more sets, or precise comparison, use combination bars.

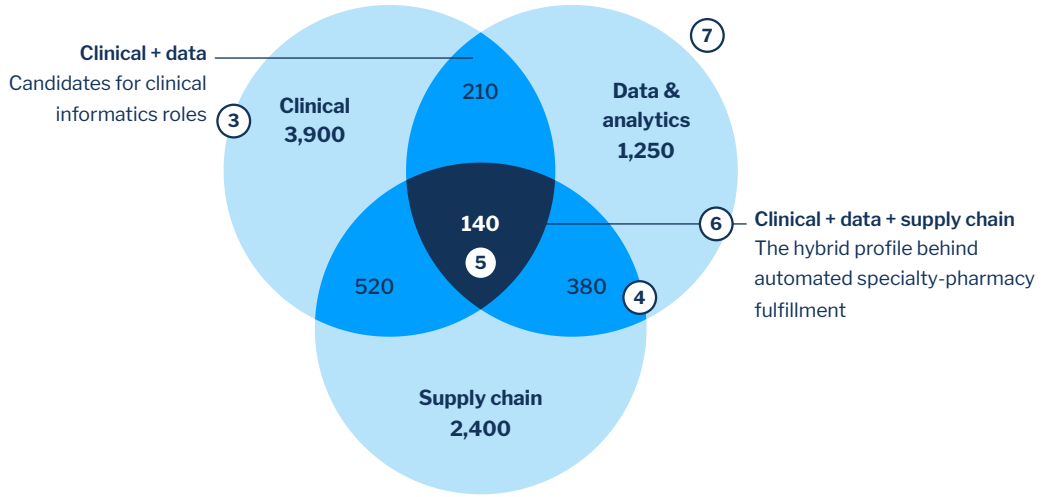
McKesson spec

- One blue that darkens with each level of overlap
- Equal, regular circles; exact fills, no transparency
- Exclusive counts in each region; descriptions outside
- No region under 10 people

1. **Insight headline**
   The overlap that matters, against the total.
2. **Subtitle**
   What the sets are, who is counted, when.
3. **Single sets, lightest**
   Bold name, count beneath.
4. **Two-way overlaps**
   A darker blue; the count only.
5. **The center, darkest**
   Navy with white text.
6. **Descriptions outside**
   A bold title and a thin leader line.
7. **Equal circles**
   Not scaled to headcount; say so.
8. **Footnote and source**
   Define the sets; counts are exclusive.

Only 140 of 8,800 employees with a verified skill profile combine clinical, data and supply-chain expertise (1)

Employees by verified skill area, U.S. regular employees, skills inventory as of Mar 2026 (headcount) (2)



Verified skill = manager-validated in the skills inventory. Circles are equal and not scaled to headcount; region counts are exclusive.

Source: Skills inventory (synthetic data for illustration), as of 31 Mar 2026; U.S. regular employees (8)

Venn best practice

Keep the shapes simple and let the shading carry the logic.

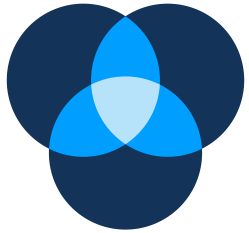
Darker as sets overlap. An overlap lighter than its sets reads backwards.

Exact fills, one hue. Translucent circles in different colors blend into muddy, off-brand tones.

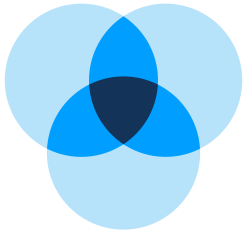
Three sets at most. Four ellipses make fifteen regions. Switch to combination bars.

Text outside. Sentences inside the circles hide the shading and can't be read.

Shading overlaps

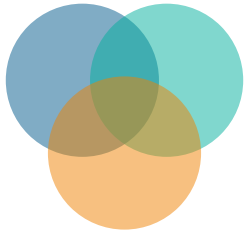


Try to avoid
Overlaps lighter.

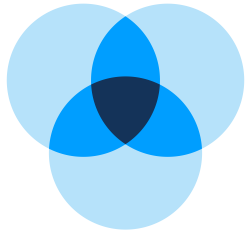


Recommended
Darker with each level.

Fills



Try to avoid
Blended hues.

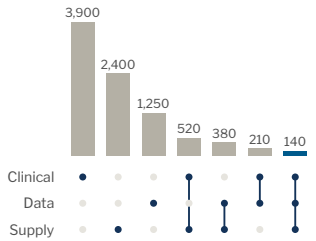


Recommended
Exact blues.

Four or more sets

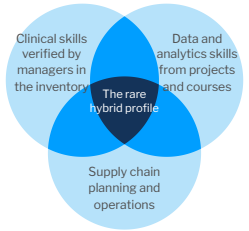


Try to avoid
Overlapping ellipses.

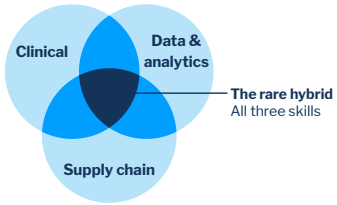


Recommended
Combination bars.

Using text



Try to avoid
Sentences inside.



Recommended
Outside, with a leader.

Build relationship charts

These charts are not native everywhere. Where a tool can't draw one, the cookbook says what to do instead.

sankey.md · parallel-coordinates.md · venn.md

Before you send it

- [ ] Sankey flows conserved; exits are named nodes
- [ ] One color per transfer type; story in blue
- [ ] Axes scaled and oriented the same way
- [ ] Venn overlaps darker; counts exclusive
- [ ] No flow, line or region under 10 people

PowerPoint and Excel	**Sankey:** no native chart. Build it in Python or HTML, export an SVG, then Insert › Pictures and *Convert to Shape* to edit labels; a linear pipeline is a native Funnel. **Parallel coordinates:** normalize in a helper table, Insert › 2-D Line, *Switch Row/Column* so measures are the categories, vertical gridlines as axes, focus series moved last. **Venn:** three Ovals; *Merge Shapes › Intersect* for each overlap; fills B6E2FA, 009EFF, 143359.
Power BI	No built-in Sankey: the AppSource Sankey visual only if your tenant has approved it; group small flows into "Other" in Power Query first. Parallel coordinates as a native Line chart with the measure on the X axis and a normalized DAX measure. No Venn visual: show the combinations as a sorted bar.

Python	Plotly `go.Sankey` for flows; matplotlib with `mckviz` for parallel coordinates and the Venn (polygon intersections, exact fills).
	`link_color = {"progressed": mv.rgba(C.focus, 0.55),  # one color per transfer type`
	`"dropped": mv.rgba(C.context, 0.45), "not_placed": mv.rgba(C.negative, 0.65)}`
	`go.Sankey(node=dict(color=C.total, thickness=6),`
	`link=dict(source=src, target=dst, value=people,`
	`color=[link_color[t] for t in kind]))`

HTML	Plotly `Sankey` with semi-transparent links; `Parcoords` lets readers brush and reorder axes; the Venn as filled `path` shapes with hover off.
R	`ggalluvial::geom_alluvium()` for Sankey-style flows; `geom_line()` over min–max scaled long data for parallel coordinates; `sf::st_intersection()` for Venn regions.

GEO-SPATIAL

Choropleth map

Geo-spatial charts show where: a pattern across geography, or values at precise locations.

A choropleth shades each area by a **rate**: postings per candidate, attrition, share eligible to retire. Use it when the pattern only makes sense on a map.

US maps use an equal-area Albers projection, with Alaska and Hawaii moved into insets.

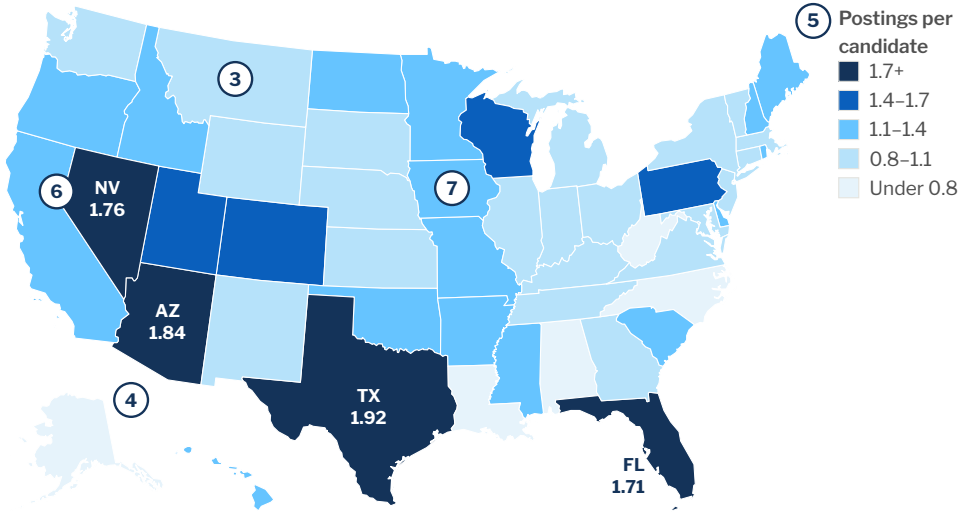
McKesson spec

- Sequential blue in 3–5 even classes
- Light base, thin white boundaries
- Label only the areas the headline names

1. **Insight headline**
   Names the states and the threshold.
2. **Subtitle**
   A rate, with its unit, scope and period.
3. **Equal-area projection**
   Albers: each state at its true relative size.
4. **Insets**
   Alaska shrunk and moved, Hawaii moved; both keep their color.
5. **Classed key**
   Five even classes, open-ended, high on top.
6. **Values on the few**
   Only the 1.7+ states carry a number.
7. **Quiet structure**
   Thin white borders; no frame, no graticule.
8. **Footnote and source**
   Define the rate and the classes.

Four Sun Belt states have 1.7+ open pharmacy technician postings per candidate (1)

Open pharmacy technician postings per qualified candidate, by state, Q2 2026 (ratio) (2)



Postings per candidate = open postings ÷ qualified candidates in the state. Classes are even-width; the top and bottom are open-ended. Alaska not to scale.

Source: External job-postings and candidate-profile data (synthetic data for illustration), Q2 2026 (8)

GEO-SPATIAL

Bubble map

Bubbles on a quiet base map show counts at locations: headcount by site, open requisitions by metro, hires by campus.

Bubble **area** carries the count. Color can carry a second measure with the focus rule: blue for the sites the headline names, grey for the rest.

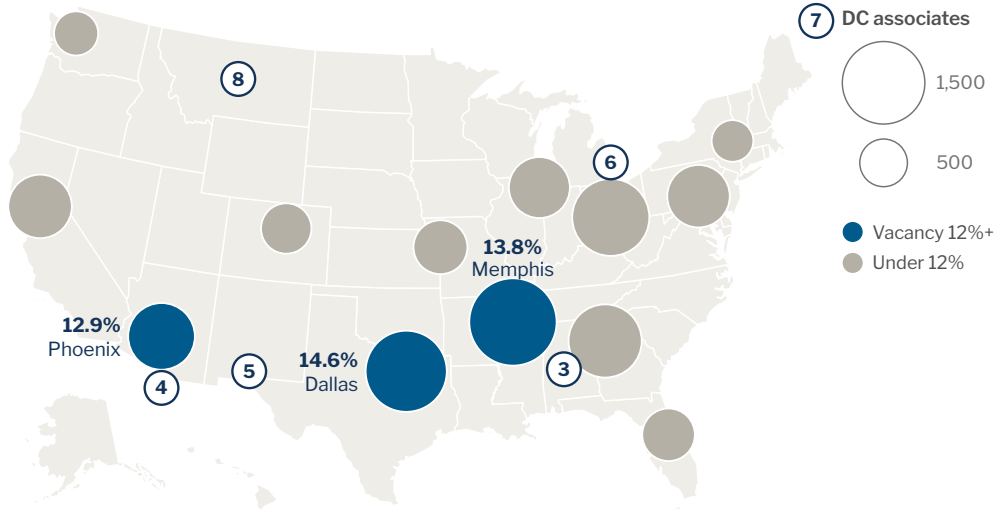
McKesson spec

- Area ∝ value; largest drawn first, white outlines
- Largest bubble about 40–56 px across
- Label the focus sites: bold value, then name
- Drop sites under 10 people and footnote them

1. **Insight headline**
   The sites, the threshold, what's at stake.
2. **Subtitle**
   Says what area and color mean.
3. **Area = associates**
   Scaled by area, never radius.
4. **Focus in blue**
   The sites at 12%+ vacancy.
5. **Direct labels**
   Bold value and site name, for the focus only.
6. **Grey context**
   Every other site; overlaps stay readable.
7. **Size and color key**
   Two reference circles, two swatches.
8. **Quiet base**
   Sand states, white borders: the map is context.
9. **Footnote and source**
   Define vacancy; note what's suppressed.

Vacancy tops 12% at three southern distribution centers that employ 35% of our DC associates (1)

Distribution center associates by site (bubble area) and vacancy rate, Mar 2026 (2)



Vacancy rate = open associate requisitions ÷ budgeted positions. 13 sites shown at their city; one satellite site with fewer than 10 associates not shown.

Source: Workday HCM and ATS (synthetic data for illustration), as of 31 Mar 2026 (9)

GEO-SPATIAL

World map

For global questions: where to source talent, where a capability center could go, where our people work.

Use an **equal-area** projection such as Equal Earth. Mercator inflates Canada, Russia and Greenland and shrinks India and Africa, so a rate looks bigger the further it is from the equator.

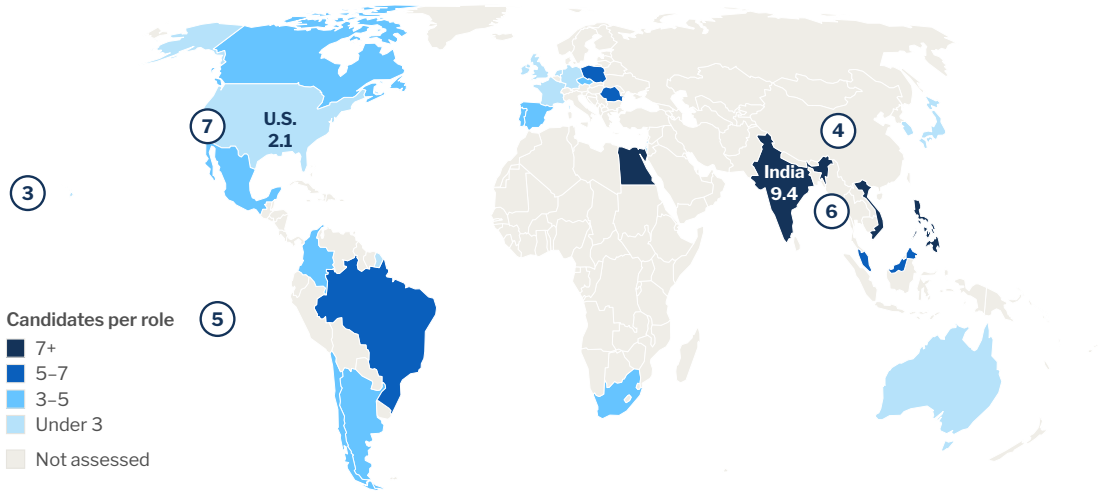
McKesson spec

- Equal Earth (or Robinson where a tool offers no equal-area option); Antarctica left out
- Countries without data in base sand, and keyed
- No palest class: it would read as "no data"
- Rates as shades; counts as bubbles

1. **Insight headline**
   The standout country, compared with home.
2. **Subtitle**
   A rate, the countries in scope, the period.
3. **Equal Earth**
   True relative areas; curved edges, no frame.
4. **Not assessed**
   Base sand, so missing never looks like low.
5. **Classed key**
   Four even classes plus "Not assessed".
6. **The standout**
   Named, with its value, in place.
7. **The comparison**
   The U.S. value the headline uses.
8. **Footnote and source**
   Define the rate and the scope.

India offers 9.4 qualified Data & AI candidates per open role, more than four times the U.S. rate (1)

Qualified Data & AI candidates per open role, countries assessed for a capability center, Q2 2026 (ratio) (2)



Candidates per role = qualified Data & AI profiles ÷ open Data & AI postings. 27 countries assessed; the rest in sand.

Source: External job-postings and professional-profile data (synthetic data for illustration), Q2 2026 (8)

GEO-SPATIAL

Zoom-in map

When the detail is smaller than a state (counties, a metro, a commute shed), zoom in. At national scale it would be a smudge.

Texas is the tightest state at 1.92. The zoom shows where inside it: the core around our Irving campus, not the outer ring.

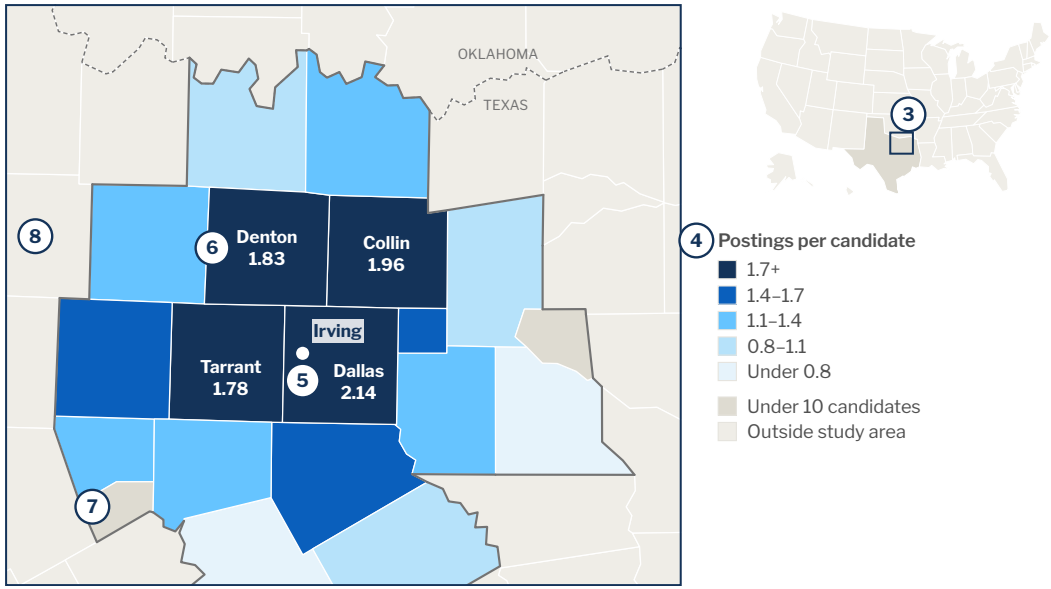
McKesson spec

- Always a locator: a small US map, the frame in navy
- The parent map's classes and colors, unchanged
- Context outside the study area in sand
- Small areas are noisy: hide any under 10 people

1. **Insight headline**
   Where inside the state, with the contrast.
2. **Subtitle**
   The same rate as the state map, by county.
3. **Locator**
   Where the zoom sits, at a glance.
4. **Same classes**
   1.7+ means the same here as on the state map.
5. **The site**
   Our campus, as a navy-ringed point.
6. **Values on the few**
   The core counties the headline names.
7. **Suppressed**
   Under 10 candidates: grey, no value.
8. **Context**
   Outside the study area in sand; state line dashed.
9. **Footnote and source**
   Define the study area and what's hidden.

Around Irving, four core counties have 1.7+ postings per candidate, while five outer counties are under 1.1 (1)

Open pharmacy technician postings per qualified candidate, by county, North Texas, Q2 2026 (ratio) (2)



Study area: the 19 counties whose center is within 70 miles of Irving; two with fewer than 10 qualified candidates not shown. Same classes as the state map.

Source: External job-postings and candidate-profile data (synthetic data for illustration), Q2 2026 (9)

GEO-SPATIAL

Map variants

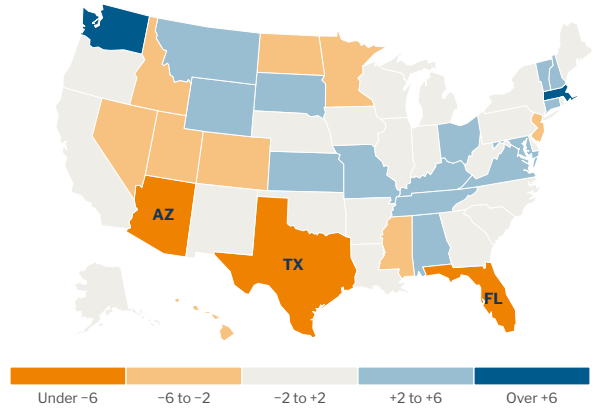
Same base, same quiet structure; the variant follows the question.

When a map is the wrong chart

- Readers need to rank or compare precisely: a sorted **bar**
- Five locations or fewer: a **bar** or a **table**
- Change over time in each region: **small multiples** of lines
- Geography is only a grouping (business units, divisions): no map at all

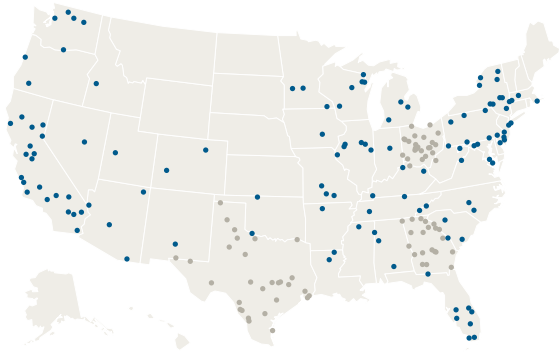
Diverging classes

When a midpoint means something. Technician pay is 6%+ below market in three states, all of them among the tightest.



Dot density

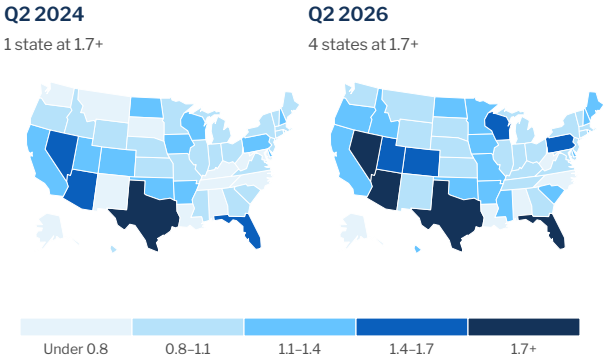
Spread and concentration. 63% of remote Customer Care agents live outside the three hub states.



1 dot = 10 agents. Grey: the three hub states (GA, OH, TX).

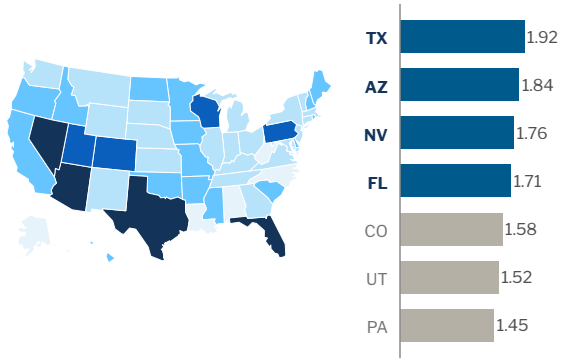
Small-multiple maps

Two periods on the same classes: 1 state at 1.7+ in 2024, 4 in 2026.



Map plus ranked bar

The map shows where; the bar shows by how much. Same blue for the same states.



GEO-SPATIAL

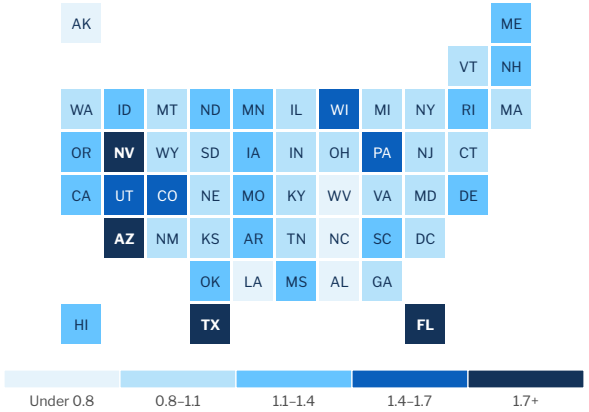
Choosing the geography

A real map is the default. Change the geography when it would hide what matters.

- Small states matter as much as big ones: a **tile grid**
- Data exists only by region: **dissolve** the states into regions
- A few areas are too small to read: an enlarged **inset**
- Detail smaller than a state: a **zoom-in** with a locator
- Countries: an equal-area **world** map

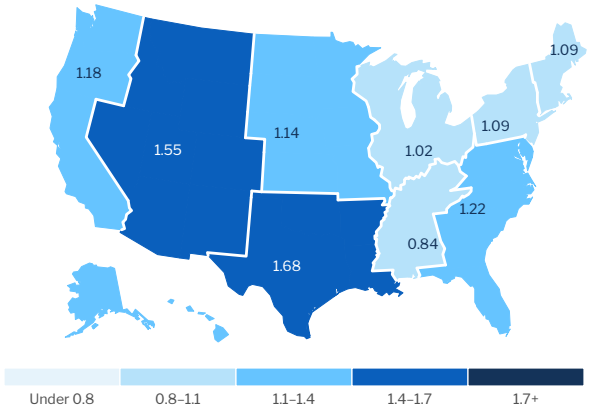
Tile grid map

Every state the same size, in roughly its place: when Rhode Island matters as much as Montana.



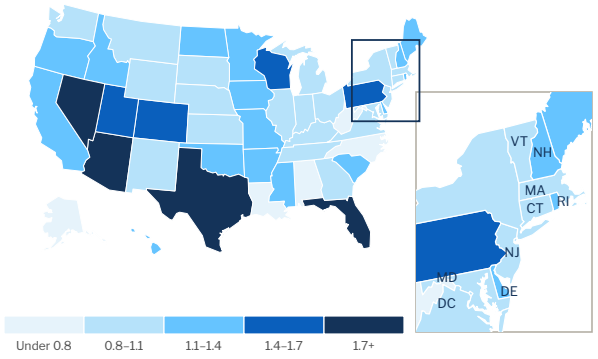
Region map

Nine Census divisions, state lines removed. West South Central is tightest at 1.68.



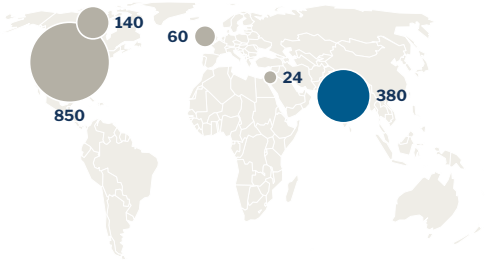
Inset for small areas

The Northeast at 3× keeps 8 small states and DC readable. Outline the source area.



World bubble map

Counts as bubbles. India holds 26% of our Data & AI people, the most outside the U.S.



Bubble area = people. Mexico and Poland (under 10 each) not shown.

GEO-SPATIAL

Maps best practice

Maps are persuasive, so they must be honest about what the color means.

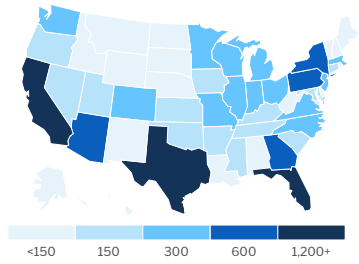
Rates, not counts. Shading headcount or postings just redraws where people live. Normalize: per candidate, per 1,000, a %. Show counts as bubbles.

Even, round classes. Three to five, no gaps or overlaps, open-ended at both ends.

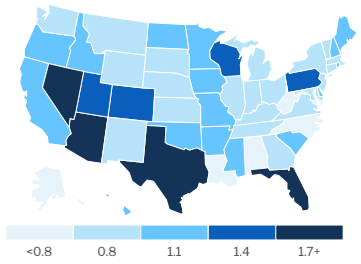
One sequential hue. A rainbow makes some hues pop for no reason. Diverging only around a meaningful midpoint.

A quiet base. Light sand, thin white boundaries. For print, darken them to grey.

Rates, not counts

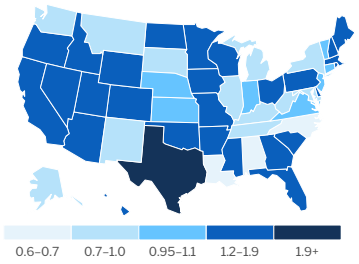


Try to avoid
Raw postings: big states win.

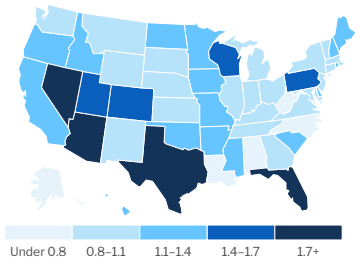


Recommended
Postings per candidate.

Class breaks

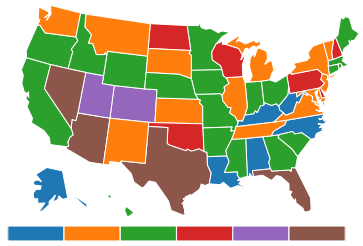


Try to avoid
Uneven, gappy, overlapping.

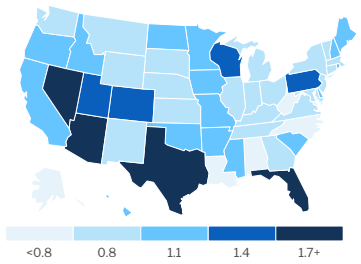


Recommended
Five even, round classes.

Color scale

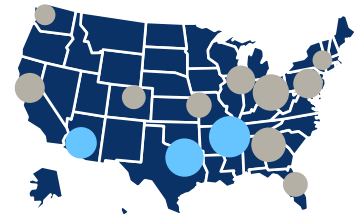


Try to avoid
Rainbow, off palette.

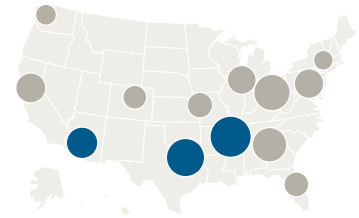


Recommended
Light to dark blue.

Base map



Try to avoid
A dark base competes.



Recommended
Sand base, data on top.

GEO-SPATIAL

Maps best practice, continued

Four more, from workforce data in particular.

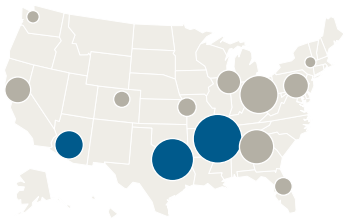
Bubbles by area. Scaling the radius makes a site twice as big look four times as big.

Label the few that matter. A number in every state is a table in the wrong shape, and the small states pile up.

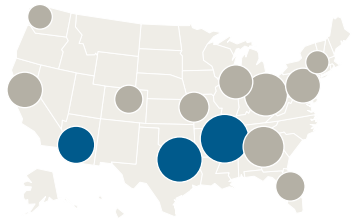
Is it really geographic? Three hubs don't need a map of fifty states.

Aggregate, then map. Never plot home addresses. Map to state, metro or site, and drop anything under 10 people, in tooltips too.

Bubble size

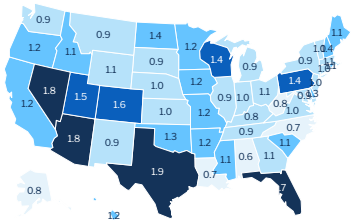


Try to avoid
Radius ∝ value.

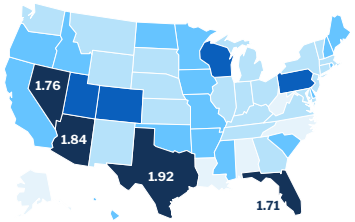


Recommended
Area ∝ value.

Labels

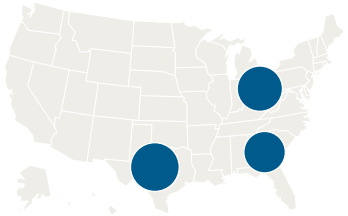


Try to avoid
A value in every state.

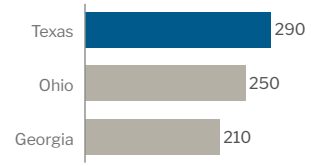


Recommended
Only the four named.

Is it geographic?

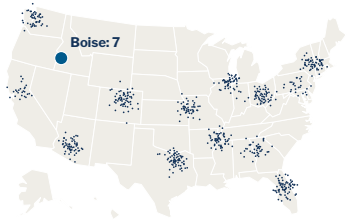


Try to avoid
A US map for three sites.

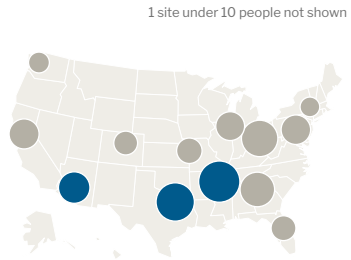


Recommended
A sorted bar.

Protect people



Try to avoid
Exact spots; a site of 7.



Recommended
Aggregated; small sites dropped.

GEO-SPATIAL

Projections and zoom-ins

Where the shapes come from matters as much as the colors on them.

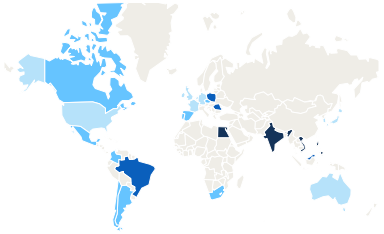
Equal-area for the world. Equal Earth, not Mercator or plain longitude and latitude: on a rate map, area is ink.

Albers for the US. Alaska and Hawaii as insets, said so in the footnote.

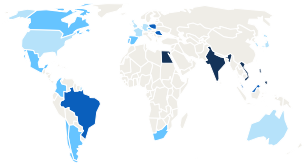
Zoom in, don't squint. County detail needs its own map, with a locator.

Keep the parent's scale. Use the same classes on the zoom; an automatic color range makes "dark" mean something new on every map.

World projection

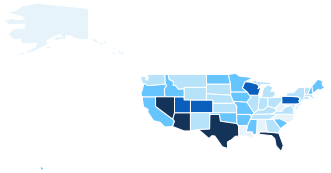


Try to avoid
Mercator: the north swells.

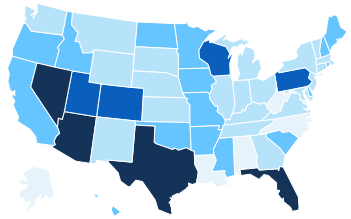


Recommended
Equal Earth.

US layout



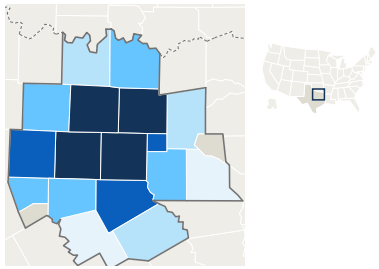
Try to avoid
Plain lon/lat: Alaska dominates.



Recommended
Albers, with insets.

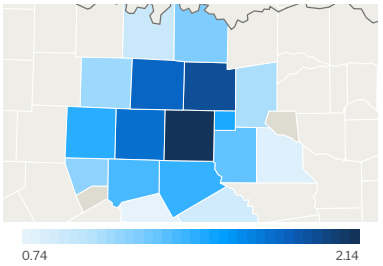
Zoom in

Try to avoid
Counties at national scale.

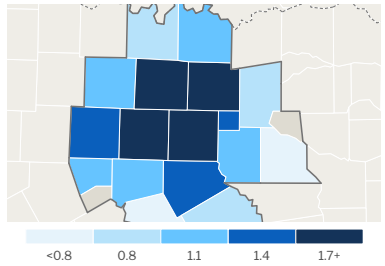


Recommended
Zoomed, with a locator.

Same scale as the parent



Try to avoid
Auto range: its own min to max.



Recommended
The state map's classes.

GEO-SPATIAL

Build maps

Choropleths are native in every tool; bubble, dot and zoom maps are easiest in Python or Plotly. Tested code and click-by-click steps are on the cookbook page.

maps.md

Before you send it

- [ ] Shaded values are rates; counts are bubbles
- [ ] 3–5 even classes; the key names the unit
- [ ] Albers for the US, Equal Earth for the world
- [ ] A zoom has a locator and the parent's classes
- [ ] No area, site or cluster under 10 people

PowerPoint and Excel	**Filled Map:** Insert › Charts › Maps › Filled Map (online: Excel geocodes through Bing). It takes countries, states, counties or postal codes. Format Data Series: Map projection *Albers* for the US or *Robinson* for the world (never Mercator), Map area *Only regions with data*, which also zooms to a state's counties. Map a class-number column (1–5) to get binned shades; draw the key as small squares. Bubble and dot maps aren't native: build them in Python and insert the SVG.
Power BI	**Azure Maps** visual (your admin must enable it): Filled map layer for rates; Bubble layer with Range scaling *Magnitude* for counts. Bin colors with a `SWITCH` measure in *Fill color › fx › Field value*. The **Shape map** visual (preview) takes your own TopoJSON, such as counties or a tile grid. Suppress small groups with `IF([Headcount] < 10, BLANK(), ...)`.
Python	The guide's maps use `guide/geo_shapes.py`: a small TopoJSON reader for the us-atlas and world-atlas files in `guide/geodata/`, drawn with matplotlib. Or Plotly: `go.Choropleth` with `fig.update_geos(scope="usa")` (Albers USA) or `projection_type="equal earth"`.
HTML	Plotly with `MCK_PLOTLY_TEMPLATE`; for a zoom, `go.Choropleth(geojson=…)` and `fitbounds="locations"`. Headline, subtitle and source go in the chart card's HTML. Tooltips may show n, but never for an area or site under 10.
R	`sf` with `tigris::shift_geometry()` and `coord_sf(crs = 5070)` for Albers with insets; `coord_sf(crs = "+proj=eqearth")` for the world. `scale_fill_manual()` on the classes; bubbles with `scale_size_area()`.

```
ax = gs.canvas(fig)                                # axes in px
m = gs.Map("states-albers-10m.json", "states", (0, 0, 480, 300), key=gs.FIPS.get)
m.draw(ax, {s: ramp[c] for s, c in cls.items()}, ec="white", lw=0.6)
x, y = m.point(-96.80, 32.78)                      # a site, in Albers USA
```

SINGLE NUMBER

Big number and KPI card

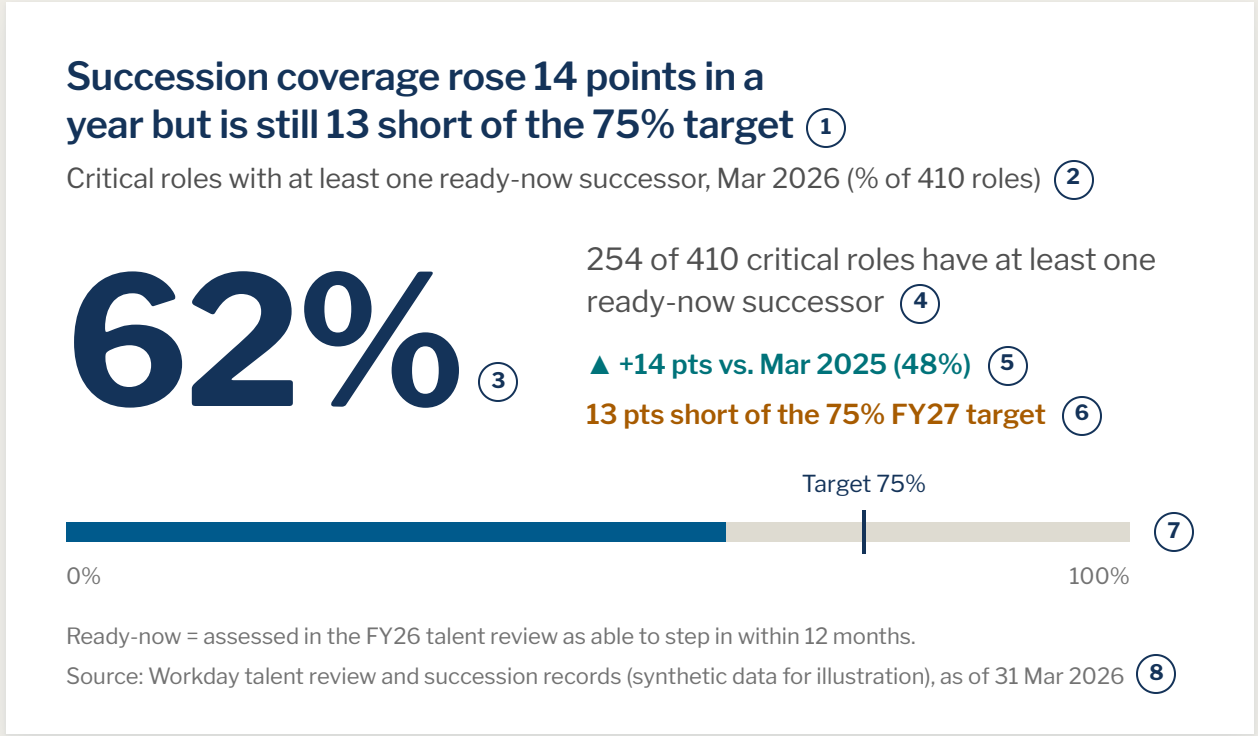
Single-number charts make one value the message, with the context and comparison that give it meaning.

Use a big number when one value is the finding: an executive summary, a slide opener, a report margin, the header row of a dashboard.

McKesson spec

- Numeral Bold 700, navy, large
- One sentence on exactly what is counted
- Always a comparison; "pts" for point changes
- Source and as-of date on the card itself

1. **Insight headline**
   The movement and the gap that remains.
2. **Subtitle**
   Metric, scope, date and the base.
3. **The number**
   Large and Bold, rounded hard.
4. **One sentence**
   Counts make the % concrete.
5. **Comparison**
   Signed, in pts, teal and ▲ because it's good news.
6. **Gap to target**
   Orange text because it's worse, said in words.
7. **Progress bar**
   0–100% track, target tick labelled.
8. **Footnote and source**
   Define the measure; date it.



SINGLE NUMBER

Progress ring

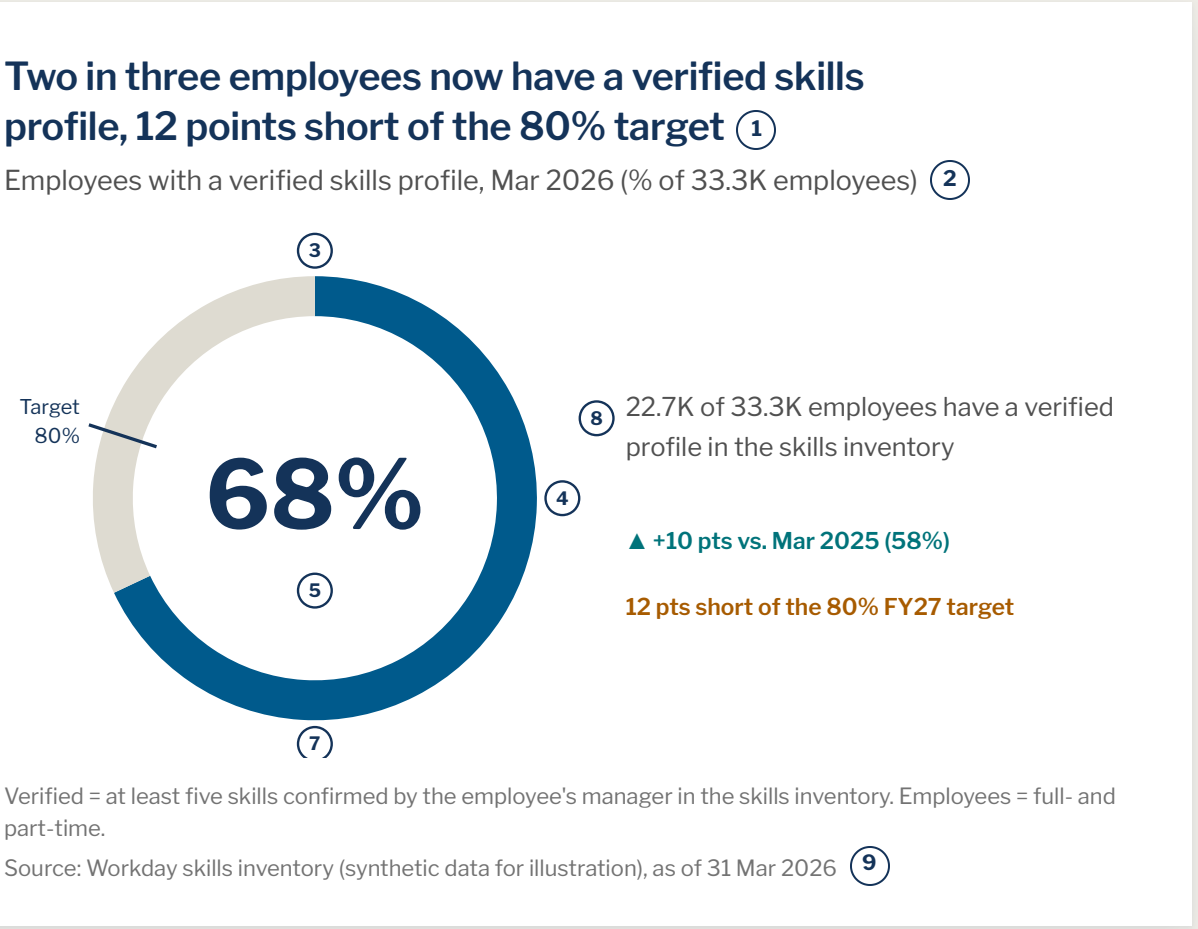
A ring shows one share of one whole moving toward a target: skills profiles completed, required learning done, plan positions filled.

It works only on a 0–100% scale. If the value can pass 100% or go negative, show the big number and the delta instead.

McKesson spec

- Starts at 12 o'clock, runs clockwise
- Blue arc, sand remainder; hole about 80%
- No rounded caps, gaps, gradients or 3D
- Target as a thin tick, labelled

1. **Insight headline**
   Where we are and how far to go.
2. **Subtitle**
   Metric, date and the base.
3. **Start at 12**
   Then clockwise, like a clock.
4. **Blue arc, sand track**
   The remainder stays visible.
5. **Number in the center**
   Bold; the only centered text.
6. **Target tick**
   Thin, navy, named.
7. **Thin stroke**
   About 18% of the radius.
8. **Sentence and comparison**
   Left-aligned beside the ring.
9. **Footnote and source**
   What "verified" means.



SINGLE NUMBER

Single number variants

The anatomy stays the same: name, number, comparison, sentence.

When a single number is the wrong chart

- The trend is the message: a **line**, or a sparkline under the number
- Many groups to compare: **bars**; eight KPI cards are a bar chart in disguise
- Several parts of a whole: a **donut** or **stacked bar**
- The base is under 10 people: don't headline it

Big number with a trend

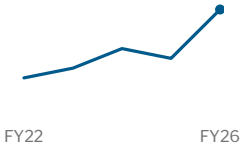
A sparkline adds the shape without a full chart: internal fill reached 29%, +5 pts in a year.

Internal fill rate, FY26

29%

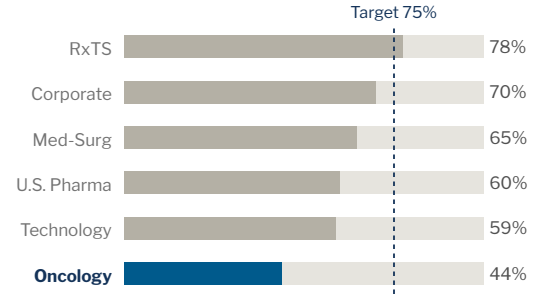
▲ +5 pts vs. FY25 (24%)

of open roles were filled by internal candidates



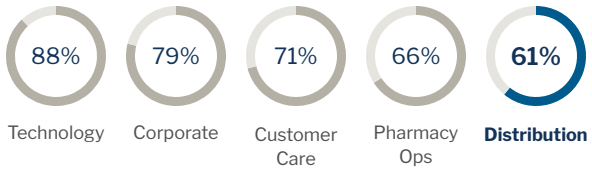
Progress bars to a target

One share per group on a shared 0–100% track. 5 of 6 units are below target; Oncology is lowest at 44%.



Circular dials

One % across a few groups, every ring 0–100%. Distribution trails at 61%.



KPI row

Three to five cards above a dashboard's main chart, each built the same way. Days to fill rose 2 days, so it's orange.

Internal fill rate	Days to fill	Succession coverage
29%	**37**	**62%**
▲ +5 pts vs. FY25	▲ +2 days vs. FY25	▲ +14 pts vs. Mar 2025

Teal = favorable, orange = unfavorable; the arrow gives the direction, the color the judgment.

SINGLE NUMBER

Single number best practice

A number alone is a fact. Context and comparison make it a finding.

Bold, large, plain. The numeral needs weight, size and space, not boxes or shadows.

Say what is counted. One sentence, with the base: "254 of 410 critical roles".

Always compare. Prior period, target or benchmark, with a sign. Percentage-point changes are "pts", never "%".

No speedometers. Gauges spend ink on a dial and invite red and green. A ring or a bar with a target does the job.

Bold, large, plain



Try to avoid
Small, thin, boxed, shadowed.



Recommended
Bold numeral, white space.

Say what is counted



Try to avoid
62% of what?

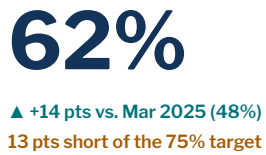


Recommended
One sentence with the base.

Always compare

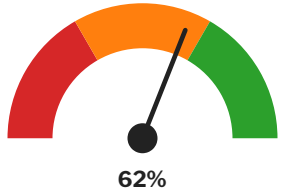


Try to avoid
"+14%" against nothing.

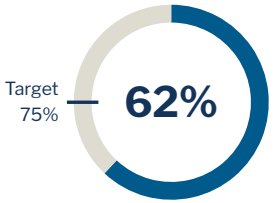


Recommended
Pts vs. a date; gap to target.

No speedometers



Try to avoid
Dial, needle, traffic lights.



Recommended
A thin ring with a target.

SINGLE NUMBER

Single number best practice, continued

Four more, from dashboards and scorecards.

Color by judgment, not direction. Teal when the change is good, orange when it's worse. Days to fill going up is bad news; pair the color with ▲▼ or words.

Round hard. 62%, 4.3K, $1.2M. Decimals suggest a precision the data doesn't have.

Rings are for 0–100%. Growth, indexes and variances get a big number and a delta.

One anatomy per row. Same order, weight and alignment in every card.

Color by judgment

Internal fill rate
29% ▲ +5 pts

Days to fill
37 ▲ +2 days

Try to avoid
Up = green, whatever it means.

Internal fill rate
29% ▲ +5 pts

Days to fill
37 ▲ +2 days

Recommended
Teal good, orange worse.

Round hard

61.9512%
+13.90244 vs. prior

Try to avoid
False precision.

62%
▲ +14 pts vs. Mar 2025

Recommended
Whole points.

Rings are for 0–100%



Data & AI headcount

Try to avoid
A ring that laps itself.

Data & AI headcount since 2021
+22%
vs. +96% for market Data & AI postings

Recommended
Big number, with a benchmark.

One anatomy per row

Internal fill rate	DAYS TO FILL	*Succession coverage*
29%	**37**	62%
▲ +5 pts vs. FY25		

Internal fill rate	Days to fill	Succession
29%	**37**	**62%**
▲ +5 pts	▲ +2 days	▲ +14 pts

Try to avoid
Every card different.

Recommended
Name, number, change.

SINGLE NUMBER

Build a KPI card

A big number is type, not a chart: build it from text wherever you can, so it stays sharp and editable.

kpi-big-number.md

Before you send it

- [ ] Number, sentence and headline agree (62% × 410 ≈ 254)
- [ ] A comparison with a sign; "pts" for points
- [ ] Colors by favorability, plus ▲▼ or words
- [ ] Rings only for 0–100% of one whole
- [ ] No KPI based on fewer than 10 people

PowerPoint and Excel	**Big number:** a text box in Libre Franklin *Bold*, 54 pt, Navy `#143359`, left inset 0; the sentence below at 14 pt; the comparison in Medium, teal `#00737A` or orange `#A75B00`. Align left and group. **Ring:** a Doughnut of value and remainder, Angle of first slice 0°, Hole Size 80%, fills `#005A8C` / `#DEDBD1`, no borders, title, legend or labels; lay the number over the center.
Power BI	**Card (new)**: the measure as the callout value (raise it to 40–54; the theme makes it Bold, Segoe UI Bold in the service), *Reference labels* for the comparison. The **KPI** visual adds trend and target: set Good `#00AF9D`, Bad `#EF8200` and the direction per metric. Rings: a Donut with Inner radius ≈ 80 and a transparent Card on top.

Python	matplotlib `Wedge` for the ring; Plotly `go.Indicator(mode="number+delta")` for a KPI row, with `increasing` / `decreasing` colors set by favorability.
	`KPI = mv.TOKENS["typography"]["roles"]["kpi"]  # Bold 700`
	`ax.text(0, 0, f"{value}%", fontsize=KPI["size_pt"] * 0.75,`
	`  fontweight=KPI["weight"], color=C.headline)`
	`txt = f"{mv.signed(value - prior)} pts vs. Mar 2025"  # true minus, explicit +`

HTML	Plain HTML text beats a chart library here: `<p class="mck-kpi">` from `mck-viz.css` sets `--mck-kpi-size` (56 px) and `--mck-kpi-weight` (700). Color deltas with `var(--mck-semantic-positive-text)` and `var(--mck-semantic-negative-text)`.
R	`annotate("text", ..., fontface = "bold")` for the number; a ring with `geom_col()` + `coord_polar(theta = "y", start = 0, direction = 1)`.

Presentations: build it up

The same data needs different treatment on a live slide, a page, a phone or a dashboard. Design for where it will be seen.

In a live meeting, don't reveal a finished chart in one click. Build it in two to five steps so the audience learns the chart before you give them the point.

1 Set the scene. Axes, labels and the data, all grey.

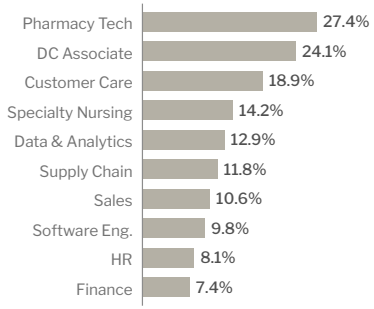
2 Highlight the insight. Color what the headline names.

3 Show the full picture. Add the benchmark and the callout.

Keep the headline, axis and order identical on every step; only the emphasis changes.

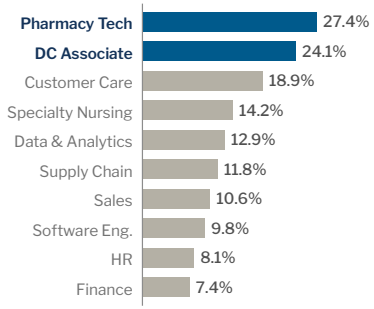
STEP 1 · SET THE SCENE

Everything the reader needs to read the chart, nothing to interpret yet.



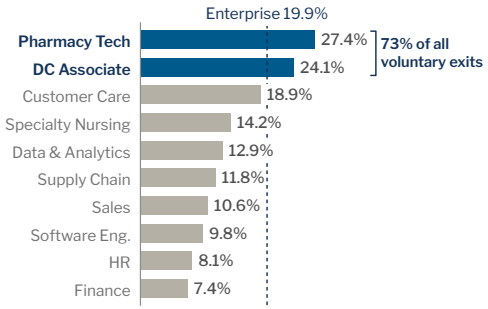
STEP 2 · HIGHLIGHT

The two families the headline names turn blue.



STEP 3 · FULL PICTURE

Benchmark and callout: 73% of all voluntary exits.



Presentations: clarity first

A slide is on screen for less than a minute. There is no time to explore it.

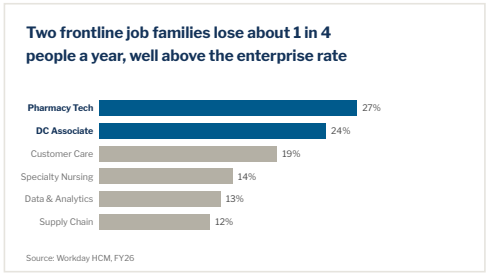
- With one chart, the slide title **is** its headline (Bold)
- One message per chart; at most two charts per slide, never a chart wall
- Short text beside the chart, or none
- Native, editable charts from the starter deck, not pictures of charts
- In a dimmed room, use the dark slide: navy background, light-blue focus

Chart wall



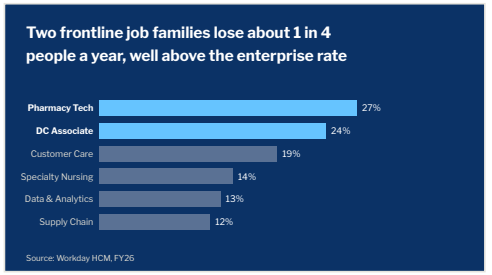
Try to avoid
A label title, four charts and a column of bullets.

One message



Recommended
Headline as title, one chart, a source line.

Dimmed room



Recommended
The same slide on navy; focus in light blue.

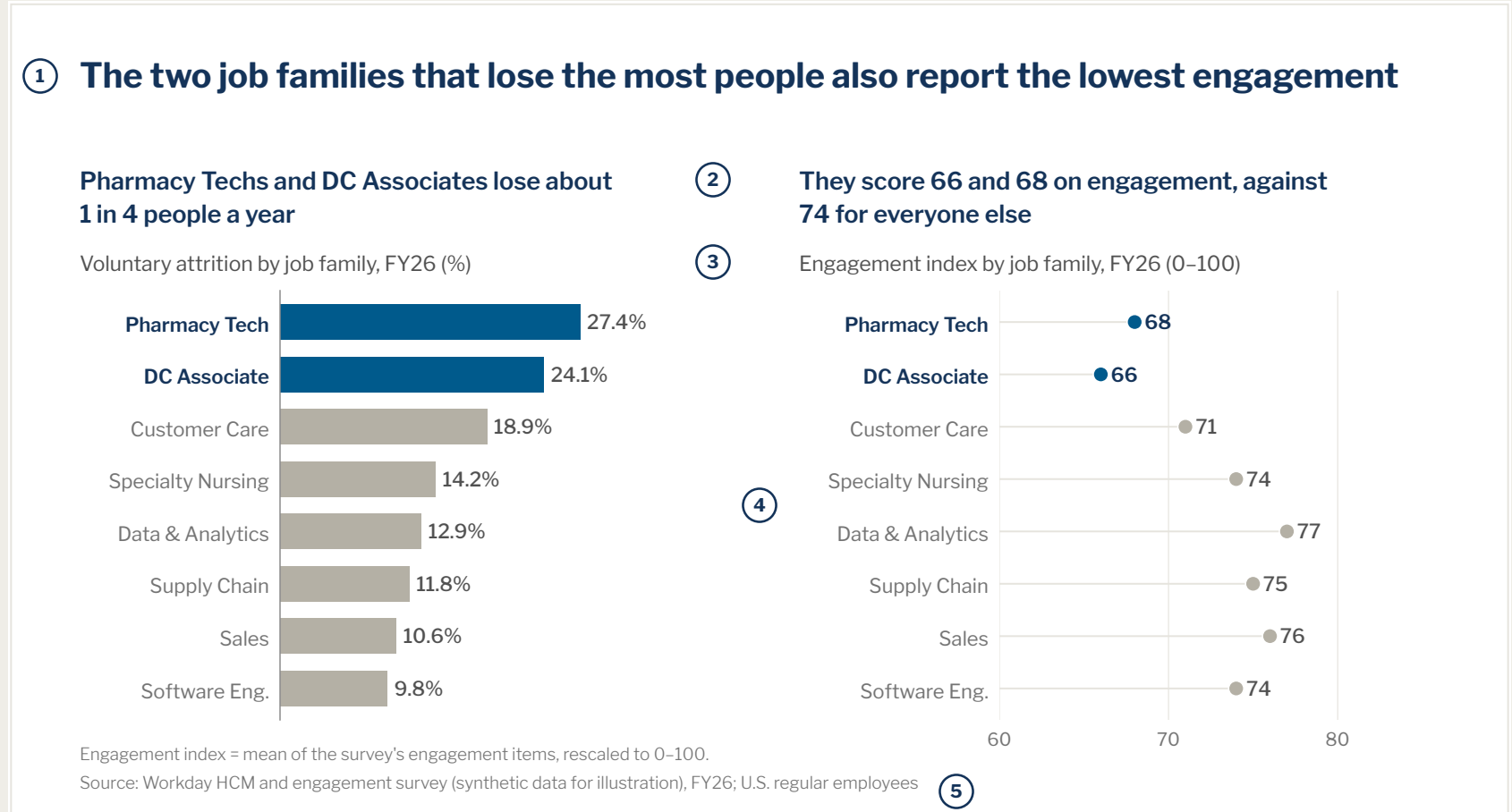
FORMATS

Two charts on a slide

One chart per slide is the default. When the message needs two pieces of evidence, the titles form a hierarchy.

1. **Slide title, Bold**
   The one message both charts support.
2. **Chart headline, SemiBold**
   Each chart's own finding: still an insight, never a label like "Engagement by job family".
3. **Subtitle per chart**
   Metric, unit, scope and period, as always.
4. **Aligned and consistent**
   Headlines on one top line; same rows, same order, same highlight.
5. **One source line**
   Shared at the bottom when the charts share sources.

Two is the limit. A third chart means a second slide.



FORMATS

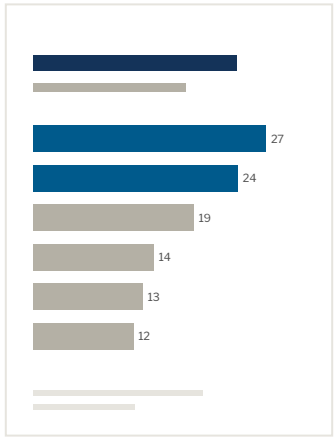
Reports and print

On paper the reader sets the pace, so charts can carry more context: full footnotes, more categories, a second chart.

- Fit the page grid: full page, two-column or one column
- Export vector (PDF, SVG), or PNG at 300 dpi
- Nothing below 9 pt; check the printed size, not the screen
- Grey must survive a black-and-white printer: never rely on color alone

Featured

One chart takes a page, for the exhibit the report is built around.



Editorial

Charts sit in the two- or three-column grid next to the text that discusses them.



Inline

A small chart inside one text column drives home a single point.



FORMATS

Web and mobile

A phone is narrow and read at arm's length. Adapt the chart, don't just shrink it.

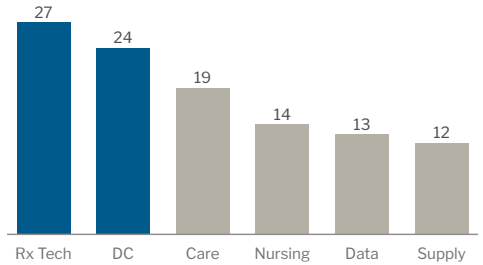
Resize and reorient. Design for portrait; columns become horizontal bars.

Adjust the hierarchy. When end labels don't fit, use a compact key above the chart and move the insight to the top.

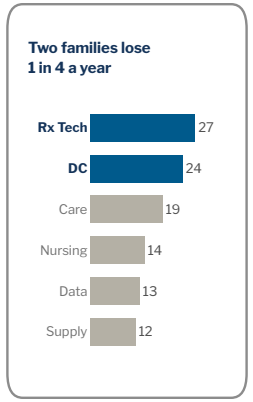
Reduce complexity. One highlighted line, the rest faint; the insight next to it.

Ensure legibility. Fewer text elements, shorter headline, nothing below 11 px.

Resize and reorient

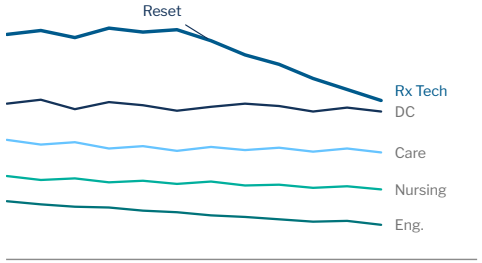


Desktop

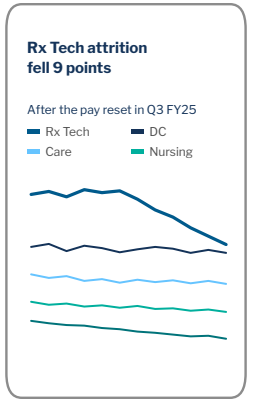


Phone

Adjust the hierarchy

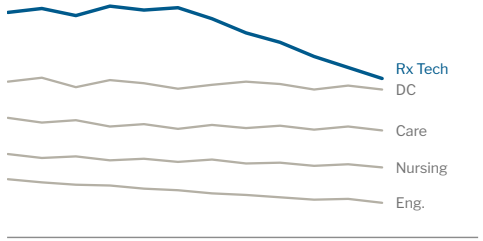


Desktop

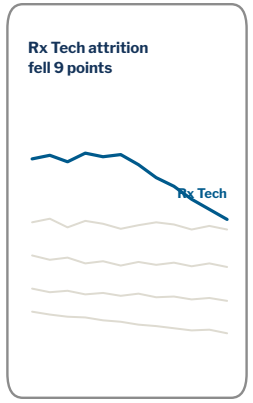


Phone

Reduce complexity



Desktop



Phone

Test at 390 px

In HTML, the starter card (`templates/html/chart-card.html`) already swaps line-end labels for a top key below 600 px. Check every web chart at 390 px wide, the width of a typical phone, before it ships.

In Power BI, build a separate **mobile layout** for each page people will open on a phone.

FORMATS

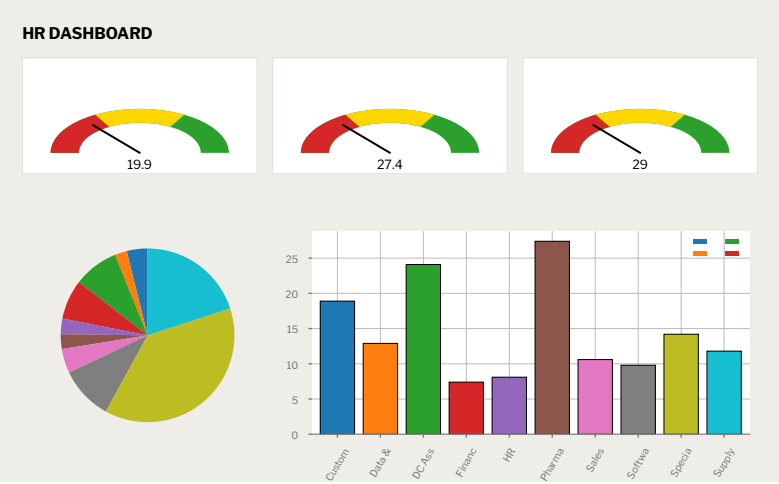
Dashboards (Power BI and Fabric)

Dashboards are exploratory, but the same rules apply: grey context, one blue focus, insight titles, minimum group sizes.

- One question per page; the title states the current answer (a DAX measure can write it)
- KPI cards top-left: the number, its label, and the comparison
- Highlights driven by measures, not by clicking bars
- Minimum n enforced in the measures, so tooltips and drill-throughs inherit it
- No gauges, pies of many slices, or rainbow defaults

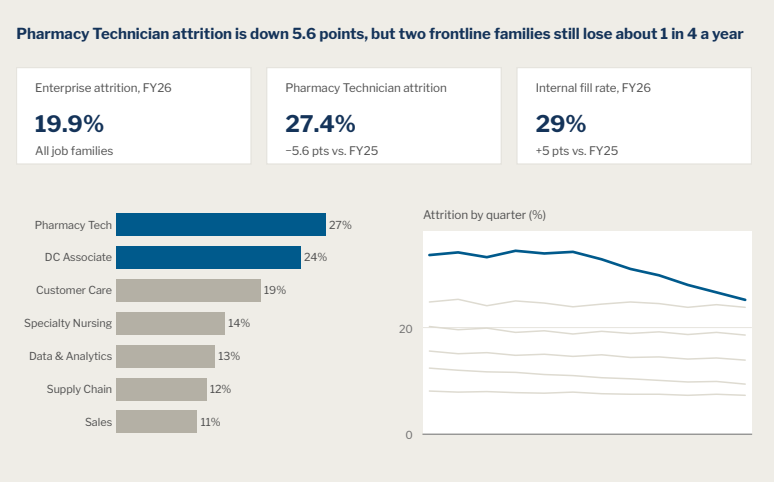
Set-up: docs/powerbi-setup.md and the theme in `dist/powerbi/`.

Tool defaults



Try to avoid
Gauges, a rainbow pie and columns, a label title. Nothing says what matters.

Built to answer a question



Recommended
An insight title, three KPI cards with comparisons, a highlighted bar and the trend.

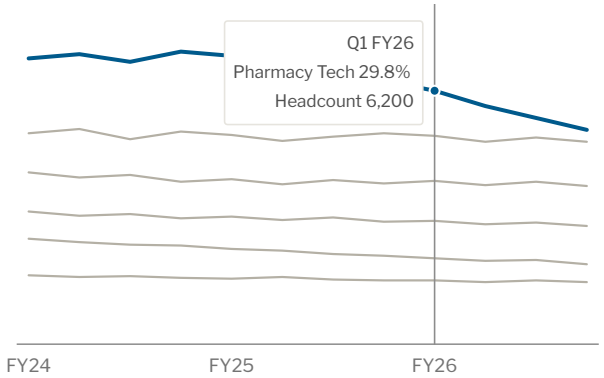
Interactivity with a purpose

Interaction lets readers explore at their own pace, but every control must earn its place.

- The default view tells the story without any clicks
- Each interaction answers a question the reader will have
- Tooltips add detail; they never carry the main point
- Filters and tooltips respect the minimum group size

Cursor tracking

A rule and a tooltip follow the cursor on time series, where reading values off the axis is hard.



Interaction	Use it for
Hover	Detail on one mark: the value, n, definition
Cursor tracking	Exact values on line charts
Toggles and filters	Switching series or groups on and off
Sliders	Scenario inputs in SWP models: attrition, hiring volume
Range selection	Narrowing time or values on an axis
Zoom	Dense time series
Play and scrub	Change over time, when motion is the point

FORMATS

Animation and scrolly-telling

Motion should help the viewer follow the data, never decorate it.

Sequence on purpose. Axes first, then data, then the insight: the build order is the reading order.

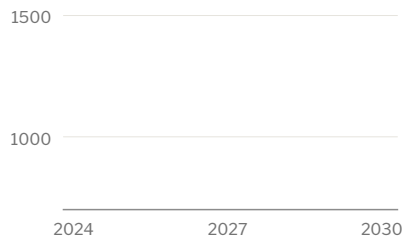
Move naturally. Ease in and out; lines draw left to right; bubbles grow from their center. No overshoot or bounce.

Say what you see. Narration and text match what is moving on screen, one point at a time.

Scrolly-telling: the chart stays pinned and updates as the text scrolls. On phones use inline snapshots, and never hijack the scroll.

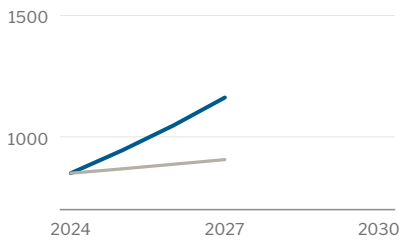
FRAME 1

The frame: axes and labels.



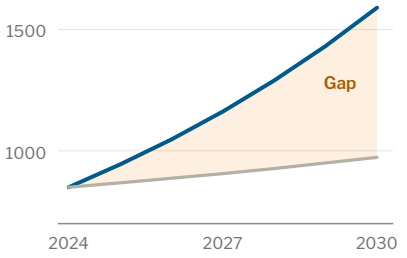
FRAME 2

Lines draw in from the left.



FRAME 3

The gap shades in once both lines land.



Avoid motion without purpose. If nothing changes in the data, nothing should move.

Avoid distraction. One moving thing at a time; split complex charts into short sequences.

Exits mirror entrances. Leave the way you arrived, so the viewer can predict it.

FORMATS

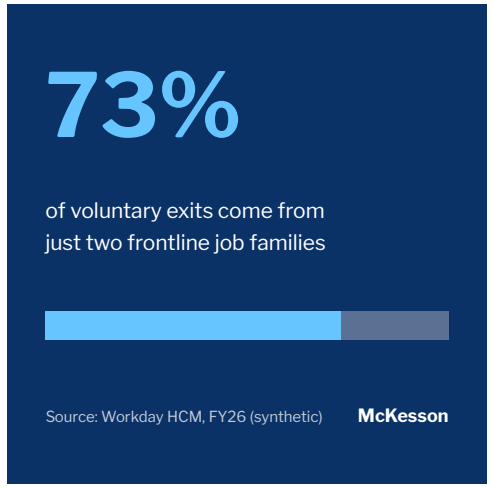
Social, infographics and icons

Single-asset graphics are seen out of context, on a phone, in a second.

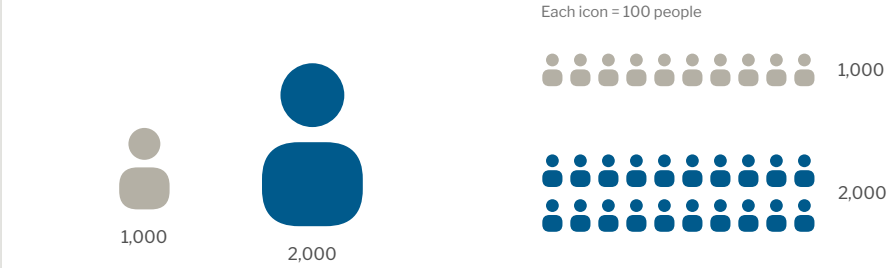
- One insight, readable without zooming
- Self-contained: headline, key number, source and brand mark
- Three-tone palettes from the tokens: navy with white and light blue
- Icons only when they help; if it's clear without one, leave it out

Use icons as tally units (one icon = a fixed number of people), never scaled up to show a bigger value.

Social card



Icons for quantities



Try to avoid
Doubling the value doubles height and width: the icon looks four times bigger.

Recommended
Repeat a unit icon: 10 versus 20.

CHEAT SHEET

Cheat sheet

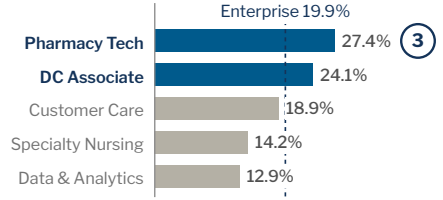
Check before you send

- [ ] The headline is an insight, and every number in it is true
- [ ] Subtitle: metric, unit, scope and period
- [ ] Grey context; at most 2 highlights, matching the headline
- [ ] Values labelled; value axis, gridlines and legend deleted
- [ ] Bars start at zero; no dual axes
- [ ] Sorted by value, or in a natural order
- [ ] True minus (−), explicit + on changes, "pts" for point changes
- [ ] Metric defined; source names system, date and population
- [ ] No group under **10 people**, even in hover text
- [ ] Nothing overlaps or clips; nothing below 9 pt

1 · EVERY CHART, ONE ANATOMY

Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate (1)

Voluntary attrition by job family, FY26 (%) (2)



Source: Workday HCM (synthetic), FY26 (4)

(1) Headline = the insight · (2) Subtitle = the measure · (3) Grey context, blue focus, values on the bars · (4) Source

2 · WRITE THE HEADLINE LAST

Who or what + verb + number + so what. Compute every number from the data.

LABEL	Attrition by job family
VAGUE	Attrition varies across job families
INSIGHT	Two frontline job families lose about 1 in 4 people a year

3 · GREY FIRST, THEN COLOR THE POINT

Context everything else	#B4B0A5
Focus what the headline names	#005A8C
Accent (navy) the one exception	#143359
Light blue a series, in the uniform pattern	#66C4FF
Orange worse only: below target, shrinking	#EF8200
Total totals, targets (dashed)	#143359

At most 2 highlights. Orange never highlights; it means worse. Amber is never used.

4 · TYPE (LIBRE FRANKLIN, 16:9 SLIDE)

Slide title	Bold	24 pt
Chart headline	SemiBold	18 pt
Subtitle	Regular	14 pt
Data labels	Medium	12 pt
Axis, categories	Regular	11 pt
Footnote, source	Light	9 pt
Big number	Bold	54 pt

One chart: the slide title is its headline. Two: each its own SemiBold insight headline.

5 · PICK THE CHART FROM THE QUESTION

- How big? Sorted bar
- Made of what? 100% bar, waterfall
- Over time? Line
- Who moved? Slope, dumbbell
- How spread? Strip, histogram
- Related? Scatter
- Flows? Sankey · Where? Map
- One number? Big number

Full version: the chart chooser.

6 · NEVER

- Red vs. green, rainbows, or color as the only cue
- 3D, shadows, gradients, borders
- Truncated bars or dual axes
- A legend where labels fit
- More than 2 charts on a slide
- Pictures of charts: use native, editable charts

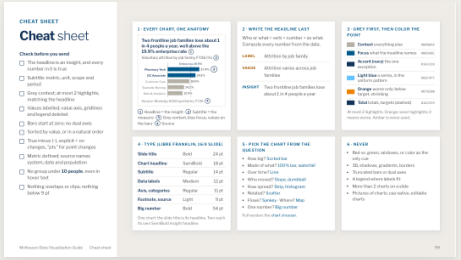
TOOLKIT

The starter kit

Everything in this guide is built into files you can start from, so the right answer is the default answer.

Every color, font size and chart constant comes from one file, `tokens/tokens.json`. Change it there, run the build, and every template, theme and this guide update together.

Pin these two. The cheat sheet and the chart chooser are pages of this guide (top bar), and print as one-page PDFs.



Cheat sheet · PDF

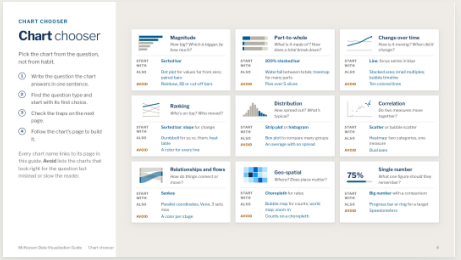


Chart chooser · PDF

File	For	Where
PowerPoint template	New decks: theme, fonts, layouts	`dist/office/`
Starter deck	Four native charts to copy	`dist/office/`
Chart workbook	Palette and Excel charts	`dist/office/`
Power BI theme	Report colors and fonts	`dist/powerbi/`
Web theme	CSS variables, Plotly, Vega-Lite	`dist/web/`
mckviz	Python: matplotlib and Plotly	`python/`
R theme	ggplot2	`dist/r/`
Agent rules	Coding agents in any repo	`dist/agents/`
Libre Franklin	The typeface (SIL OFL)	`fonts/`

TOOLKIT

Set up once

Ten minutes per tool, and every chart you make afterwards starts on-brand.

Detailed steps, troubleshooting and screenshots:

- docs/office-setup.md
- docs/powerbi-setup.md
- AGENTS.md (Python, HTML, R)

Fonts (everyone)	Install the Libre Franklin TTFs from `fonts/LibreFranklin/` (right-click › Install for all users). Without them, Office and Power BI fall back to Arial or Segoe UI.
PowerPoint and Excel	Copy `dist/office/Theme Colors/McKesson Data Viz.xml` to `%APPDATA%\Microsoft\Templates\Document Themes\Theme Colors\` and the Theme Fonts file to `…\Theme Fonts\`. Then **Design › Variants › Colors / Fonts** (PowerPoint) or **Page Layout › Colors / Fonts** (Excel). New decks start from `McKesson-DataViz.potx`.
Power BI	**View › Themes › Browse for themes** and pick `dist/powerbi/mckesson-dataviz-theme.json`. Viewers without Libre Franklin see Segoe UI, so leave room in labels.

Python	`git clone https://github.com/amccown-mck/Data-Viz-Guide  # then, in it:`
	`pip install -e "./python[examples]"   # editable, so the fonts are found`
	`import mckviz as mv`
	`mv.use()          # fonts, matplotlib style, Plotly template "mck"`
	`C = mv.colors     # C.focus, C.context, C.accent, C.negative ...`

HTML	Link `dist/web/mck-viz.css` for the CSS variables and chart-card styles; add `mck-plotly-template.js` for Plotly or use `vega-lite-config.json` as the Vega-Lite `config`. Start from `templates/html/chart-card.html`.
R	`source("dist/r/theme_mck.R")`, then `ff <- mck_font()` and `theme_mck(base_family = ff)`. Render with `ragg`.

TOOLKIT

Check before you send

Run this list on every chart that leaves the team, whoever or whatever made it.

In Python, `mv.lint(fig)` checks the mechanical items for you: titles and source present, at most six colors, palette only (amber is an error), no red–green, bars from zero, labels versus axis, legends, dual axes, pies, rotated or tiny text. It also reminds you to check any orange carries meaning.

No tool can tell whether the headline is true or the chart fits the question. That part needs a person, or an agent told to check.

Rendered from docs/qa-checklist.md, so it always matches.

MESSAGE

- [ ] Headline is a **full-sentence insight**, ≤ 2 lines, no final period, balanced line breaks.
- [ ] Every claim in the headline is **visibly supported** (recompute counts, multipliers, "all/most/none").
- [ ] Subtitle states **metric, unit, scope and period**, and nothing opinionated.
- [ ] The chart type matches the question (choosing-a-chart.md).
- [ ] One message per chart; ≤ 2 charts per slide.

CLUTTER

- [ ] Direct labels instead of a legend where possible; legend (if any) top-left, in data order.
- [ ] No borders, 3D, shadows, gradients, heavy gridlines, tick marks, rotated axis titles.
- [ ] Value axis removed when values are labelled.
- [ ] Nothing overlaps: labels, callouts, legend, headline, reference lines.

ACCESSIBILITY AND DELIVERY

- [ ] Text contrast ≥ 4.5:1; fills < 3:1 carry labels.
- [ ] Alt text / text summary provided (headline + key numbers).
- [ ] Right format: native Office chart, SVG/PDF or 300 dpi PNG for print, 200 dpi PNG for screens, or responsive HTML.
- [ ] Mobile/dark variants checked if the chart will be shown that way.

COLOR AND EMPHASIS

- [ ] Highlight follows the headline: grey context, blue focus, navy accent; **≤ 2 highlighted** items. Orange appears only where something is worse (negative, below target).
- [ ] Categorical colors (if any) come from the fixed order, ≤ 6, consistent across the deck.
- [ ] No red–green contrasts; nothing encoded by color alone.
- [ ] Focus category labels are bolded/colored to echo the highlight.

TYPOGRAPHY AND LAYOUT

- [ ] Libre Franklin (or approved fallback); role sizes per typography-and-layout.md; nothing below 9 pt / 11 px.
- [ ] Headline, subtitle, key, footnote and source **share one left edge**.
- [ ] Sentence case; consistent terminology (FTE vs. headcount; fiscal vs. calendar year).

DATA INTEGRITY

- [ ] Bars/columns start at **zero** (waterfall exception is marked and footnoted).
- [ ] Bubble **area** ∝ value; pie/donut sums to 100% and has ≤ 5 segments.
- [ ] No dual axes (or justified and clearly labelled).
- [ ] Sorted by value, or in a natural order (time, bands, levels).
- [ ] Small multiples share a scale.
- [ ] Numbers rounded sensibly; units in subtitle; true minus (−); consistent K/M/B.

SOURCE AND ETHICS

- [ ] Footnotes define metrics and caveats.
- [ ] Source line names the system(s), **as-of date** and **population**.
- [ ] No group smaller than the **minimum n** (default 10), including in tooltips, filters and drill-downs.
- [ ] Protected attributes used only for approved analyses, aggregated.
- [ ] Forecasts/estimates are labelled and show a range or scenario.

TOOLKIT

For coding agents

Claude, Copilot and other agents follow the same rules as people, automatically.

In this repo, agents read `AGENTS.md` on their own (Claude through `CLAUDE.md`, Copilot through `.github/copilot-instructions.md`, plus per-language rules for Python, web and R files).

In any other repo, copy one file: `dist/agents/mck-viz-rules.md`. It is generated from the tokens, so its colors are always current.

Ask for a chart like this

"Make a McKesson chart of voluntary attrition by job family for FY26 from `examples/data/attrition_by_job_family.csv`, for a leadership deck. Highlight the families above the enterprise rate."

What comes back

The `mck-chart` skill walks the five steps (question, chart type, build, headline, QA), runs `mv.lint`, looks at the rendered chart, and returns the file, a chart brief and alt text.

The chart brief

```
Question:    Where is attrition concentrated?
Mode:        explanatory     Medium: slide
Chart:       sorted bar (cookbook/bar.md)
Focus:       2 families > enterprise rate
Headline:    Two frontline job families lose
             about 1 in 4 people a year ...
Subtitle:    Voluntary attrition by job
             family, FY26 (%)
Source:      Workday HCM (synthetic), 31 Mar 2026
Min n check: smallest group 700 (>= 10)
QA:          pass
```